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RESEARCH DESIGN

A research design is a master plan specifying the methods and procedures for collecting and analyzing the data. It is a strategy or blueprint that plans the action for carrying through the research project data.

A research design involves a series of rational decision-making choices depending upon the various options available to the researchers. Broadly it is composed of different elements like: the purpose of the study, the unit of analysis, time dimension, mode of observation, sampling design, observation tools, data processing, and data analysis. Let us look at each one of these elements.

1. Purpose of the Study

From the perspective of purpose of the study, a research can be exploratory, descriptive, and explanatory (the distinctions we have already discussed). As we have already covered a number of steps in the research process, at this stage it is assumed that we are pretty sure about what we are looking for whereby we have gone much beyond the stage of an exploratory study (all studies have elements of exploration in them).

Beyond the exploratory stage now we are entering into the formal stage of delineating the plan for data collection, data processing, and data analysis. Here our focus is on whether our study is going to be a *descriptive* or *explanatory*. The essential difference between descriptive and explanatory studies lies in their objectives. **If the research is concerned with finding out who, what, where, when, or how much, then the study is descriptive.** If it is concerned with learning why – that is how one variable produces changes in another – it is causal. Research on crime as such is descriptive when it measures the types of crimes committed, how often, when, where, and by whom. **In an explanatory study, we try to explain relationships among variables – for instance, why the crime rate is higher in locality A than in locality B. Every explanatory study in the beginning is likely to be descriptive as well.**

Methodological rigor increases as one moves from exploratory study to explanatory study, which may encompass hypothesis testing involving multiple methods of data collection, sophistications in sampling designs, formulation of instruments of data collection, data processing, and data analysis. Since the purpose of the study is likely to determine how rigorous the research design is likely to be, therefore, the researcher would decide very early on about the purpose of his/her study.

Within the explanatory study, researcher may further decide about the type of investigation i.e. causal versus correlational. The researcher must decide whether a causal or correlational study is needed to find an answer to the issue at hand. The former is done when it is necessary to establish a definitive cause-and-effect relationship. If the researcher just wants a mere identification of important factors “associated with” the problem, then a correlational study is called for. Whether the study is basically a correlational or causal will help in deciding about the mode of observation – survey study or an experimental study.

2. Unit of Analysis

The unit of analysis refers to the level of aggregation of the data collected during the subsequent data analysis stage. If, for instance, the problem statement focuses on how to raise the motivational levels of employees in general, then we are interested in individual employees in the organization and would have to find out what we can do to raise their motivation. Here the unit of analysis is the **individual**. We will be looking at the data gathered from each individual and treating each employee’s response as an individual data source.

If the researcher is interested in studying two-person interactions, then several two-person groups (also known as dyads) will become the unit of analysis. Analysis of husband-wife interactions in families and

supervisor-subordinate relationship at the work place, teacher-student relationship in the educational institution are good examples of dyads as unit of analysis.

If the problem statement is related to group effectiveness, the unit of analysis would be at group level. In other words, even though we may gather relevant data from all individuals comprising, say six groups, we would aggregate the individual data into group data so as to see the differences among six **groups**. If we compare different departments in the organization, then data analysis will be done at the department level – that is, the individuals in the department will be treated as one unit – and comparisons made treating the department as a unit of analysis.

The research question determines the unit of analysis. Keeping the research question in view, it is necessary to decide on the unit of analysis since the data collection methods, sample size, and even the variables included in the framework may sometimes be determined or guided by the level at which the data are aggregated for analysis.

Units of analysis in a study are typically also the *units of observation*. Thus, to study voting intentions, we would interview (observe) individual voters. Sometimes, however, we “observe” our units of analysis indirectly. For example, we might ask husbands and wives their individual voting intentions, for purpose of distinguishing couples who agree and disagree politically. We might want to find out whether political disagreements tend to cause family disharmony, perhaps. In this case, our unit of analysis would be families, though the *unit of observation* would be the individual wives and husbands.

3. Time Dimension

Do we make the observations more or less at one time or over a long period, former called as cross-sectional studies and the latter as longitudinal studies. While planning the strategy for data collection the time dimension may be an important component.

Cross-Sectional Studies are carried out **once** and represent a snapshot of one point in time. Data are collected just once, perhaps over a period of days or weeks or months, in order to answer the research question.

Longitudinal Studies are repeated over an extended period. The advantage of longitudinal studies is that it can track changes over time. For example, the researcher might want to study employees’ behavior before and after a change in the top management, so as to know what effects the change accomplished. Here, because data are gathered at two different points in time, the study is not cross-sectional or of the one-shot kind, but is carried longitudinally across a period of time. Such studies, as when data on the dependent variable are gathered at two or more points in time to answer the research question, are called longitudinal. Longitudinal studies can be *panel studies* and *cohort studies* which were discussed earlier.

4. Researcher Control of Variables

In terms of researcher’s ability to manipulate variables, we can differentiate between experimental and ex post facto design. **In an experiment, the researcher attempts to control and/or manipulate the variables in the study.** It is enough that we can cause variables to be changed or held constant in keeping with our research objectives. Experimental design is appropriate when one wishes to discover whether certain variables produce effects in other variables. Experimentation provides the most powerful support possible for hypothesis of causation.

Experimental studies can be contrived and non-contrived. Research can be done in the natural environment where work proceeds normally (i.e. in non contrived setting) or in artificial, contrived setting. **Correlational studies are invariably conducted in non contrived settings, whereas most rigorous causal studies are done in contrived lab settings.** **Correlational studies doe in organizations are called**

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field studies. Studies conducted to establish cause-and-effect relationship using the same natural environment are called **field experiments**. Here the researcher does not interfere with the natural occurrence of events in as much as independent variable is manipulated.

Experiments done to establish cause and effect relationship beyond the possibility of the least doubt require the creation of an artificial, contrived environment in which all the extraneous factor are strictly controlled. Similar subjects are chosen carefully to respond to certain manipulated stimuli. These studies are referred to as **lab experiments**.

With an **ex post facto** design, investigators have no control over the variables in the sense of being able to manipulate them. They can only report what has happened or what is happening. It is important that the researchers using this design not influence the variables; to do so introduces bias. The researcher is limited to holding factors constant by judicious selection of subjects according to strict sampling procedures and by statistical manipulation of findings. **Survey research is an example of such study.**

5. Choice of Research Design: Mode of Observation

There could be number of ways to collect the data depending upon whether the study is quantitative or qualitative, descriptive or explanatory, cross-sectional or longitudinal, and contrived or non-contrived, the researcher decides about the mode of observation. **The modes could be like: survey, experiment, communication analysis (content analysis) field observation, case study, focus group discussion.**

6. Sampling Design

The basic idea of sampling is that by selecting some of the elements in population, we may draw conclusions about the entire population. A population element is the subject on which the measurement is being taken. It is the unit of analysis. Sampling has its own advantages and disadvantages. Depending upon the nature of the study the researchers decides about following appropriate type of sampling design.

7. Observation Tools

Observation tool mostly used by social researchers are: questionnaire, interview schedule, Interview guide, and check list. In the research design, the researcher will specify the tools of data collection along the logic justifying the appropriateness of the selected tool.

8. Field Data Collection

Depending upon the mode of observation, the researcher will outline the procedure for field operations. The researcher will try to look after the questions like: How the data will be collected? Who will be responsible for the collections of data? What training will be imparted to the field functionaries? How will the quality control of data be maintained?

9. Data Processing and Data Analysis

In the research design the researcher is required to tell how the data shall be processed (manually, mechanically), and analysis plans explicated. In case the qualitative data are to be quantifies the procedures should be spelled out. The procedures for the construction of score Indexes, if any, should be explained.

The research design should also say something about the analysis plan, the use of statistics, and the inferences to be drawn.

Survey Research: An Overview

Surveys require asking people, who are called **respondents**, for information, using either verbal or written questions. Questionnaires or interviews are utilized to collect data on the telephone, face-to-face, and through other communication media. The more formal term **sample survey** emphasizes that the purpose of contacting respondents is to obtain a representative sample of the target population. Thus, a **survey** is defined as a method of gathering **primary data** based on communication with a representative sample of individuals.

Steps in Conducting a Survey

The survey researcher follows a deductive approach. He or she begins with a theoretical or applied research problem and ends with empirical measurement and data analysis. Once a researcher decides that survey is an appropriate method, basic steps in a research project can broadly be divided into six sub-steps.

1. **Develop the hypothesis; decide on type of survey (mail, interview, telephone); write survey questions (decide on response categories, design lay out).** The researcher develops an instrument – a survey questionnaire or interview schedule – that he or she uses to measure variables. Respondents read the questions themselves and mark answers on a *questionnaire*. An *interview schedule* is a set of questions read to the respondent by an interviewer, who also records the responses. To simplify the discussion, we will use only the term *questionnaire*.
2. **Plan how to record data; pilot test survey instrument.** When preparing the questionnaire, the researcher thinks ahead to how he or she will record and organize data for analysis. The questionnaire is pilot tested on a small set of respondents similar to those in the final survey.
3. **Decide on target population; get sampling frame; decide on sample size; select the sample.**
4. **Locate respondents; conduct interviews; carefully record data.** The researcher locates sampled respondents in person, by telephone, or by mail. Respondents are given information and instructions on completing the questionnaire or interview.
5. **Enter data into computers; recheck all data; perform statistical analysis on data.**
6. **Describe methods and findings in research report; present findings to others for critique and evaluation.**

SURVEY RESEARCH

Research Design can be classified by the *approach* used to gather primary data. There are really two alternatives. We can *observe* conditions, behavior, events, people, or processes. Or we can *communicate* with people about various topics, including participants' attitudes, motivations, intentions, and expectations.

The **communication approach** involves surveying people and recording their responses for analysis. The great strength of the survey as a primary data collecting approach is its versatility. What media do we use for communicating with people? The traditional face to face communication (interview) for conducting surveys is still in vogue. Nevertheless, the digital technology is having a profound impact on the society as well as on research. Its greatest impact is on the creation of new forms of communication media.

Human Interactive Media and Electronic Interactive Media

When two people engage in conversation, human interaction takes place. **Human interactive media** are personal forms of communication. One human being directs a message to and interacts with another individual (or a small group). When they think of interviewing, most people envision this type of face-to-face dialogue or a conversation on telephone.

Electronic interactive media allows researchers to reach a large audience, to personalize individual messages, and to interact with members of the audience using digital technology. To a large extent electronic interactive media users are controlled by the users themselves. In the context of surveys, respondents are not passive audience members. They are actively involved in a two-way communication when electronic interactive media are utilized.

The Internet, the medium that is radically altering many organizations' research strategies, provides a prominent example of the new electronic interactive media.

Non-Interactive Media

The traditional questionnaire received by mail and completed by the respondent does not allow a dialogue or exchange of information for immediate feedback. Self-administered questionnaires printed on paper are also non-interactive.

CHOOSING A COMMUNICATION MEDIA

Once the researcher has determined that surveying is the appropriate data collection approach, various means may be used to secure information from individual. A research can conduct a survey by personal interview, telephone, mail, computer, or a combination of these media.

Personal Interviewing

A personal interview (i.e. face to face communication) is a two way conversation initiated by an interviewer to obtain information from a respondent. The differences in the roles of the interviewer and the respondent are pronounced. They are typically strangers, and the interviewer generally controls the topics and patterns of discussion. The consequences of the event are usually insignificant for the respondent. The respondent is asked to provide information and has little hope of receiving any immediate or direct benefit from this cooperation.

Personal interviews may take place in a factory, in a homeowner's doorway, in an executive's office, in a shopping mall, or in other settings.

Advantages of Personal Interviewing:

The face-to-face interaction between interviewer and respondent has several characteristics that help researchers obtain complete and precise information. Personal interviews offer many advantages.

1. The Opportunity for Feedback

Personal interviews allow for feedback. For example, an employee who is reluctant to provide sensitive information about his workplace may be reassured by the interviewer that his answers will be strictly confidential. The interviewer may also provide feedback in clarifying any questions an employee or any other respondent has about the instructions or questions. Circumstances may dictate that at the conclusion of the interview, the respondent be given additional information concerning the purpose of the study (part of debriefing). This is easily accomplished in personal interview.

2. Probing Complex Questions

An important characteristic of personal interview is the opportunity to follow up, by probing. If a respondent's answer is brief or unclear, the researcher may ask for a clearer or more comprehensive explanation. Probing implies the verbal prompts made by the interviewer when the respondent must be motivated to communicate his or her answer more fully. Probing encourages respondents to enlarge on, clarify, or explain answers. Probing becomes all the more important when the questions don't have structured response categories. The complex question that cannot easily be asked in telephone or mail surveys can be handled by skillful interviewers.

3. Length of Interview

If the research objective requires an extremely lengthy questionnaire, personal interviews may be the only alternative. Generally, telephone interviews last fewer than 10 minutes, whereas a personal interview can be much longer, perhaps more than an hour. A rule of thumb for mail questionnaire is that it should not be more than six pages.

4. High Completion Rate

The social interaction between a well-trained interviewer and a respondent in personal interview increases the likelihood that the respondent will answer all items on the questionnaire. The respondent who grows bored with a telephone interview may terminate the interview at his or her discretion simply by hanging up the phone. A respondent's self administration of a mail questionnaire requires more effort. Rather than writing a long explanation, the respondent may fail to complete some of the questions on the self administered questionnaire. This will be an **item non-response** – that is, failure to provide an answer to a question. It is less likely to happen with an experienced interviewer and in a face to face situation.

5. Props and Visual Aids

Interviewing respondents face to face allows an investigator to show them a new product sample, a sketch of proposed office, or some other visual aid. The respondents can even taste samples of different products and can give their evaluations. Such an evaluation cannot be done in telephone interview or mail survey.

6. High Participation Rate

While some people are reluctant to participate in a survey, the presence of an interviewer generally increases the percentages of people willing to complete the interview. Respondents are not required to do any reading or writing – all they have to do is to talk. Most people enjoy sharing information and

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insights with friendly and sympathetic interviewers. Certainly, in personal interviews there is a higher rate of participation rate of the respondents compared with mail surveys and telephone interviews.

7. Observation of the Non-Verbal Behavior

In a personal interview, the interviewer can catch the facial expressions, body movements, and, depending upon the goals of the study, the environment of the respondent. Such observations may supplement the verbal information.

8. Non-Literates can participate in Study

Since the respondent has neither to read nor to write, therefore, an illiterate or a functionally illiterate person can also take part in the survey study.

9. Interviewer can Prescreen Respondent

In order to ensure that the respondent fits the sampling criteria, the interviewer can do some prescreening of the respondent. In personal interview the interviewer makes it sure that only the relevant respondent provides the information. In case of mail survey we are not sure who actually filled out the questionnaire, but in personal interview, the interview may be able to have some control over the environment of the information providers. In case there are other people around, he may make an excuse from other because he is interested in the true opinion of the sampled person.

10. CAPI – Computer Assisted Personal Interviewing

With the use of such modern technology the responses of the respondents can be entered into a portable microcomputer to reduce error and cost.

Disadvantages of Personal Interviewing:

1. High Cost

Personal interviews are generally more expensive than mail, internet, and telephone surveys. The geographic proximity of respondents, the length of the questionnaire, and the number of people who are non-respondents because they could not be contacted all influence the cost of the personal interviews. The training of the field interviewers, supervision, and other logistical support cost may add up the total cost of the study. People usually estimate the cost of personal interviews is usually 15 times higher than the mail survey

2. Scarcity of Highly Trained Interviewers

In case of a big study (especially a sponsored study) there shall be a need of highly trained interviewers, who are not easily available. Using unqualified and untrained interviewers are likely to have a negative effect on the quality of the data and the subsequent generalizations.

3. Lack of Anonymity of Respondent

Because the respondent in a personal interview is not anonymous therefore he/she may be reluctant to provide confidential information to another person. Though the interviewer provides all the assurance for the confidentiality of the information (by not asking the name or address) but the mere fact the respondent has been located, therefore he/she may not trust.

4. Callbacks – a Labor Intensive Work

When the person selected to be in the sample cannot be contacted on the first visit, a systematic procedure is normally initiated to call back at another time. **Callbacks** or attempts to re-contact individuals selected for the sample are the major means to reducing non-response error. It is a labor intensive work and definitely increases the cost.

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5. Interviewer Influence

There is some evidence that the demographic characteristics of the interviewer influence respondents' answers. Respondent's sex, age, and physical appearance can have an effect on the responses of the respondent.

6. Interviewer Bias

Interviewer's personal likings and dis-likings, the environment, and cultural biases can affect the understanding of the responses, its recording, and its interpretation.

7. No Opportunity to Consult

The interview may take place anywhere – place of work, in the shopping mall, at home – the respondent may be unable to consult record, incase he/she has to do so for any specific question.

8. Less Standardized Wording

Despite the fact that the questions have been printed and have a specified order, these questions are read by the interviewer. The interviewers intentionally or unintentionally may not be able to use the standardized wording which may bias the data. Similarly the order of the questions may be altered.

9. Limitations in Respondents' Availability and Accessibility

Some executive officers or VIPs may not be available or accessible to interviewers. Some of them may not be willing to talk to strangers for security reasons.

10. Some Neighborhoods are Difficult to Visit

Just for security reasons some neighborhoods may not allow outsiders to enter the premises. Even the formal permission may be denied because these residents don't want to contact any strangers.

Door to Door Interviews

These are the personal interviews conducted at respondent's home or place of work. It is likely to provide more representative sample of the population than mail questionnaire. Some people may prefer to give a verbal response rather than in writing. People who do not have telephones, who have unlisted numbers, or who are otherwise difficult to contact may be reached through door to door interviews.

Door to door interview may exclude individuals living in multiple dwelling units with security systems, such as high rise apartment dwellers, or executives who are too busy to grant a personal interview during business hours.

People, who are at home and willing to participate, especially if interviews are conducted in the day time, are somewhat more likely to be stay-at-home "moms" or retired people. These and other variables related to respondents' tendencies to stay at home may affect participation.

Intercept Interviews in Malls and Other High-Traffic Areas

Personal interviews conducted in shopping malls are referred to as mall intercept interviews. Interviewers generally stop and attempt to question shoppers at a central point within the mall or at the entrance. These are low cost. No travel is required to the respondent's home – instead the respondent

comes to the interviewer, and thus many interviews can be conducted quickly. The incidence of refusal is high, however, because individuals may be in a hurry.

In mall intercept interviews the researcher must recognize that he or she should not be looking for representative sample of the total population. Each mall will have its own customer characteristics of customers.

Personal interviews in the shopping mall may be appropriate when demographic factors are not likely to influence the survey's findings or when target group is a special population segment, such as the parents of children of bike-riding age.

End 20 Lectures

INTERCEPT INTERVIEWS IN MALLS AND OTHER HIGH-TRAFFIC AREAS

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Telephone Interviewing

Telephone interviewing has been a mainstay of commercial survey research. The quality of data obtained by telephone may be comparable to that collected in personal interviews. Respondents may even be more willing to provide detailed and reliable information on a variety of personal topics over the telephone than in personal interviews. Telephone surveys can provide representative samples of general population in most industrialized countries.

Central Location Interviewing

Research agencies and interviewing services typically conduct all telephone interviews from central location. WATS (Wide-Area Telecommunications Service) lines, provided by long distance telephone service at fixed rates, allow interviewers to make unlimited telephone calls throughout the entire country or within a specific geographic area. Such central location interviewing allows firms to hire staffs of professional interviewers and to supervise and control the quality of interviewing more effectively. When telephone interviews are centralized and computerized, the research becomes even more cost-effective.

Computer-Assisted Telephone Interviewing (CATI)

Advances in computer technology allow responses to telephone interviews to be entered directly into a computer in a process known as **computer assisted telephone interviewing (CATI)**. Telephone interviewers are seated at computer terminals. A monitor displays the questionnaire, one question at a time, along with pre-coded possible responses to each question. The interviewer reads each question as it is shown on the screen. When the respondent answers, the interviewer enters the response into the computer, and it is automatically stored in the computer's memory when the computer displays the next question on the screen. A computer assisted telephone interviewing requires that answers to the questions be highly structured. A lot of computer programming facilitates telephone interviewing.

The Strengths of Telephone Interviewing:

1. High Speed

The speed of data collection is a major advantage of telephone interviewing. For example, union officials who wish to survey members' attitudes toward a strike may conduct a telephone survey during the last few days of the bargaining process. Whereas data collection with mail or personal interviews

can take several weeks, hundreds of interviews can be conducted literally overnight. When the interviewer enters the respondents' answers directly into a computerized system, data processing can be done even faster.

2. Saves Cost

As the cost of personal interviews continues to increase, telephone interviews are becoming relatively inexpensive. It is estimated the cost of telephone interviewing is less than 25% of the door to door personal interviews.

3. Callbacks

An unanswered call, a busy signal, or a respondent who is not at home requires a callback. Telephone callbacks are substantially easier and less expensive than personal interview callbacks.

4. Can Use Computerized Random Digit Dialing

Use of

5. Expanded Geographic Area Coverage without Increasing the Cost

6. uses fewer but highly Skilled Interviewers

7. Reduced Interviewer Bias

8. Better Access to hard-to-reach respondents through repeated callbacks

In some neighborhoods, people are reluctant to allow stranger to come inside their house, or even stop on the doorstep. The same people, however, may be preferably willing to cooperate with a telephone survey request. Likewise, interviewers may be somewhat reluctant to conduct face-to-face interviews in certain neighborhoods, especially during the evening hours. Telephone interviewing avoids these problems.

9. Use Computer Assisted Telephone Interviewing (CATI)

Responses can be directly entered into computer file to reduce error and cost.

Weaknesses of Telephone Interviewing

1. Absence of Face-to-Face Contact

Telephone interviews are more impersonal than face-to-face interviews. Respondents may answer embarrassing or confidential questions more willingly in a telephone interview than in a personal interview. People may be more comfortable to answer sensitive and threatening questions through mail surveys.

Absence of face-to-face contact can be a liability. The interviewer and the respondent don't see each other what they are doing (Responding still responding when he/she is thinking and not speaking. Has the interviewer finished recording the information)?

2. Response Rate is lower than for Personal Interviews

Some individuals refuse to participate in telephone interviews. Telephone researchers can run into several roadblocks when trying to obtain executives' cooperation at work. Participants find it easier to terminate a phone interview.

3. Lack of Visual Medium

Since visual aids cannot be utilized in telephone interview, research that requires visual material cannot be conducted by phone..

4. Limited Duration

Length of the interview is limited. Respondents who feel they have spent too much time in the interview will simply hang up. (a good rule of thumb is to plan telephone interviews to be approximately 10 minutes long).

5. Many Numbers are unlisted or not working

6. Less Participant Involvement

Telephone surveys can result in less thorough responses, and those interviewed by phone find the experience to be less rewarding than a personal interview. Participants report less rapport with telephone interviewers than with personal interviewers.

7. Distracting Physical Environment

Multiple phones distract the interview situation which may affect the quality of the data.

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Self-Administered Questionnaires

The self administered questionnaire has become ubiquitous in modern living. Service evaluations of hotels, restaurants, car dealerships, and transportation providers furnish ready examples. Often a short questionnaire is left to be completed by the participants in a convenient location or is packed with the product. Self-administered mail questionnaires are delivered not only through postal services, but also via fax and courier service. Other modalities include computer-delivered and intercept studies.

Mail Questionnaire

A mail survey is a self administered questionnaire sent to respondents through the mail. This paper-and-pencil method has several advantages and disadvantages.

Advantages of Mail Questionnaire

1. Geographic Flexibility

Mail questionnaires can reach a geographically dispersed sample simultaneously and at a relatively low cost because interviewers are not required. Respondents in isolated areas or those who are otherwise difficult to reach (executives) can be contacted more easily by mail.

2. Sample Accessibility

Researchers can contact participants who may otherwise be inaccessible. Some people, such as major corporate executives and physicians, are difficult to reach in person or by phone, as gatekeepers limit access. But the researchers can often access these special participants by mail or computer.

3. Self-Administered Questionnaires save Time

Self-administered questionnaires can be widely distributed to a large number of employees, so organizational problems may be assessed quickly and inexpensively. Questionnaires may be administered during group meetings as well as in the class rooms. The researcher can establish rapport with the respondents, can stay there for any clarifications, and may also be for any debriefing.

4. Saves Cost

Mail questionnaires are relatively inexpensive compared to personal interviews and telephone surveys. However, these may not be so cheap. Most include a follow-up mailing, which requires additional postage and printing of additional questionnaires.

5. Respondent Convenience

Mail surveys and self administered questionnaires can be filled out whenever the respondent has time. Thus there is a better chance that respondents will take time to think about their response. Many hard-to-reach respondents place high value on responding to surveys at their own convenience and are best contacted by mail. In some situations, particularly in organizational research, mail questionnaires allow respondents time to collect facts (such as records of absenteeism) that they may not be able to recall without checking. In the case of household surveys, the respondents may provide more valid and factual information by checking with family members compared with if they are giving a personal interview.

6. Anonymity

Mail surveys are typically perceived as more impersonal, providing more anonymity than the other communication modes, including other methods for distributing self administered questionnaires. Absence of interviewer can induce respondents to reveal sensitive or socially undesirable information.

7. Standardized Questions

Mail questionnaires are highly standardized, and the questions are quite structured.

Disadvantages of Mail Questionnaire

1. Low Response Rate

Mail questionnaire has very low rate of return of the filled questionnaires.

2. Low Completion Rate

There are chances that respondents leave many questions as unanswered, either because they did not understand the question or they shied away.

3. Increases Cost

The researcher keeps on waiting for the return. When enough response is not there, then the reminders are sent, and again there is a waiting time. With the reminders copies of the questionnaires are sent. All this adds to the cost of the study.

4. Interviewer's Absence

Respondent may have different interpretations of the questions. Due to the absence of the interviewer, the respondents are unable to get any help for needed clarifications.

5. No Control on Question Order

In a self administered/mail questionnaire, the respondent usually reads the whole of the questionnaire prior to answering the questions. The latter questions may influence the answers to the earlier questions; thereby it is likely to bias the data. In interview the questionnaire remains in the hands of the

interviewer, and the respondent does not know what question is likely to follow. Therefore, in interview there is a control in the question order.

6. Cannot Use Lengthy Questionnaire

Mail questionnaires vary considerably in length, ranging from extremely short postcard questionnaires to lengthy, multi-page booklets requiring respondents to fill thousands of answers. Lengthy questionnaires are usually avoided by the respondents. A general rule of thumb is that it should not exceed six pages.

7. No Control over the Environment

The researcher does not know about who filled the questionnaire

8. Cannot Catch the Non-Verbal Behavior

9. Non-Literates cannot participate

For participation in the mail/self administered questionnaire related studies, the respondents have to be educated up to a certain level. Hence the non-educated people are in a way excluded from the study.

Increasing Response Rate

Here are some guidelines for increasing the response rate. Response rate is the number of questionnaires returned or completed, divided by the total number of eligible people who were contacted or asked to participate in the survey.

Cover Letter

The cover letter that accompanies the questionnaire or is printed on the first page of the questionnaire is an important means of inducing a reader to complete and return the questionnaire. In the letter tell why **the study is important**, **who is sponsoring the study**, **how was the respondent selected**, assuring the anonymity of the respondent could help in establishing rapport and **motivating the respondent to respond**.

A personalized letter addressed to a specific individual shows the respondent that he or she is important. Including an individually typed letter on letterhead versus printed form is an important element in increasing the response rate in mail surveys.

Money Helps

The respondent's motivation for returning a questionnaire may be increased by offering monetary incentives or premiums. Although pens, lottery tickets, and variety of premiums have been used, monetary incentives appear to be the most effective and least biasing incentive. It attracts the attention and creates a sense of obligation. Money incentive works for all income categories.

Interesting Questions

The topic of the research and thus the point of the questions cannot be manipulated without changing the problem definition. However, certain interesting questions can be added to the questionnaire, perhaps in the beginning, to stimulate the respondents' interest and to induce cooperation.

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LESSON 22

SELF ADMINISTERED QUESTIONNAIRES (CONTINUED)

Increasing Response Rate

Here are some guidelines for increasing the response rate. Response rate is the number of questionnaires returned or completed, divided by the total number of eligible people who were contacted or asked to participate in the survey.

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Follow-Ups

Follow-up implies the communication of the message to respondents through different means for the return of questionnaire. After responses from the first wave of mailing begin to trickle-in, most studies use follow-up, reminder for getting the response. A follow-up may include a duplicate questionnaire or may merely be a reminder to return the original questionnaire. Multiple contacts almost always increase response rates. The more attempts made to reach people, the greater the chances of their responding.

Preliminary Notification

Advance notification, by either letter or telephone, that a questionnaire will be arriving has been successful in increasing the response rates in some situations. Advance notices that go out close to the questionnaire mailing time produce better results than those sent too far in advance. This technique presupposes a certain level of development of the country where such facilities are available. Even otherwise, it depends upon the nature of the study as well as the type of respondents selected for the study.

Survey Sponsorship

Sponsorship of the study makes a difference for motivating the respondents to return the questionnaires. It depends upon the goodwill of the sponsoring agency that can activate/deactivate the respondent to fill the questionnaire and return it. There is some evidence that "official" and "respected"

sponsorship increases the response rate. Sponsorship by well-known and prestigious organizations, such as universities or government agencies, may significantly influence response rates.

Return Envelopes

The inclusion of a stamped, self addressed envelop encourages response because it simplifies questionnaire return.

Postage

The existing evidence shows that expedited delivery is very effective in increasing response rate. First class or third class mail, stamped mail or metered mail may make a difference.

Personalization

Personalization of the mailing has no clear-cut advantage in terms of improved response rates. Neither personal inside addresses nor individually signed cover letters significantly increased response rates; personally typed cover letters proved to be somewhat effective.

Size, Reproduction, and Color

The size of the paper, the printing, and color may have some effect, though not significant, on the response rate. It is recommended to use the A-4 size paper and while sending it do not fold it. The attractive printing may be another factor influencing the return rate. If questionnaire has different parts, the use of different colors of paper may motivate the respondents to take interest in the study and return the questionnaire.

The manipulation of one or two techniques independently of all others may do little to stimulate response. May be the researcher has to make use of all the possible techniques simultaneously, so that the response rate could be increased. Such an effort is referred to as Total Design Effort (TDE).

E-Mail Surveys

Questionnaires can be distributed via e-mail. E-mail is relatively new method of communication, and many individuals cannot be reached this way. However, certain projects lend themselves to, such as internal surveys of employees or satisfaction surveys of retail buyers who regularly deal with an organization via e-mail.

The benefits of an e-mail include speed of distribution, lower distribution and processing cost, faster turnaround time, more flexibility, and less handling of paper questionnaires.

Many respondents may feel that they can be more candid in e-mail than in person or on telephone, for the same reason they are candid on other self administered questionnaires.

In many organizations the employees know that their e-mails are not secure, that “eves-dropping” by a supervisor could occur. Further maintaining the respondent’s anonymity is difficult, because a reply to an e-mail message typically includes the sender’s address. Researchers designing e-mail surveys should assure respondents that their responses will be confidential.

Not all e-mail systems have the same capacity: some handle color and graphics well; others are limited to text. The extensive differences in the capabilities of respondents’ computers and email software limit the types of questions and the layout of the questionnaire.

Internet Surveys

An internet survey is a self-administered questionnaire posted on a Web site. Respondents provide answers to questions displayed on screen by highlighting a phrase, clicking an icon, or keying in an answer. Like any other survey, Internet surveys have both advantages and disadvantages.

Advantages of Internet Surveys

Speed and Cost Effectiveness: Internet survey allow the marketers to reach a large audience (possible a global one), to personalize the individual messages, and to secure confidential answers quickly and cost effectively. The computer to computer self administered questionnaires eliminate the cost of paper, postage, data entry, and other administrative costs. Once an Internet questionnaire has been developed, the incremental cost of reaching additional respondents is marginal. Hence samples can be larger than with interviews or other types of self-administered questionnaires.

Visual Appeal and Interactivity: Surveys conducted on Internet can be interactive. The researcher can use more sophisticated lines of questioning based on the respondents' prior answers. Many of this interactive survey utilize color, sound, and animation, which may help to increase the respondents' cooperation and willingness to spend more time answering questions. The Internet is an excellent medium for the presentation of visual materials, such as photographs or drawings of product prototypes, advertisements, and movie trailers.

Respondent Participation and Cooperation: Participation in some Internet surveys occurs because computer users intentionally navigate to a particular Web site where questions are displayed. In some instances individuals expect to encounter a survey at a Web site; in other cases it is totally unexpected.

Accurate Real-Time Data Capture: The computer to computer nature of Internet surveys means that each respondent's answers are entered directly into the researcher's computer as soon as the questionnaire is submitted. In addition, the questionnaire software may be programmed to reject improper data entry.

Real-time data capture allows for real-time data analysis. A researcher can review up-to-the –minute sample size counts and tabulation data from an Internet survey in real time.

Callbacks: When the sample for Internet survey is drawn from a consumer panel, it is easy to recontact those who have not yet completed the questionnaire. Computer software can also identify the passwords of those respondents who completed only a portion of the questionnaire and send those people customized messages.

Personalized and Flexible Questioning: There is no interviewer in Internet surveys but the respondent interacts directly with the software on a Web site. In other words the computer program asks questions in sequence determined by a respondent's previous answer. The questions appear on the computer screen, and answers are recorded by simply pressing a key clicking an icon, thus immediately entering the data into the computer's memory. This ability to sequence the question based on previous responses is a major advantage of computer-assisted surveys.

Respondent Anonymity: Respondents are more likely to provide sensitive information when they can remain anonymous. The anonymity of the Internet encourages respondents to provide honest answers to sensitive questions.

Most respondents do not feel threatening to enter information into the computer because of the absence of the interviewer. They may be assured that no human will ever see their individual responses.

Response Rate: Response rate can be increased by sending e-mail friendly reminders.

Disadvantages of Internet Surveys

All People cannot Participate: Many people in the general public cannot access the Internet. And, all people with Internet access do not have the same level of technology. Many lack powerful computers or software that is compatible with advanced features programmed into many Internet questionnaires. Some individuals have minimum computer skills. They may not know how to navigate through and provide answers to an Internet questionnaire.

No Physical Incentive: Unlike mail surveys, Internet surveys do not offer the opportunity to send a physical incentive to the respondent.

SELECTING THE APPROPRIATE SURVEY RESEARCH DESIGN

The choice of communication method is not as complicated as it might appear. By comparing the research objectives with the strengths and weaknesses of each method, the researcher will be able to choose one that is suited to the needs. Nevertheless, there is no “best” form of survey. Each has advantages and disadvantages. A researcher who must ask highly confidential questions may conduct a mail survey, thus trading off the speed of data collection to avoid any possibility of interviewer bias.

To determine the appropriate technique, the researcher must ask questions such as “Is the assistance of an interviewer necessary? Are respondents likely to be interested in the issues being investigated? Will cooperation be easily attained? How quickly the information is needed? Will the study require a long complex questionnaire? How large is the budget?” The criteria – cost, speed, anonymity, and the like – may be different for each project.

If none of the choices turns out to be a particularly good fit, it is possible to combine the best characteristics of two or more alternatives into a *mixed mode*. Although this decision will incur the costs of the combined modes, the flexibility of tailoring a method to the unique need of the project is often an acceptable trade off.

22 Lectures End

LESSON 23

TOOLS FOR DATA COLLECTION

Broadly there are tools of data collection as part of communication surveys. These are:

1. Interview schedule
2. Questionnaire
3. Interview Guide

As discussed earlier interview schedule and questionnaires both are predesigned list of questions used for communication with the respondents. In the case of *interview schedule*, the list of questions remains in the hands of the interviewer who asks questions from the respondent, gets his/her response, and records the responses. *Questionnaire* is also a list of questions, which is handed over to the respondent, who reads the questions and records the answers himself. For purposes of convenience *questionnaire* will refer to both interview schedule as well as questionnaire.

Interview guide is list of topics that are to be covered during the course of interview. Interview guide is used for purposes of an in-depth interviewing. Questions on the topics are formulated on the spot. Most of the questions are open ended. The interviewer may not use the same wording for each respondent; the number of questions may be different; the sequence of questions may also be different.

23 Guidelines for Questionnaire Design

A survey is only as good as the questions it asks. Questionnaire design is one of the most critical stages in the survey research process. While common sense and good grammar are important in question writing, more is required in the art of questionnaire design. To assume that people will understand the questions is common error. People may not simply know what is being asked. They may be unaware of topic of interest, they may confuse the subject with something else, or the question may not mean the same thing to every respondent. Respondents may simply refuse to answer personal questions. Further, properly wording the questionnaire is crucial, as some problems may be minimized or avoided altogether if a skilled researcher composes the questions.

A good questionnaire forms an integrated whole. The researcher weaves questions together so they flow smoothly. He or she includes introductory remarks and instructions for clarification and measures each variable with one or more survey questions.

What should be asked?

The problem definition will indicate which type of information must be collected to answer the research question; different types of questions may be better at obtaining certain type of information than others.

1. Questionnaire Relevancy

A questionnaire is relevant if no unnecessary information is collected and if the information that is needed to solve the problem is obtained.

Asking the wrong or an irrelevant question is a pitfall to be avoided. If the task is to pinpoint compensation problems, for example, questions asking for general information about morale may be inappropriate. To ensure information relevancy, the researcher must be specific about data needs, and there should be a rationale for each item of information.

2. Questionnaire Accuracy

Once the researcher has decided what should be asked, the criterion of accuracy becomes of primary concern. Accuracy means that the information is reliable and valid. While experienced researchers believe that one should use simple, understandable, unbiased, unambiguous, and nonirritating words. Obtaining accurate answer from respondents is strongly influenced by the researcher's ability to design a questionnaire that facilitates recall and that will motivate the respondent to cooperate. Therefore avoid

jargon, slang, and abbreviations. The respondents may not understand some basic terminology. Respondents can probably tell the interviewer whether they are married, single, divorced, separated, or widowed, but providing their “marital status” may present a problem. Therefore, asking somebody about his/her *marital status* while the person may not understand the meaning of marital status is likely to mess up the information. Words used in the questionnaire should be readily understandable to all respondents.

3. Avoid Ambiguity, Confusion, and Vagueness.

Ambiguity and vagueness plague most question writers. A researcher might make implicit assumptions without thinking of respondents’ perspectives. For example, the question, “what is your income?” could mean weekly, monthly, or annual; family or personal; before taxes or after taxes; for this year or last year; from salary or from all sources. The confusion causes inconsistencies in how different respondents assign meaning to and answer the question.

Another source of ambiguity is the use of indefinite words or response categories. Consider the words such as *often*, *occasionally*, *usually*, *regularly*, *frequently*, *many*, *good*, *fair*, and *poor*. Each of these words has many meanings. For one person frequent reading of *Time* magazine may be reading six or seven issues a year; for another it may be two issues a year. The word *fair* has great variety of meanings; the same is true for many indefinite words.

4. Avoid Double-Barreled Questions

Make each question about one and only one. A double barreled question consists of two or more questions joined together. It makes the respondent’s answer ambiguous. For example, if asked, “Does this company have pension and health insurance benefits?” a respondent at the company with health insurance benefits only might answer either yes or no. The response has an ambiguous meaning and the researcher cannot be certain of the respondent’s intentions. When multiple questions are asked in one question, the results may be exceedingly difficult to interpret.

5. Avoid Leading Questions

Make respondents feel that all responses are legitimate. Do not let them aware of an answer that the researcher wants. A leading question is the one that leads the respondent to choose one response over another by its wording. For example, the question, “you don’t smoke, do you?” leads respondents to state that they do not smoke. “Don’t you think that women should be empowered?” In most cases the respondent is likely to agree with the statement.

6. Avoid Loaded Questions

Loaded questions suggest a socially desirable answer or are emotionally charged. “Should the city government repair all the broken streets?” Most of the people are going to agree with this question simply because this is highly socially desirable. A question which may be challenging the traditionally set patterns of behavior may be considered as emotionally charged i.e. it is loaded with such material which may hit the emotions of the people. Look at some behaviors associated with masculinity in Pakistani society. Let us ask a husband “Have you ever been beaten up by your wife?” Straight away this question may be considered to be a challenge to the masculinity of the person. Hence it may be embarrassing for the person to admit such an experience. Therefore, even if the husband was beaten up by his wife, he might give a socially desirable answer.

7. Avoid Burdensome Questions that may Tax the Respondent’s Memory

A simple fact of human life is that people forget. Researchers writing questions about past behavior or events should recognize that certain questions may make serious demand on the respondent’s memory.

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“How did you feel about your brother when you were 6 years old?” It may be very difficult to recall something from the childhood.

8. Arrange Questions in a Proper Sequence

The order of question, or the question sequence, may serve several functions for the researcher. If the opening questions are interesting, simple to comprehend, and easy to answer, respondent's cooperation and involvement can be maintained throughout the questionnaire. If respondent's curiosity is not aroused at the outset, they can become disinterested and terminate the interview.

Sequencing specific questions before asking about broader issues is a common cause of question order bias. In some situations it may be advisable to ask general question before specific question to obtain the freest opinion of the respondent. This procedure, known as **funnel technique**, allows the researcher to understand the respondent's frame of reference before asking specific questions about the level of respondent's information and intensity of his or her opinions.

9. Use Filter Question, if Needed

Asking a question that doesn't apply to the respondent or that the respondent is not qualified to answer may be irritating or may cause a biased response. Including filter question minimizes the chance of asking questions that are inapplicable. Filter question is that question which screens out respondents not qualified to answer a second question. For example the researcher wants to know about the bringing up of one's children. “How much time do you spend playing games with your oldest child?” What if the respondent is unmarried? Even if the respondent is married but does not have the child. In both these situations the question is inapplicable to him/her. Before this question the person may put a filter question whether or not the respondent is married.

10. Layout of the questionnaire

There are two format or layout issues: the overall physical layout of the questionnaire and the format of questions and responses.

Good layout and physical attractiveness is crucial in mail, Internet, and other self-administered questionnaires. For different reason it is also important to have a good layout in questionnaires designed for personal and telephone interviews.

Give each question a number and put identifying information on questionnaire. Never cramp questions together or create a confusing appearance.

Make a cover sheet or face sheet for each, for administrative use. Put the time and date of the interview, the interviewer, the respondent identification number, and interviewer's comments and observations on it. Give interviewers and respondents instructions on the questionnaire. Print instructions in a different style from question to distinguish them.

Layout is important for mail questionnaires because there is no friendly interviewer to interact with the respondent. Instead the questionnaire's appearance persuades the respondents. In mail surveys, include a polite, professional cover letter on letterhead stationery, identifying the researcher and offering a telephone number for any questions. Always end with “Thank you for your participation.”

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LESSON 24

PILOT TESTING OF THE QUESTIONNAIRE

Pilot testing also called pre-testing means small scale trial run of a particular component; here we are referring to pilot testing of the questionnaire.

Conventional wisdom suggests that pre-testing not only is an established practice for discovering errors but also is useful for extra training the research team. Ironically, professionals who have participated in scores of studies are more likely to pretest an instrument than is a beginning researcher hurrying to complete a project. Revising questions five or more times is not unusual. Yet inexperienced researchers often underestimate the need to follow the design-test-revise process.

It is important to pilot test the instrument to ensure that the questions are understood by the respondents and there are no problems with the wording or measurement. Pilot testing involves the use of a small number of respondents to test the appropriateness of the questions and their comprehension. Usually, the draft questionnaire is tried out on a group that is selected on a convenience and that is similar in makeup to the one that ultimately will be sampled. Making a mistake with 25 or so subjects can avert the disaster of administering an invalid questionnaire to several hundred individuals. Hence the main purpose of pilot testing is to identify potential problems with the methods, logistics, and the questionnaire.

Administering a questionnaire exactly as planned in the actual study often is not possible. For example, mailing out a questionnaire might require several weeks. Pre-testing a questionnaire in this manner might provide important information on response rate, but it may not point out why questions were skipped or why respondents found certain questions ambiguous or confusing. The ability of personal interviewer to record requests for additional explanation and to register comments indicating respondent's difficulty with question sequence or other factors is the primary reason why interviewers are often used for pretest work.

What aspects to be evaluated during pilot testing?

1. Reactions of Respondents:

The reactions of the respondents can be looked at from different angles. The researcher may be familiar with the local culture; still getting the first hand experience is always useful. Going to the field, contacting the people, and their reactions to the different aspects of research may be a learning experience.

- **Availability of study population timing.** In case we are doing interviewing then pre-testing might help to find out the most appropriate time when the respondent shall be available. The researcher can plan the interviewing accordingly.
- **Acceptability of the questions asked.** An important purpose of pre-testing is to discover participants' reaction to the questions. If the participants do not find the experience stimulating when an interviewer is physically present, how will they react on the phone, or in the self administered mode? Pre-testing should help to discover where repetitiveness or redundancy is bothersome or what topics were not covered that the participant expected. An alert interviewer will look for questions or even sections that the participant perceives to be sensitive or threatening or topics about which the participant knows nothing.
- **Pre-testing** will also provide the opportunity to see the acceptability of the wording of the questions in the local cultural context. Some of the issues may be discussed openly while for others people use a disguised language. If people consider the use of certain phrases as offensive, then it is high time to change the wording.
- **Willingness of the respondents to co-operate.** Field testing of the questionnaire will give the idea about the level of cooperation the research team is likely to get from the respondents, particularly if they have to interview them.

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2. Discovering errors in the instrument:

- **Do the tools provide you the information? Reliability. Suitability for analysis.** Tabulation of the results /of a pretest helps determine whether the questionnaire will meet the objectives of the research. A preliminary analysis often illustrates that although respondents can easily comprehend and answer a given question, it is an inappropriate question because it does not help solving the issue. The information may not be suitable for analysis.
- **Time taken/needed to interview/conduct the observation.** Pre-testing can indicate the time taken for interview or to conduct the observation. Too long questionnaires may not be recommended and, therefore, need modification. It can also help in estimating average time being taken to collect information from a respondent. Such an exercise can help in budget estimations.
- **If there is any need to revise the format of the tool.** Question arrangement can play a significant role in the success of the instrument. May be we should start with stimulating questions and place sensitive questions last. Such a situation might be handled through pre-testing. Therefore, pre-testing may help in putting questions in proper **sequence, using acceptable wording, doing appropriate translation, question spacing, structuring of answers, coding system, and needing instructions for interviewers (probing).**

3. Sampling procedure can be checked:

- **The extent to which instructions given are followed.** Field functionaries are given the instructions for following a sampling procedure. Depending upon the type of sampling to be followed, the field worker must follow the guidelines otherwise the quality of the study will be hampered. During the pre-testing one could see not only the extent to which the instructions are being followed but also locate the problems in carrying out those instructions. Also what could be the solutions to those problems?
- **How much time is needed to locate the respondents?** By following the instructions how easy it is to locate the respondents, and how much time is needed to do that activity. It could help in calculating the overall time for data collection, having relevancy for budgeting these resources.

4. Staffing and activities of research team can be checked:

- **How successful the training has been?** Pre-testing can be seen as a period of extra training. The pre-testing exercise can provide a good opportunity to make an evaluation of the achievement of the objectives of training. For any deficiencies additional training may be provided.
- **What is the work output of each member?** The researcher can calculate the average output of each fieldworker and accordingly calculate the number of workers needed to finish the work on time. It can also help in making the budget estimates.
- **How well the research team works together?** It is a good opportunity to observe the kind of coordination the research team has. The integrated work is likely to affect the efficiency of the team. Any shortcomings could be looked after.
- **Is the logistical support adequate?** Of course we are leaving the field functionaries in isolation. They shall be in need of other logistical support like the transportation, boarding, lodging, guidance and supervision. Some of these aspects could also be appraised during the pre-testing.

5. Procedure for data processing and analysis can be evaluated:

- **Make dummy tables.** See how can we tabulate the data and use the appropriate statistics for purposes of interpretations.

LESSON 25

INTERVIEWING

A personal interviewer administering a questionnaire door to door, a telephone interviewer calling from a central location, an observer counting pedestrians in a shopping mall, and others involved in the collection of data and the supervision of that process are all **fieldworkers**. The activities they perform vary substantially. The supervision of data collection for a mail survey differs from the data collection in an observation study. Nevertheless there are some basic issues in all kinds of fieldwork. Just for convenience, in this session we shall focus on the interviewing process conducted by personal interviewers. However, many of the issues apply to all fieldworkers, no matter what their specific setting.

Who conducts the fieldwork?

Data collection in a sponsored study is rarely carried out by the person who designs the research project. For a student, depending upon the sample size, data collection is usually done by the student himself/herself. However, the data collection stage is crucial, because the research project is no better than the data collected in the field. Therefore, it is important that the research administrator selects capable people who may be entrusted to collect the data.

There are Field Interviewing Services, who specialize in data gathering. These agencies perform door-to-door surveys, central location telephone interviewing, and other forms of fieldwork for fee. These agencies typically employ field supervisors who oversee and train interviewers, edit questionnaires completed in the field, and confirm that the interviews have been conducted.

Whether the research administrator hires in-house interviewers or selects a field interviewing service, it is desirable to have fieldworkers meet certain job requirements. Although the job requirements for different types of surveys vary, normally interviewers should be healthy, outgoing, honest, accurate, responsible, motivated, and of pleasing appearance – well groomed and properly dressed. An essential part of the interviewing process is establishing rapport with the respondent.

In-House Training

After personnel are selected, they must be trained. The training that the interviewer will receive after being selected by a company may vary from virtually no training to one week program. Almost always there will be a **briefing session** on the particular project.

The objective of training is to ensure that the data collection instrument is administered uniformly by all fieldworkers. The goal of training session is to ensure that each respondent is provided with common information. If the data are collected in a uniform manner from all respondents, the training session will have been success.

More extensive training programs are likely to cover the following topics:

1. How to make initial contact with the respondent and secure the interview?
2. How to ask survey questions?
3. How to probe?
4. How to record responses? How to terminate the interview?

The Role of the Interviewer

Survey research interviewing is a specialized kind of interviewing. As with most interviewing, its goal is to obtain accurate information from another person.

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The survey interview is a social relationship. Like other social relationships, it involves social roles, norms, and expectations. The interview is a short-term, secondary social interaction between two strangers with the explicit purpose of one person's obtaining specific information from the other. The social roles are those of the interviewer and the interviewee or respondent. Information is obtained in a structured conversation in which the interviewer asks prearranged questions and records answers, and the respondent answers.

The role of interviewer is difficult. They obtain cooperation and build rapport, yet remain neutral and objective. They encroach on respondents' time and privacy for information that may not benefit the respondents. They try to reduce embarrassment, fear, and suspicion so that respondents feel comfortable revealing information. They explain the nature of the survey research or give hints about social roles in an interview. Good interviewers monitor the pace and direction of the social interaction as well as content of the answers and the behavior of the respondents.

Survey interviewers are nonjudgmental and do not reveal their opinions, verbally or nonverbally. If the respondent asks for an interviewer's opinion, he or she politely redirects the respondent and indicate that such questions are inappropriate.

Stages of an Interview

Making Initial Contact and Securing the Interview

The interview proceeds through stages, beginning with introduction and entry. Interviewers are trained to make appropriate opening remarks that will convince the person that his or her cooperation is important.

Asslaam-o-Alaykum, my name is _____ and I am working for a National Survey Company. We are conducting a survey concerning "women empowerment." I would like to get a few of your ideas.

For the initial contact in a telephone interview, the introduction might be:

Asslaam-o-Alaykum, my name is _____. I am calling from Department of Social Research, Virtual University.

By indicating that telephone call is a long distance, interviewers attempt to capitalize on the fact that most people feel a long distance call is something special, unusual, or important. Giving one's personal name personalizes the call.

Personal interviewers may carry a letter of identification that will indicate that the study is bona fide research project and not a salesman's call. The name of the research agency is used to assure the respondent that the caller is trustworthy.

Asking the Questions

The purpose of the interview is, of course, to have the interviewer ask questions and record the respondent's answers. Training in the art of stating questions can be extremely beneficial, because interviewer bias can be a source of considerable error in survey research.

There are five major principles for asking questions:

- Ask the questions exactly as they are worded in the questionnaire.
- Read each question very slowly.
- Ask the question in the order in which they are presented in the questionnaire.

- Repeat questions that are misunderstood or misinterpreted.

Although interviewers are generally trained in these procedures, when working in the field many interviewers do not follow them exactly. Do not take shortcuts when the task becomes monotonous. Interviewers may shorten questions or rephrase unconsciously when they rely on their memory of the question rather than reading the question as it is worded.

If the respondents do not understand a question, they will usually ask for some clarification. The recommended procedure is to repeat the question, or if the respondent does not understand a word, the interviewer should respond with “just whatever it means to you.

Often the respondents volunteer information relevant to a question that is supposed to be asked at a later point in the research. In this situation the response should be recorded under the question that deals specifically with that subject. Then rather than skip the question that was answered out of sequence, the interviewers should be trained to say something like “We have briefly discussed this, but let me ask you” By asking every question, the interviewer can be sure that complete answers are recorded.

Probing

Probing means the verbal prompts made by field worker when the respondent must be motivated to communicate his or her answer or to enlarge on, clarify or explain an answer. Probing may be needed for two types of situations. First, it is necessary when the respondent must be motivated to enlarge on, clarify, or explain his or her answer. The interviewer must encourage the respondent to clarify or expand on answers by providing a stimulus that will not suggest the interviewer’s own ideas. The ability to probe with neutral stimuli is the mark of an experienced interviewer. Second, probing may be necessary in situations in which the respondent begins to ramble or lose track of the question. In such cases the respondent must be led to focus on specific content of the interview and to avoid irrelevant and unnecessary information. Probing is also needed when the interviewer recognizes an irrelevant or inaccurate answer.

The interviewer has several possible probing tactics to choose from, depending on the situation:

- *Repetition of the question.* The respondent who remains completely silent may not have understood the question or may not have decided how to answer it. Mere repetition may encourage the respondent to answer in such cases. For example, if the question is “What is there that you do not like about your supervisor?” and the respondent does not answer, the interviewer may probe: “just to check, is there anything you do not like about your supervisor?”
- *An expectant pause.* If the interviewer believes the respondent has more to say, the “silent probe,” accompanied by an expectant look may motivate the respondent to gather his/her thoughts and give a complete response.
- *Repetition of the respondent’s reply.* As the interviewer records the response, he or she may repeat the respondent’s reply verbatim. This may stimulate the respondent to expand on the answer.
- *Neutral questions or comments.* Asking neutral question may indicate the type of information that the interviewer is seeking. For example, if the interviewer believes that the respondent’s motives should be clarified, he or she might ask, “Why do you feel that way?” If the interviewer feels that there is a need to clarify a word or phrase, then he/she might ask, “What do you mean by _____?”

Recording the Responses

The rules for recording responses to closed ended questions vary with the specific question. The general rule, however, is to place a check in the box that correctly reflects the respondent’s answer.

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The general instructions for recording answers to open-ended response questions is to record the answer verbatim, a task that is difficult for most people. Some of these suggestions are:

- Record responses during the interview.
- Use the respondent's own words.
- Do not summarize or paraphrase the respondent's answer.
- Include everything that pertains to the question objectives.
- Include all your probes.

Terminating the Interview

Fieldworkers should not close the interview before all the information has been secured. The interviewer whose departure is hasty will not be able to record those spontaneous comments respondents sometimes offer after all formal questions have been asked. Avoiding hasty departures is also a matter of courtesy.

Fieldworkers should also answer to the best of their ability any questions the respondent has concerning the nature and purpose of the study. Always leave by observing the local cultural customs. "Don't burn your bridges." Because the fieldworker may be required to re-interview the respondent at some future time, he or she should leave the respondent with positive feeling about having cooperated in a worthwhile undertaking. It is extremely important to thank the respondent for his or her cooperation.

The interviewer then goes to a quiet and private place to edit the questionnaire and record other details such as the date, time, and place of interview; a thumbnail sketch of the respondent and interview situation, the respondent's attitude; and any unusual circumstances. The interviewer also records personal feelings and anything that was suspected.

Principles of Interviewing

The Basics

Have integrity and be honest. This is the cornerstone of all professional inquiry, regardless of its purpose.

Have patience and tact. Interviewers ask for information from people they do not know. Thus all the rules of human relations that apply to inquiry situations – patience, tact, courtesy – apply "in spades" to interviewing.

Have attention to accuracy and detail. Among the greatest interviewing "sins" are inaccuracy and superficiality, for the professional analyst can misunderstand, and in turn mislead, a client. Do not record the answer unless you fully understand it yourself. Probe for clarification and detailed full answers.

Exhibit a real interest in the inquiry at hand, but keep your opinions to yourself. Impartiality is imperative.

Be a good listener. Some interviewers talk too much, wasting time when respondents could be supplying more pertinent facts or opinions on the topic.

Keep the inquiry and respondents' responses confidential. Do not discuss the studies you are doing with relatives, friends, or associates. Never quote one respondent's opinion to another.

Respect others' rights. Survey research depends on the goodwill of others to provide information. There should be no coercion. Impress on prospective respondents that their cooperation is important and valuable.

Interview Bias

- Information obtained during interview should be as free as possible of bias.
- Bias could be introduced by the interviewer, interviewee, or the situation. Interviewer bias falls into six categories:

Interviewer Bias

1. Interviewer could bias the data if proper rapport is not established – Errors by the respondent – forgetting, embarrassment, misunderstanding, or lying because of the presence of others.
2. Unintentional errors or interviewer sloppiness – contacting the wrong person, misreading a question, omitting questions, reading questions in the wrong order, recording wrong answer, or misunderstanding the respondent.
3. Intentional subversion by the interviewer – purposeful alteration of answers, omission or rewording of questions, or choice of an alternative respondent.
4. Influence due to the interviewer's expectations about a respondent's appearance, living situation, or other answers.
5. Failure of an interviewer to probe or to probe properly.
6. Influence on the answers due to the interviewer's appearance, tone, attitude, reactions to answers, or comments made outside of the interview schedule.

Interviewee Bias تعصب، بد گمانی، طرفداری

- Errors made by the respondent –

 1. Interviewees can bias the data when they do not come out with their true opinion but provide information that they think what the interviewer expects of them or would like to hear.
 2. They do not understand the question, they may feel difficult or hesitant to clarify.
 3. Some interviewees may be turned off because of the personal liking, or the dress of the interviewer, or the manner in which questions are put. So they may not provide truthful answers.
 4. Some may provide socially undesirable answers.

Situational Bias

- Situational biases in terms of:

 1. Non-participants – Unwillingness or inability to participate. Bias the sample.
 2. Trust levels and rapport established by different interviewers. Elicit answers of different degrees of openness.
 3. The physical setting of the interview. Respondent may not feel comfortable to be interviewed at work.

Some Tips for Interviewing

- Know the culture of the people in advance.
- Appearance – wear acceptable dress.
- Pleasantness and flexibility.
- Carry the letter of authority.
- Establish credibility and rapport. Motivating individuals to respond.
- Familiarity with the questionnaire.
- Following the question wording/ question order
- Recording responses exactly.
- Probing for responses.
- Closing the interview. No false promises. Also don't burn your bridges.
- Edit the questionnaire in the first available opportunity.

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LESSON 26

SAMPLE AND SAMPLING TERMINOLOGY

A **sample** is a subset, or some part, of a larger whole. A larger whole could be anything out of which sample is taken. That 'whole' could be a bucket of water, a bag of sugar, a group of organizations, a group of students, a group of customers, or a group mid-level managers in an organization. A complete group of entities sharing some common set of characteristics is population. In other words, the totality out of which sample is drawn is referred to as **population**.

Why sample?**1. Saves Cost, Labor, and Time**

Applied research projects usually have budget and time constraints. Since sample can save financial cost as well as time, therefore, to go for sample study is pragmatic.

Of course, a researcher investigating a population with an extremely small number of population elements may elect to conduct a study on the total population rather than a sample because cost, labor, and time constraints are relatively insignificant.

Although sample study cuts costs, reduces labor requirements, and gathers vital information quickly, yet there could be other reasons.

2. Quality Management/supervision

Professional fieldworkers are a scarce commodity. In a large study rather than employing less qualified staff it may be advisable to do a sample study and employ highly qualified professional fieldworkers. It can certainly affect the quality of the study. At the same time it may be easier to manage a small group and produce quality information. Supervision, record keeping, training, and so forth would all be more difficult in a very large study.

3. Accurate and Reliable Results

Another major reason for sampling is that samples, if properly selected, are sufficiently accurate in most of the cases. If the elements of a population are quite similar, only a small sample is necessary to accurately portray the characteristics of interest. Most of us have had blood samples taken from the finger, the arm, or another part of body. The assumption is that blood is sufficiently similar through out the body, the characteristics of the blood can be determined on the basis of sample.

When the elements of population are highly homogenous, samples are highly representative of the population. Under these circumstances almost any sample is as good as another.

Samples may be more accurate than census. In a census study of large population there is a greater likelihood of non-sampling errors. In a survey mistakes may occur that are unrelated to the selection of people in the study. For example, a response may be coded incorrectly, or the keyboard operator might make data entry error. Interviewer mistakes, tabulation errors, and other non-sampling errors may increase during census because of the increased volume of work. In sample increased accuracy is possible because the fieldwork and tabulation of the data can be closely supervised than would be possible in a census. In field survey, a small, well trained, closely supervised group may do a more careful and accurate job of collecting information than a large group of nonprofessional interviewers trying to contact everyone.

4. Sampling may be the Only Way

Many research projects, especially those in quality control testing, require the destruction of the items being tested. If the manufacturer of firecrackers wished to find out whether each product met a specific production standard, there would be no product left after testing. Similarly, consider the case of electric

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bulbs. In testing the life of bulbs, if we were to burn every bulb produced, there would be none left to sell. This is destructive sampling.

5. Determine the Period of Study

Interviewing every element of a large population without sampling requires lot of time, may be a year or more. In such a long period study, even the seasonal variation may influence the response pattern of the respondents. For example, if the study was aimed at measuring the level of unemployment in a given large city, the unemployment rate produced by the survey data would not refer to the city as of the beginning of interviewing or as of the end. Researcher may be forced to attribute the unemployment to some hypothetical date, representing to the midpoint of the study period. Hence it will be difficult to determine the exact timing to which the data of the study pertains.

Sampling Terminology

There are a number of technical terms used in books on research and statistics which need explanation. Some of the important terms are:

Element

An element is that unit about which information is collected and which provides the basis of analysis. Typically, in survey research, elements are people or certain types of people. It is that unit about which information is collected and that provides the basis of analysis. It can be a person, groups, families, organizations, corporations, communities, and so forth.

Population

A population is the theoretically specified aggregation of study elements. It is translating the abstract concept into workable concept. For example, let us look at the study of “college students.” Theoretically who are the college students? They might include students registered in government colleges and/or private colleges, students of intermediate classes and/or graduate classes, students of professional colleges and/or non-professional colleges, and many other variations. In this way the pool of all available elements is population.

Target Population

Out of the conceptual variations what exactly the researcher wants to focus on. This may also be called a target population. Target population is the complete group of specific population elements relevant to the research project. Target population may also be called *survey population* i.e. that aggregation of elements from which the survey sample is actually selected.

At the outset of the sampling process, it is vitally important to carefully define the target population so the proper source from which the data are to be collected can be identified. In our example of ‘college students’ finally we may decide to study the college students from government institutions located in Lahore, who are studying social sciences, who are aged 19 years of age, and hailing from rural areas.

Sampling

The process of using a small number of items or parts of a larger population to make conclusions about the whole population. It enables the researchers to estimate unknown characteristics of the population.

Sampling Frame

In actual practice the sample will be drawn from a list of population elements that is often different from the target population that has been defined. A sampling frame is the list of elements from which the

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sample may be drawn. A simple example could be listing of all college students meeting the criteria of target population and who are enrolled on the specified date.

A sampling frame is also called the *working population* because it provides the list that can be worked with operationally. In our example, such a list could be prepared with help of the staff of the selected colleges.

Sampling Frame Error

A sampling frame error occurs when certain sample elements are excluded or when the entire population is not accurately represented in the sampling frame. The error that occurs when certain sample elements are not listed or available and are not represented in the sampling frame.

Sampling Unit

A sampling unit is that element or set of elements considered for selection in some stage of sampling. Sampling may be done in single stage or in multiple stages. In a simple, single-stage sample, the sampling units are the same as the elements. In more complex samples, however, different levels of sampling units may be employed. For example, a researcher may select a sample of *Mohallahs* in a city, and then select a sample of households from the selected *Mohallahs*, and finally may select a sample of adults from the selected households. The sampling units of these three stages of sampling are respectively *Mohallah*, households, and adults, of which the last of these are the elements. More specifically, the terms “primary sampling units,” “secondary sampling units,” and “final sampling units” would be used to designate the successive stages.

Observation Unit

An observation unit, or unit of data collection, is an element or aggregation of elements from which the information is collected. Often the unit of analysis and unit of observation are the same – the individual person – but this need not be the case. Thus the researcher may interview heads of household (the observation units) to collect information about every member of the household (the unit of analysis).

Parameter

A parameter is the summary description of a given variable in a population. The mean income of all families in a city and the age distribution of the city’s population are parameters. An important part of survey research involves the estimation of population parameters on the basis of sample observation.

Statistic

A statistic is the summary description of a given variable in a survey sample. Thus the mean income computed from the survey sample and the age distribution of that sample are statistics. Sample statistics are used to make estimates of the population parameters.

Sampling Error

Probability sampling methods seldom, if ever, provide statistics exactly equal to the parameters that they are used to estimate. Probability theory, however, permits us to estimate the error to be expected for a given sample (more information to be sought from professional in Statistics).

LESSON 27

PROBABILITY AND NON-PROBABILITY SAMPLING

There are several alternative ways of taking a sample. The major alternative sampling plans may be grouped into **probability techniques** and **non-probability techniques**. **In probability sampling every element in the population has a known nonzero probability of selection.** The simple random is the best known probability sample, in which each member of the population has an equal probability of being selected. Probability sampling designs are used when the representativeness of the sample is of importance in the interest of wider generalisability. **When time or other factors, rather than generalisability, become critical, non-probability sampling is generally used.**

In non-probability sampling the probability of any particular element of the population being chosen is unknown. The selection of units in non-probability sampling is quite arbitrary, as researchers rely heavily on personal judgment. It should be noted that there are no appropriate statistical techniques for measuring random sampling error from a non-probability sample. Thus projecting the data beyond the sample is statistically inappropriate. Nevertheless, there are occasions when non-probability samples are best suited for the researcher's purpose.

Types of non-probability sampling:

In non-probability sampling designs, the elements in the population do not have any probabilities attached to their being chosen as sample subjects. This means that the findings from the study of the sample cannot be confidently generalized to the population. However the researchers may at times be less concerned about generalisability than obtaining some preliminary information in a quick and inexpensive way. Sometimes non-probability could be the only way to collect the data.

Convenience Sampling

Convenience sampling (also called *haphazard or accidental sampling*) refers to sampling by obtaining units or people who are most conveniently available. **For example, it may be convenient and economical to sample employees in companies in a nearby area, sample from a pool of friends and neighbors.** The person-on-the street interview conducted by TV programs is another example. TV interviewers go on the street with camera and microphone to talk to few people who are convenient to interview. **The people walking past a TV studio in the middle of the day do not represent everyone (homemakers, people in the rural areas).** Likewise, TV interviewers select people who look “normal” to them and avoid people who are unattractive, poor, very old, or inarticulate.

Another example of haphazard sample is that of a newspaper that asks the readers to clip a questionnaire from the paper and mail it in. Not everyone reads the newspaper, has an interest in the topic, or will take the time to cut out the questionnaire, and mail it. Some will, and the number who do so may seem large, but the sample cannot be used to generalize accurately to the population.

Convenience samples are least reliable but normally the cheapest and easiest to conduct. Convenience sampling is most often used during the exploratory phase of a research project and is perhaps the best way of getting some basic information quickly and efficiently. Often such sample is taken to test ideas or even to gain ideas about a subject of interest.

Purposive Sampling

Depending upon the type of topic, the researcher lays down the criteria for the subjects to be included in the sample. Whoever meets that criteria could be selected in the sample. The researcher might select such cases or might provide the criteria to somebody else and leave it to his/her judgment for the actual selection of the subjects. That is why such a sample is also called as **judgmental or expert opinion sample**. For example a researcher is interested in studying students who are enrolled in a course on

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research methods, are highly regular, are frequent participants in the class discussions, and often come with new ideas. The criteria has been laid down, the researcher may do this job himself/herself, or may ask the teacher of this class to select the students by using the said criteria. In the latter situation we are leaving it to the judgment of the teacher to select the subjects. Similarly we can give some criteria to the fieldworkers and leave it to their judgment to select the subjects accordingly. In a study of working women the researcher may lay down the criteria like: the lady is married, has two children, one of her child is school going age, and is living in nuclear family.

Quota Sampling

A sampling procedure that ensures that certain characteristics of a population sample will be represented to the exact extent that the researcher desires. In this case the researcher first identifies relevant categories of people (e.g. male and female; or under age 30, ages 30 to 60, over 60, etc) then decides how many to get in each category. Thus the number of people in various categories of sample is fixed. For example the researcher decides to select 5 males and 5 females under age 30, 10 males and 10 females aged 30 to 60, and 5 males and 5 females over age 60 for a 40 person sample. This is quota sampling.

Once the quota has been fixed then the researcher may use convenience sampling. The convenience sampling may introduce **bias**. For example, the field worker might select the individual according to his/her liking, who can easily be contacted, willing to be interviewed, and belong to middle class. Quota sampling can be considered as a form of proportionate stratified sampling, in which a predetermined proportion of people are sampled from different groups, but on a convenience basis. Speed of data collection, lower costs, and convenience are the major advantages of quota sampling compared to probability sampling. Quota sampling becomes necessary when a subset of a population is underrepresented, and may not get any representation if equal opportunity is provided to each. Although there are many problems with quota sampling, careful supervision of the data collection may provide a representative sample of the various subgroups within the population.

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Snowball sampling (also called *network*, *chain referral*, or *reputational sampling*) is a method for identifying and sampling (or selecting) cases in the network. It is based on an analogy to a snowball, which begins small but becomes larger as it is rolled on wet snow and picks up additional snow. It begins with one or a few people or cases and spreads out on the basis of links to these initial cases.

This design has been found quite useful where respondents are difficult to identify and are best located through referral networks. In the initial stage of snowball sampling, individuals are discovered and may or may not be selected through probability methods. This group is then used to locate others who possess similar characteristics and who, in turn, identify others. The “snowball” gathers subjects as it rolls along.

For example, a researcher examines friendship networks among teenagers in a community. He or she begins with three teenagers who do not know each other. Each teen names four close friends. The researcher then goes to the four friends and asks each to name four close friends, then goes to those four and does the same thing again, and so forth. Before long, a large number of people are involved. Each person in the sample is directly or indirectly tied to the original teenagers, and several people may have named the same person. The researcher eventually stops, either because no new names are given, indicating a closed network, or because the network is so large that it is at the limit of what he or she can study.

Sequential Sampling

Sequential sampling is similar to purposive sampling with one difference. In purposive sampling, the researcher tries to find as many relevant cases as possible, until time, financial resources, or his or her

energy is exhausted. The principle is to get every possible case. In sequential sampling, a researcher continues to gather cases until the amount of new information or diversity is filled. The principle is to gather cases until a saturation point is reached. In economic terms, information is gathered, or the incremental benefit for additional cases, levels off or drops significantly. It requires that the researcher continuously evaluates all the collected cases. For example, a researcher locates and plans in-depth interviews with 60 widows over 70 years old who have been living without a spouse for 10 or more years. Depending on the researcher's purposes, getting an additional 20 widows whose life experiences, social background, and worldview differ little from the first 60 may be unnecessary.

Theoretical Sampling

In theoretical sampling, what the researcher is sampling (e.g. people, situation, events, time periods, etc.) is carefully selected, as the researcher develops grounded theory. A growing theoretical interest guides the selection of sample cases. The researcher selects cases based on new insights they may provide. For example, a field researcher may be observing a site and a group of people during week days. Theoretically, the researcher may question whether the people act the same at other times or when other aspects of site change. He or she could then sample other time periods (e.g. nights and weekends) to get more full picture and learn whether important conditions are the same.

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LESSON 28

TYPES OF PROBABILITY SAMPLING

Probability samples that rely on random processes require more work than nonrandom ones. A researcher must identify specific sampling elements (e.g. persons) to include in the sample. For example, if conducting a telephone survey, the researcher needs to try to reach the specific sampled person, by calling back several times, to get an accurate sample.

Random samples are most likely to yield a sample that truly represents the population. In addition, random sampling lets a researcher statistically calculate the relationship between the sample and the population – that is the size of *sampling error*. A non-statistical definition of the sampling error is the deviation between sample result and a population parameter due to random process.

Simple Random Sample

The simple random sample is both the easiest random sample to understand and the one on which other types are modeled. In simple random sampling, a research develops an accurate sampling frame, selects elements from sampling frame according to mathematically random procedure, then locates the exact element that was selected for inclusion in the sample.

After numbering all elements in a sampling frame, the researcher uses a list of random numbers to decide which elements to select. He or she needs as many random numbers as there are elements to be sampled: for example, for a sample of 100, 100 random numbers are needed. The researcher can get random numbers from a random number table, a table of numbers chosen in a mathematically random way. Random-number tables are available in most statistics and research methods books. The numbers are generated by a pure random process so that any number has an equal probability of appearing in any position. Computer programs can also produce lists of random number.

A random starting point should be selected at the outset.

Random sampling does not guarantee that every random sample perfectly represents the population. Instead, it means that most random samples will be close to the population most of the time, and that one can calculate the probability of a particular sample being inaccurate. A researcher estimates the chance that a particular sample is off or unrepresentative by using information from the sample to estimate the sampling distribution. The sampling distribution is the key idea that lets a researcher calculate sampling error and confidence interval.

Systematic Random Sample

Systematic random sampling is simple random sampling with a short cut for random selection. Again, the first step is to number each element in the sampling frame. Instead of using a list of random numbers, researcher calculates a *sampling interval*, and the interval becomes his or her own quasi random selection method. The sampling interval (i.e. 1 in K where K is some number) tells the researcher how to select elements from a sampling frame by skipping elements in the frame before one for the sample.

Sampling intervals are easy to compute. We need the sample size and the population size. You can think of the sample interval as the inverse of the sampling ratio. The sampling ratio for 300 names out of 900 will be $300/900 = .333 = 33.3$ percent. The sampling interval is $900/300 = 3$

Begin with a random start. The easiest way to do this is to point blindly at a number from those from the beginning that are likely to be part of the sampling interval.

When the elements are organized in some kind of cycle or pattern, the systematic sampling will not give a representative sample.

Stratified Random Sample

When the population is heterogeneous, the use of simple random sample may not produce representative sample. Some of the bigger strata may get over representation while some of the small ones may entirely be eliminated. Look at the variables that are likely to affect the results, and stratify the population in such a way that each stratum becomes homogeneous group within itself. Then draw the required sample by using the table of random numbers. Hence in stratified random sampling a sub-sample is drawn utilizing simple random sampling within each stratum. (Randomization is not done for quota sampling).

There are three reasons why a researcher chooses a stratified random sample: (1) to increase a sample's statistical efficiency, (2) to provide adequate data for analyzing the various subpopulations, and (3) to enable different research methods and procedures to be used in different strata.

1. Stratification is usually more efficient statistically than simple random sampling and at worst it is equal to it. **With the ideal stratification, each stratum is homogeneous internally and heterogeneous with other strata.** This might occur in a sample that includes members of several distinct ethnic groups. In this instance, stratification makes a pronounced improvement in statistical efficiency.

Stratified random sampling provides the assurance that the sample will accurately reflect the population on the basis of criterion or criteria used for stratification. This is a concern because occasionally simple random sampling yields a disproportionate number of one group or another, and the sample ends up being less representative than it could be.

Random sampling error will be reduced with the use of stratified random sampling. Because each group is internally homogeneous but there are comparative differences between groups. More technically, a smaller standard error may result from stratified sampling because the groups are adequately represented when strata are combined.

2. It is possible when the researcher wants to study the characteristics of a certain population subgroups. Thus if one wishes to draw some conclusions about activities in different classes of student body, stratified sampling would be used.
3. Stratified sampling is also called for when different methods of data collection are applied in different parts of the population. This might occur when we survey company employees at the home office with one method but must use a different approach with employees scattered over the country.

Stratification Process

The ideal stratification would be based on the primary variable (the dependent variable) under study. The criterion is identified as an efficient basis for stratification. The criterion for stratification is that it is a characteristic of the population elements known to be related to the dependent variable or other variables of interest. The variable chosen should increase homogeneity within each stratum and increase heterogeneity between strata.

Next, for each separate subgroup or stratum, a list of population elements must be obtained. Serially number the elements within each stratum. Using a table of random numbers or some other device, a separate simple random sample is taken within each stratum. Of course the researcher must determine how large a sample must be drawn from each stratum.

Proportionate versus Disproportionate

If the number of sampling units drawn from each stratum is in proportion to the relative population size of the stratum, the sample is proportionate stratified sampling. Sometime, however, a disproportionate stratified sample will be selected to ensure an adequate number of sampling units in every stratum.

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In a disproportionate, sample size for each stratum is not allocated in proportion to the population size, but is dictated by analytical considerations.

Cluster Sampling

The purpose of cluster sampling is to sample economically while retaining the characteristics of a probability sample. Groups or chunks of elements that, ideally, would have heterogeneity among the members within each group are chosen for study in cluster sampling. This is in contrast to choosing some elements from the population as in simple random sampling, or stratifying and then choosing members from the strata, or choosing every n th case in the population in systematic sampling. When several groups with intra-group heterogeneity and inter-group homogeneity are found, then a random sampling of the clusters or groups can ideally be done and information gathered from each of the members in the randomly chosen clusters.

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Cluster samples offer more heterogeneity within groups and more homogeneity among and homogeneity within each group and heterogeneity across groups.

Cluster sampling addresses two problems: researchers lack a good sampling frame for a dispersed population and the cost to reach a sampled element is very high. A cluster is unit that contains final sampling elements but can be treated temporarily as a sampling element itself. Researcher first samples clusters, each of which contains elements, then draws a second a second sample from within the clusters selected in the first stage of sampling. In other words, the researcher randomly samples clusters, and then randomly samples elements from within the selected clusters. He or she can create a good sampling frame of clusters, even if it is impossible to create one for sampling elements. Once the researcher gets a sample of clusters, creating a sampling frame for elements within each cluster becomes more manageable. A second advantage for geographically dispersed populations is that elements within each cluster are physically closer to each other. This may produce a savings in locating or reaching each element.

A researcher draws several samples in stages in cluster sampling. In a three-stage sample, stage 1 is random sampling of big clusters; stage 2 is random sampling of small clusters within each selected big cluster; and the last stage is sampling of elements from within the sampled within the sampled small clusters. First, one randomly samples the city blocks, then households within blocks, then individuals within households. This can also be an example of **multistage area sampling**.

The unit costs of cluster sampling are much lower than those of other probability sampling designs. However, cluster sampling exposes itself to greater biases at each stage of sampling.

Double Sampling

This plan is adopted when further information is needed from a subset of the group from which some information has already been collected for the same study. A sampling design where initially a sample is used in a study to collect some preliminary information of interest, and later a sub-sample of this primary sample is used to examine the matter in more detail, is called double sampling.

What is the Appropriate Sample Design?

A researcher who must make a decision concerning the most appropriate sample design for a specific project will identify a number of sampling criteria and evaluate the relative importance of each criterion before selecting a sample design. The most common criteria

Degree of Accuracy

Selecting a representative sample is, of course, important to all researchers. However, the error may vary from project to project, especially when cost saving or another benefit may be a trade-off for reduction in accuracy.

Resources

The costs associated with the different sampling techniques vary tremendously. If the researcher's financial and human resources are restricted, this limitation of resources will eliminate certain methods. For a graduate student working on a master's thesis, conducting a national survey is almost always out of the question because of limited resources. Managers usually weigh the cost of research versus the value of information often will opt to save money by using non-probability sampling design rather than make the decision to conduct no research at all.

Advance Knowledge of the Population

Advance knowledge of population characteristics, such as the availability of lists of population members, is an important criterion. A lack of adequate list may automatically rule out any type of probability sampling..

National versus Local Project

Geographic proximity of population elements will influence sample design. When population elements are unequally distributed geographically, a cluster sampling may become more attractive.

Need for Statistical Analysis

The need for statistical projections based on the sample is often a criterion. Non-probability sampling techniques do not allow researcher to use statistical analysis to project the data beyond the sample.

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LESSON 29

DATA ANALYSIS

Once the data begins to flow in, attention turns to data analysis. If the project has been done correctly, the analysis planning is already done. Back at the research design stage or at least by the completion of the proposal or the pilot test, decisions should have been made about how to analyze the data.

During the analysis stage several interrelated procedures are performed to summarize and rearrange the data. The goal of most research is to provide information. There is a difference between raw data and information.

Information refers to a body of facts that are in a format suitable for decision making, whereas **data** are simply recorded measures of certain phenomenon. The raw data collected in the field must be transformed into information that will answer the sponsor's (e.g. manager's) questions. The conversion of raw data into information requires that the data be edited and coded so that the data may be transferred to a computer or other data storage medium.

If the database is large, there are many advantages to utilizing a computer. Assuming a large database, entering the data into computer follows the coding procedure.

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Editing

Occasionally, a fieldworker makes a mistake and records an improbable answer (e.g., birth year: 1843) or interviews an ineligible respondent (e.g., someone too young to qualify). Seemingly contradictory answers, such as “no” to automobile ownership but “yes” to an expenditure on automobile insurance, may appear on a questionnaire. There are many problems like these that must be dealt with before the data can be coded. Editing procedures are conducted to make the data ready for coding and transfer to data storage.

Editing is the process of checking and adjusting the data for omissions, legibility, and consistency. Editing may be differentiated from coding, which is the assignment of numerical scales or classifying symbols to previously edited data.

The purpose of editing is to ensure the completeness, consistency, and readability of the data to be transferred to data storage. The editor's task is to check for errors and omissions on the questionnaires or other data collection forms.

The editor may have to reconstruct some data. For instance, a respondent may indicate weekly income rather than monthly income, as requested on the questionnaire. The editor must convert the information to monthly data without adding any extraneous information. The editor “should bring to light all hidden values and extract all possible information from a questionnaire, while adding nothing extraneous.”

Field Editing

In large projects, field supervisors are often responsible for conducting preliminary field edits. The purpose of field editing the same day as the interview is to catch technical omissions (such as a blank page), check legibility of the handwriting, and clarify responses that are logically or conceptually inconsistent. If a daily field editing is conducted, a supervisor who edits completed questionnaires will frequently be able to question the interviewers, who may be able to recall the interview well enough to correct any problems. The number of “no answers,” or incomplete answers can be reduced with a rapid follow-up simulated by a field edit. The daily edit also allows fieldworkers to re-contact the respondent to fill in omissions before the situation has changed. The field edit may also indicate the need for further training of interviewers.

In-House Editing

Although almost simultaneous editing in the field is highly desirable, in many situations (particularly with mail questionnaires), early reviewing of the data is not possible. In-house editing rigorously investigates the results of data collection.

Editing for Consistency:

The in-house editor's task is to ensure that inconsistent or contradictory responses are adjusted and that answers will not be a problem for coders and keyboard punchers. Consider the situation in which a telephone interviewer has been instructed to interview only registered voters that requires voters to be 18 years old. If the editor's reviews of a questionnaire indicate that the respondent was only 17 years of age, the editor's task is to eliminate this obviously incorrect sampling unit. Thus, in this example, the editor's job is to make sure that the sampling unit is consistent with the objectives of the study.

Editing requires checking for logically consistent responses. The in-house editor must determine if the answers given by a respondent to one question are consistent with those given to other, related questions. Many surveys utilize filter questions or skip questions that direct the sequence of questions, depending upon respondent's answer. In some cases the respondent will have answered a sequence of questions that should not have been asked. The editor should adjust these answers, usually to "no answer" or "inapplicable," so that the responses will be consistent.

Editing for Completeness: In some cases the respondent may have answered only the second portion of a two-part question. An in-house editor may have to adjust the answers to the following question for completeness.

Does your organization have more than one Internet Web site? Yes ____ No. ____

If a respondent checked neither "yes" nor "No", but indicated three Internet Web sites, the editor may check the "yes" to ensure that this answer is not missing from the questionnaire.

Item Non-response: It is a technical term for an unanswered question on an otherwise complete questionnaire. Specific decision rules for handling this problem should be meticulously outlined in the editorial instructions. In many situations the decision rule will be to do nothing with the unanswered question: the editor merely indicates item non response by writing a message instructing the coder to record a "missing value" or blank as the response. However, in case the response is necessary then the editor uses the **plug value**. The decision rule may be to "plug in" an average or neutral value in each case of missing data. A blank response in an interval scale item with a mid point would be to assign the mid point in the scale as the response to that particular item. Another way is to assign to the item the mean value of the responses of all those who have responded to that particular item. Another choice is to give the item the mean of the responses of this particular respondent to all other questions measuring the variables. Another decision rule may be to alternate the choice of the response categories used as plug values (e.g. "yes" the first time, "no" the second time, "yes" the third time, and so on).

The editor must also decide whether or not an entire questionnaire is "usable." When a questionnaire has too many (say 25%) answers missing, it may not be suitable for the planned data analysis. In such a situation the editor simply records the fact that a particular incomplete questionnaire has been dropped from the sample.

Editing Questions Answered out of Order: Another situation an editor may face is the need to rearrange the answers to an open-ended response to a question. For example, a respondent may have provided the answer to a subsequent question in his answer to an earlier open-ended response question. Because the respondent had already clearly identified his answer, the interviewer may have avoided asking the subsequent question. The interviewer may have wanted to avoid hearing "I have already answered that earlier" and to maintain rapport with the respondent and therefore skipped the question. To make the response appear in the same order as on other questionnaires, the editor may remove the out-of-order answer to the section related to the skipped question.

Coding

Coding involves assigning numbers or other symbols to answers so the responses can be grouped into limited number of classes or categories. The classifying of data into limited categories sacrifices some data detail but is necessary for efficient analysis. Nevertheless, it is recommended that try to keep the data in raw form so far it is possible. When the data have been entered into the computer you can always ask the computer to group and regroup the categories. In case the data have been entered in the compute in grouped form, it will not be possible to disaggregate it.

Although codes are generally considered to be numerical symbols, they are more broadly defined as the rules for interpreting, classifying, and recording data. Codes allow data to be processed in a computer. Researchers organize data into fields, records, and files. A **field** is a collection of characters (a character is a single number, letter of the alphabet, or special symbol such as the question mark) that represent a single type of data. A **record** is collection of related fields. A **file** is a collection of related records. File, records, and fields are stored on magnetic tapes, floppy disks, or hard drives.

Researchers use a coding procedure and codebook. A **coding procedure** is a set of rules stating that certain numbers are assigned to variable attributes. For example, a researchers codes males as 1 and females as 2. Each category of variable and missing information needs a code. A **codebook** is a document (i.e. one or more pages) describing the coding procedure and the location of data for variables in a format that computers can use.

When you code data, it is very important to create a well-organized, detailed codebook and make multiple copies of it. If you do not write down the details of the coding procedure, or if you misplace thee codebook, you have lost thee key to the data and may have to recode the data again.

Researchers begin thinking about a coding procedure and a codebook before they collect data. For example a survey researcher pre-codes a questionnaire before collecting thee data. Pre-coding means placing the code categories (e.g. 1 for male, 2 for female) on the questionnaire. Sometimes to reduce dependence on codebooks, researchers also place the location in the computer format on the questionnaire.

If the researcher does not pre-code, his or her first step after collecting and editing of data is to crate a codebook. He or she also gives each case an identification number to keep track of the cases. Next, the researcher transfers the information from each questionnaire into a format that computers can read.

Code Construction

When the question has a fixed-alternative (closed ended) format, the number of categories requiring codes is determined during the questionnaire design stage. The codes 8 and 9 are conventionally given to “don’t know” (DK) and “no answer” (NA) respectively. However, many computer program fields recognize a blank field or a certain character symbol, such as a period (.), as indicating a missing value (no answer).

There are two basic rules for code construction. First, the coding categories should be **exhaustive** – that is, coding categories should be provided for all subjects or objects or responses. With a categorical variable such as sex, making categories exhaustive is not a problem. However, when the response represents a small number of subjects or when the responses might be categorized in a class not typically found, there may be a problem.

Second, the coding categories should also be **mutually exclusive** and **independent**. This means that there should be no overlap between the categories, to ensure that a subject or response can be placed in only one category. This frequently requires that an “other” code category be included, so that the

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categories are all inclusive and mutually exclusive. For example, managerial span of control might be coded 1, 2, 3, 4, and “5 or more.” The “5 or more” category ensures everyone a place in a category.

When a questionnaire is highly structured, pre-coding of the categories typically occurs before the data are collected. In many cases, such as when researchers are using open-ended response questions, a framework for classifying responses to questions cannot be established before data collection. This situation requires some careful thought concerning the determination of categories after editing process has been completed. This is called post-coding or simply *coding*. The purpose of **coding open-ended response questions** is to reduce the large number of individual responses to a few general categories of answers that can be assigned numerical scores. Code construction in these situations necessarily must reflect the judgment of the researcher. A major objective in code-building process is to accurately transfer the meaning from written answers to numeric codes.

Code Book

A book identifying each variable in a study and its position in the data matrix. The book is used to identify a variable's description, code name, and field. Here is a sample:

<u>Q/V No.</u>	<u>Field/ col. No.</u>	<u>Code values</u>
• --	1-5	Study number
• -	6	City
•		1 = Lahore
•		2 = Rawalpindi
•		3 = Karachi
•	7 -9	Interview No.
• Sex	10	1 = Male
•		2 = Female
• Age	11-12	Actual
• Education	13	1 = Non literate
		2 = Literate

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Production Coding

Transferring the data from the questionnaire or data collection form after the data have been collected is called production coding. Depending upon the nature of the data collection form, codes may be written directly on the instrument or on a special coding sheet.

Data Entries

Use of scanner sheets for data collection may facilitate the entry of the responses directly into the computer without manual keying in the data. In studies involving highly structured paper questionnaires, an **Optical scanning system** may be used to read material directly to the computer's memory into the computer's memory. Optical scanners process the marked-sensed questionnaires and store the answers in a file.

Cleaning Data

The final stage in the coding process is the error checking and verification, or “**data cleaning**” stage, which is a check to make sure that all codes are legitimate. Accuracy is extremely important when coding data. Errors made when coding or entering data into a computer threaten the validity of measures and cause misleading results. A researcher who has perfect sample, perfect measures, and no errors in gathering data, but who makes errors in the coding process or in entering data into a computer, can ruin a whole research project.

LESSON 30

DATA TRANSFORMATION

Data transformation is the process of changing data from their original form to a format that is more suitable to perform a data analysis that will achieve the research objectives. Researchers often modify the values of a scalar data or create new variables. For example many researchers believe that response bias will be less if interviewers ask consumers for their year of birth rather than their age, even though the objective of the data analysis is to investigate respondents' age in years. This does not present a problem for the research analyst, because a simple data transformation is possible. The raw data coded at birth year can be easily transformed to age by subtracting the birth year from the current year.

Collapsing or combining categories of a variable is a common data transformation that reduces the number of categories. For example five categories of Likert scale response categories to a question may be combined like: the “strongly agree” and the “agree” response categories are combined. The “strongly disagree” and the “disagree” response categories are combined into a single category. The result is the collapsing of the five-category scale down to three.

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Creating new variables by re-specifying the data numeric or logical transformations is another important data transformation. For example, Likert summated scale reflects the combination of scores (raw data) from various attitudinal statements. The summative score for an attitude scale with three statements is calculated as follows:

$$\text{Summative Score} = \text{Variable 1} + \text{Variable 2} + \text{Variable 3}$$

This calculation can be accomplished by using simple arithmetic or by programming a computer with a data transformation equation that creates the new variable “summative score.”

The researchers have created numerous different scales and indexes to measure social phenomenon. For example scales and indexes have been developed to measure the degree of formalization in bureaucratic organization, the prestige of occupations, the adjustment of people in marriage, the intensity of group interaction, the level of social activity in a community, and the level of socio-economic development of a nation.

Keep it in mind that every social phenomenon can be measured. Some constructs can be measured directly and produce precise numerical values (e.g. family income). Other constructs require the use of surrogates or proxies that indirectly measure a variable (e.g. job satisfaction). Second, a lot can be learned from measures used by other researchers. We are fortunate to have the work of thousands of researchers to draw on. It is not always necessary to start from a scratch. We can use a past scale or index, or we can modify it for our own purposes. The process of creating measures for a construct evolves over time. Measurement is an ongoing process with constant change; new concepts are developed, theoretical definitions are refined, and scales or indexes that measure old or new constructs are improved.

Indexes and Scales

Scales and indexes are often used interchangeably. One researcher's scale is another's index. Both produce ordinal- or interval- level measures of variable. To add to the confusion, scale and index techniques can be combined in one measure. Scales and indexes give a researcher more information about variables and make it possible to assess the quality of measurement. Scales and indexes increase reliability and validity, and they aid in **data reduction**; that is condense and simplify the information that is collected.

A **scale** is a measure in which the researcher captures the intensity, direction, level, or potency of a variable construct. It arranges responses or observation on a continuum. A scale can use single indicator or multiple indicators. Most are at the ordinal level of measurement.

An index is a measure in which a researcher adds or combines several distinct indicators of a construct into a single score. This composite score is often a simple sum of multiple indicators. It is used for content or convergent validity. Indexes are often measured at the interval or ratio level.

Researchers sometimes combine the features of scales and indexes in a single measure. This is common when a researcher has several indicators that are scales. He or she then adds these indicators together to yield a single score, thereby an index.

Unidimensionality: It means that all the items in a scale or index fit together, or measure a single construct. Unidimensionality says: If you combine several specific pieces of information into a single score or measure, have all the pieces measure the same thing. (each sub dimension is part of the construct's overall content).

For example, we define the construct “feminist ideology” as a general ideology about gender. Feminist ideology is a highly abstract and general construct. It includes a specific beliefs and attitudes towards social, economic, political, family, sexual relations. The ideology's five belief areas parts of a single general construct. The parts are mutually reinforcing and together form a system of beliefs about dignity, strength, and power of women.

Index Construction

You may have heard about a consumer price index (CPI). The CPI, which is a measure of inflation, is created by totaling the cost of buying a list of goods and services (e.g. food, rent, and utilities) and comparing the total to the cost of buying the same list in the previous year. An **index** is combination of items into a single numerical score. Various components or subgroups of a construct are each measured, and then combined into one measure.

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There are many types of indexes. For example, if you take an exam with 25 questions, the total number of questions correct is a kind of index. It is a composite measure in which each question measures a small piece of knowledge, and all the questions scored correct or incorrect are totaled to produce a single measure.

One way to demonstrate that indexes are not a very complicated is to use one. Answer yes or no to the seven questions that follow on the characteristics of an occupation. Base your answers on your thoughts regarding the following four occupations: long-distance truck driver, medical doctor, accountant, telephone operator. Score each answer 1 for yes and 0 for no.

1. Does it pay good salary?
2. Is the job secure from layoffs or unemployment?
3. Is the work interesting and challenging?
4. Are its working conditions (e.g. hours, safety, time on the road) good?
5. Are there opportunities for career advancement and promotion?
6. Is it prestigious or looked up to by others?
7. Does it permit self-direction and the freedom to make decisions?

Total the seven answers for each of the four occupations. Which had the highest and which had the lowest score? The seven questions are our operational definition of the construct good occupation. Each question represents a subpart of our theoretical definition.

Creating indexes is so easy that it is important to be careful that every item in the index has face validity. Items without face validity should be excluded. Each part of the construct should be measured with at least one indicator. Of course, it is better to measure the parts of a construct with multiple indicators.

Another example of an index is college quality index. Our theoretical definition says that a high quality college has six distinguished characteristics: (1) fewer students per faculty member, (2) a highly educated faculty, (3) more books in the library, (4) fewer students dropping out of college, (5) more students who go to advanced degrees, and (6) faculty members who publish books or scholarly articles. We score 100 colleges on each item, and then add the score for each to create an index score of college quality that can be used to compare colleges.

Indexes can be combined with one another. For example, in order to strengthen the college quality index. We add a sub-index on teaching quality. The index contain eight elements: (1) average size of classes, (2) percentage of class time devoted to discussion, (3) number of different classes each faculty member teaches, (4) availability of faculty to students outside the classroom, (5) currency and amount of reading assigned, (6) degree to which assignments promote learning, (7) degree to which faculty get to know each student, and (8) student ratings of instruction. Similar sub-index measures can be created for other parts of the college quality index. They can be combined into a more global measure of college quality. This further elaborates the definition of a construct “quality of college.”

Weighting

An important issue in index construction is whether to weight items. Unless it is otherwise stated, assume that an index is un-weighted. Likewise, unless we have a good reason for assigning different weights, use equal weights. A weighted index gives each item equal weight. It involves adding up the items without modification, as if each were multiplied by 1 (or – 1 for negative items that are negative).

Scoring and Score Index

In one of our previous discussions we had tried to measure job satisfaction. It was operationalized with the help of dimensions and elements. We had constructed number of statements on each element with 5 response categories using Likert scale i.e. strongly agree, agree, undecided, disagree, and strongly disagree. We could score each of these items from 1 to 5 depending upon the degree of agreement with the statement. The statements have been both positive as well as negative. For positive statements we can score straight away from 5 to 1 i.e. strongly agree to strongly disagree. For the negative statements we have to reverse the score i.e. 1 for “strongly agree,” 2 for “agree,” 3 for “undecided” to 4 for “disagree,” and 5 for “strongly disagree.” Reason being that negative multiplied by a negative becomes positive i.e. a negative statement and a person strongly disagreeing with it implies that he or she has a positive response so we give a score of 5 in this example. In our example, let us say there were 23 statements measuring for different elements and dimensions measuring job satisfaction. When on each statement the respondent could get a minimum score of 1 and a maximum score of 5, on 23 statements a respondent could get a minimum score of (23 X 1) and a maximum score of (23 X 5) 115. In this way the score index ranges from 23 to 115, the lower end of the score index showing minimum job satisfaction and upper end as the highest job satisfaction. In reality we may not find any on the extremes, rather the respondents could be spread along this continuum. We could use the raw scores of independent and dependent variable and apply appropriate statistics for testing the hypothesis. We could also divide the score index into different categories like high “job satisfaction” and “low satisfaction” for presentation in a table. We cross-classify job satisfaction with some other variable, apply appropriate statistics for testing the hypothesis.

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LESSON 31

DATA PRESENTATION

Tables and graphs (pictorial presentation of data) may simplify and clarify the research data. Tabular and graphic representation of data may take a number of forms, ranging from computer printouts to elaborate pictographs. The purpose of each table or graph, however, is to facilitate the summarization and communication of the meaning of the data.

Although there are a number of standardized forms for presenting data in table or graphs, the creative researcher can increase the effectiveness of particular presentation. Bar charts, pie charts, curve diagrams, pictograms, and other graphic forms of presentation create a strong visual impression.

The proliferation of computer technology in business and universities has greatly facilitated tabulation and statistical analysis. Commercial packages eliminate the need to write a new program every time you want to tabulate and analyze data with a computer. SAS, Statistical Package for the Social Sciences (SPSS), SYSTAT, Epi. Info. And MINITAB is commonly used statistical packages. These user friendly packages emphasize statistical calculations and hypothesis testing for varied types of data. They also provide programs for entering and editing data. Most of these packages contain sizeable arrays of programs for descriptive analysis and univariate, bivariate, and multivariate statistical analysis.

Results with one variable

Frequency Distribution

Several useful techniques for displaying data are in use. The easiest way to describe the numerical data of one variable is with a frequency distribution. It can be used with nominal-, ordinal-, interval-, or ratio-level data and takes many forms. For example we have data of 400 students. We can summarize the data on the gender of the students at a glance with raw count or a *frequency distribution*

Table 1: Frequency distribution of students

Gender	Frequency	Percent
Male	300	75
Female	100	25
Total	400	100

We can present the same information in a graphic form. Some common types of graphic presentations are the *histograms*, *bar chart*, and *pie chart*. Bar charts or graphs are used for discrete variables. They can have vertical or horizontal orientation with small space between the bars. The terminology is not exact, but histograms are usually upright bar graphs for interval or ratio data.

Presentation of data in these forms lays emphasis on visual representation and graphical techniques over summary statistics. Summary statistics may obscure, conceal, or even misrepresent the underlying structure of the data. Therefore it is suggested that data analysis should begin with visual inspection.

The presented data has to be interpreted. The purpose of interpretation is to explain the meanings of the data so that we can make inferences and formulate conclusions. Therefore, **interpretation refers to making inferences pertinent to the meaning and implications of the research investigation and drawing conclusions. In order for interpretation, the data have to be meaningfully analyzed. For purposes of analysis the researchers use statistics.**

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The word *statistics* has several meanings. It can mean a set of collected numbers (e.g. numbers telling how many people living in a city) as well as a branch of applied mathematics used to manipulate and summarize the features of numbers. Social researchers use both types of statistics. Here, we focus on the second type – ways to manipulate and summarize numbers that represent data from research project.

Descriptive statistics describe numerical data. They can be categorized by the number of variables involved: univariate, bivariate, or multivariate (for one, two, and three or more variables). Univariate statistics describe one variable.

Researchers often want to summarize the information about one variable into a single number. They use three measures of central tendency, or measures of the center of the frequency distribution: mean, median and mode, which are often called averages (a less precise and less clear way to say the same thing). The *mode* is simply the most common or frequently occurring number. The *median* is the middle point. The *mean* also called the arithmetic average, is the most widely used measure of central tendency. A particular central tendency is used depending upon the nature of the data.

Bivariate Tables

The bivariate contingency table is widely used. The table is based on cross-tabulation (cross-classification); that is the cases are organized in the table on the basis of two variables at the same time.

A contingency table is formed by cross-tabulating the two or more variables. It is contingent because the cases in each category of a variable get distributed into each category of a second variable. The table distributes cases into categories of multiple variables at the same time and shows how the cases, by the category of one variable, are “contingent upon” the categories of the other variables.

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Constructing Percentage Tables

It is to construct a percentage table, but there are ways to make it look professional. Let us take two variables like the age of the respondents and their attitude towards “women empowerment.” Assuming that age affects the attitude towards women empowerment let us hypothesize: the lower the age, the higher the favorable attitude towards “women empowerment.” The age range of the respondents is 25 to 70, and the attitude index has three categories of “highly favorable,” “medium favorable,” and “low favorable.” The age variable has so many categories that making a table with that number becomes unwieldy and meaningless. Therefore, we regroup (recode) the age categories into three i.e. under 40 years, 40 – 60 years, and 61 + years.

Univariate table for age

• Table 2: Age of the respondents .		
• Age (Yrs.)	Frequency	Percent
• Under 40	1000	33.3
• 40 – 60	1000	33.3
• 61 +	1000	33.3
• Total	3000	100

Univariate table for attitude

- Table 3: Attitude towards women empowerment

Attitude	Frequency	Percent
Hi Favorable	1100	37
Med Favorable	1050	35
Lo Favorable	850	28
Total	3000	100

Bivariate table

- Table 4: Age by attitude towards women empowerment

Level of attitude	Age (in years)						Total	
	under 40		40 –60		61 +			
	F.	%	F.	%	F	%	F	%
Hi Favorable	600	60	300	30	200	20	1100	37
Med. Favorable	300	30	500	50	250	25	1050	35
Lo Favorable	100	10	200	20	500	50	850	28
Total	1000	100	1000	100	1000	100	3000	100

LESSON 32

THE PARTS OF THE TABLE

1. Give each table a number.
2. Give each table a title, which names variables and provides background information
3. Label the row and columns variables and give name to each of the variable categories.
4. Include the totals of the columns and rows. These are called *marginals*. They equal the univariate frequency distribution for the variable.
5. Each number or place that corresponds to the intersection of a category for each variable is a *cell of a table*.
6. The numbers with the labeled variable categories and the totals are called the *body of the table*.
7. If there is missing information, report the number of missing cases near the table to account for all original cases.

Researchers convert raw count tables into percentages to see bi-variate relationship. There are three ways to percentage a table: by row, by column, and for the total. The first two are often used and show relationship.

Is it best to percentage by row or column? Either could be appropriate. A researcher's hypothesis may imply looking at row percentages or the column percentages. Here, the hypothesis is that age affects attitude, so column percentages are most helpful. Whenever one factor in a cross-tabulation can be considered the cause of the other, percentage will be most illuminating if they are computed in the direction of the causal factor.

Reading a percentage Table: Once we understand how table is made, reading it and figuring out what it says are much easier. To read a table, first look at the title, the variable labels, and any background information. Next, look at the direction in which percentages have been computed – in rows or columns.

Researchers read percentaged tables to make comparisons. Comparisons are made in the opposite direction from that in which percentages are computed. A rule of thumb is to compare across rows if the table is percentaged down (i.e. by column) and to compare up and down in columns if the table is percentaged across (i.e. by row).

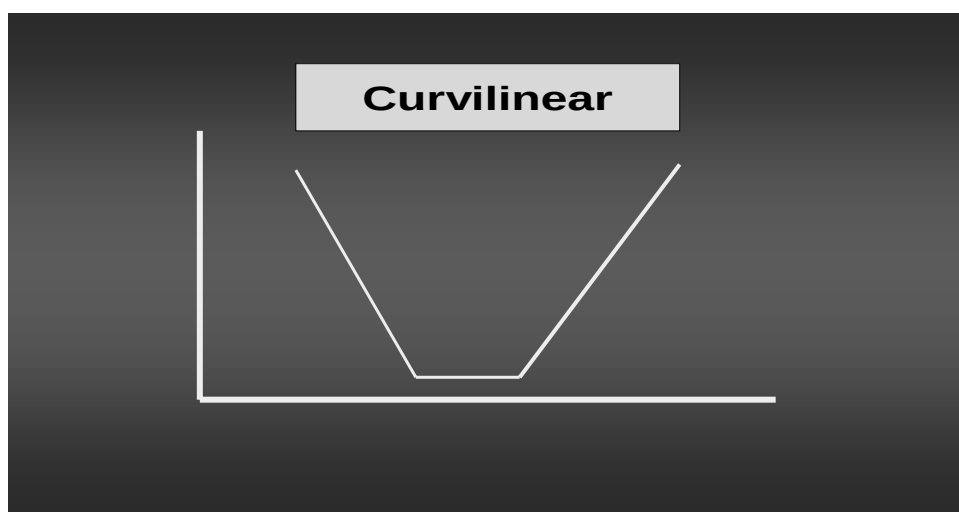
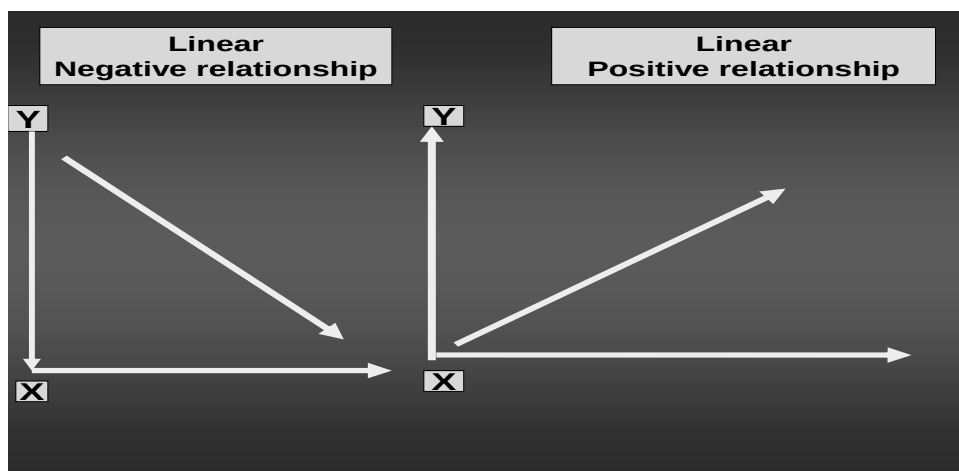
It takes practice to see a relationship in a percentaged table. If there is no relationship in a table, the cell percentages look approximately equal across rows or columns. A linear relationship looks like larger percentages in the diagonal cells. If there is curvilinear relationship, the largest percentages form a pattern across cells. For example, the largest cells might be the upper right, the bottom middle, and the upper left. It is easiest to see a relationship in a moderate-sized table (9 to 16 cells) where most cells have some cases (at least five cases are recommended) and the relationship is strong and precise.

Linear relationship

- Table 4: Age by attitude towards women empowerment

Level of attitude	Age (in years)								Total
	under 40		40 –60		61 +				
	F.	%	F.	%	F.	%	F.	%	
Hi Favorable	600	60	300	30	200	20	1100	37	
Med. Favorable	300	30	500	50	250	25	1050	28	
Lo Favorable	100	10	200	20	500	50	850	28	
Total	1000	100	1000	100	1000	100	3000	100	

- Larger percentages in the diagonal cells



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A simple way to see strong relationships is to circle the largest percentage in each row (in row percentaged tables) or columns (for column-percentaged tables) and see if a line appears.

A simple way to see strong relationship is to circle the largest percentage in applicable row or column and see if a line appears

- Table 4: Age by attitude towards women empowerment

Level of attitude	Age (in years)						Total	
	under 40		40 –60		61 +			
	F.	%	F.	%	F.	%	F.	%
Hi Favorable	600	60	300	30	200	20	1100	37
Med. Favorable	300	30	500	50	250	25	1050	35
Lo Favorable	100	10	200	20	500	50	800	28
Total	1000	100	1000	100	1000	100	3000	100

The circle-the-largest-cell rule works – with one important caveat. The categories in the percentages table must be ordinal or interval. The lowest variable categories begin at the bottom left. If the categories in a table are not ordered the same way, the rule does not work.

Statistical Control

Showing an association or relationship between two variables is not sufficient to say that an independent variable *causes* a dependent variable. In addition to temporal order and association, a researcher must eliminate alternative explanations – explanations that can make the hypothetical relationship spurious. Experimental researchers do this by choosing a research design that physically controls potential alternative explanations for results (i.e. that threaten internal validity).

In non-experimental research, a researcher controls for alternative explanations with statistics. He or she measures possible alternative explanations with *control variables*, and then examines the control variables with multivariate tables and statistics that help him or her to decide whether a bivariate relationship is spurious. They also show the relative size of the effect of multiple independent variables on dependent variable.

A researcher controls for alternative explanation in multivariate (more than two variables) analysis by introducing a third (sometimes fourth, or fifth) variable. For example, a bivariate table shows that young people show more favorable attitude towards women empowerment. But the relationship between age and attitude towards women empowerment may be spurious because men and women may have different attitudes. To test whether the relationship is actually due to gender, a researcher must control for gender; in other words, effects of gender are statistically removed. Once this is done, a researcher can see whether the bivariate relationship between age and attitude towards women empowerment remains.

A researcher controls for a third variable by seeing whether the bivariate relationship persists within categories of the control variable. For example controls for gender, and the relationship between age and attitude persists. This means that both male and females show negative association between age and attitude toward women empowerment. In other words, the control variable has no effect. When this is so, the bivariate relationship is not spurious.

If the bivariate relationship weakens or disappears after the control variable is considered, it means that the age is not real factor that makes the difference in attitude towards women empowerment, rather it is the gender of the respondents.

Statistical control is a key idea in advanced statistical techniques. A measure of association like the correlation co-efficient only suggests a relationship. Until a researcher considers control variables, the bivariate relationship could be spurious. Researchers are cautious in interpreting bivariate relationships until they have considered control variables.

After they introduce control variables, researchers talk about the *net effect* of an independent variable – the effect of independent variable “net of,” or in spite of, the control variable. There are two ways to introduce control variables: trivariate percentaged tables and multiple regression analysis.

Constructing Trivariate Tables

In order to meet all the conditions needed for causality, researchers want to “control for” or see whether an alternative explanation explains away a causal relationship. If an alternative explanation explains a relationship, then bivariate relationship is spurious. Alternative explanations are operationalize as a third variable, which are called *control variables* because they control for alternative explanation.

One way to take such third variables into consideration and see whether they influence the bivariate relationship is to statistically introduce control variables using trivariate or three variable tables. Trivariate tables differ slightly from bivariate tables; they consist of multiple bivariate tables.

A trivariate table has a bivariate table of the independent and dependent variable for each category of the control variable. These new tables are called *partials*. The number of partials depends on the number of categories in control variable. Partial tables look like bivariate tables, but they use a subset of the cases. Only cases with a specific value on the control variable are in the partial. Thus it is possible to break apart a bivariate table to form partials, or combine the partials to restore the initial bivariate table.

Trivariate tables have three limitations. First, they are difficult to interpret if a control variable has more than four categories. Second, control variables can be at any level of measurement, but interval or ratio control variables must be grouped (i.e. converted to an ordinal level), and how cases are grouped can affect the interpretation of effects. Finally, the total number of cases is a limiting factor because the cases are divided among cells in partials. The number of cells in the partials equals the number of cells in the bivariate relationship multiplied by the number of categories in the control variables. For example if the control variable has three categories, and a bivariate table has 12 cells, the partials have $3 \times 12 = 36$ cells. An average of five cases per cell is recommended, so the researcher will need $5 \times 36 = 180$ cases at minimum.

Like a bivariate table construction, a trivariate table begins with a compound frequency distribution (CFD), but it is a three-way instead of two-way CFD. An example of a trivariate table with “gender” as control variable for the bivariate table is shown here:

Partial table for males

Level of Attitude	Age (in years)							
	Under 40		40—60		61+		Total	
	F	%	F	%	F.	%	F.	%.
High	300	60	200	33	30	6	530	33
Medium	140	28	270	45	120	24	530	33
Low	60	12	130	22	350	70	540	34
Total	500	100	600	100	500	100	1600	100

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Partial table for females

Level of Attitude	Age (in years)							
	Under 40		40—60		61+		Total	
	F	%	F	%	F.	%	F.	%.
High	350	70	200	50	20	4	570	41
Medium	150	30	150	38	220	44	520	37
Low	-	-	50	12	260	52	310	22
Total	500	100	400	100	500	100	1400	100

The replication pattern is the easiest to understand. It is when the partials replicate or reproduce the same relationship that existed in the bivariate table before considering the control variable. It means that the control variable has no effect.

The specification pattern is the next easiest pattern. It occurs when one partial replicates the initial bivariate relationship but other partials do not. For example, we find a strong (negative) bivariate relationship between age of the respondents and attitude towards women empowerment. We control for gender and discover the relationship holds only for males (i.e. the strong negative relationship was in the partial for males, but not for females). This is specification because a researcher can specify the category of the control variable in which the initial relationship persists.

The interpretation pattern describes the situation in which the control variable intervenes between the original independent variable and the dependent variables.

The suppressor variable pattern occurs when the bivariate tables suggest independence but relationship appears in one or both of the partials. For example, the age of the respondents and their attitudes towards women empowerment are independent in a bivariate table. Once the control variable “gender” is introduced, the relationship between the two variables appears in the partial tables. The control variable is suppressor variable because it suppressed the true relationship; the true relationship appears in partials.

Multiple Regression Analysis

Multiple regression controls for many alternative explanations of variables simultaneously (it is rarely possible to use more than one control variable using percentaged tables). Multiple regression is a technique whose calculation you may have learnt in the course on statistics.

Note

In the preceding discussion you have been exposed to the descriptive analysis of the data. Certainly there are statistical tests which can be applied to test the hypothesis, which you may have learnt in your course on statistics.

LESSON 33

EXPERIMENTAL RESEARCH

Experimental research builds on the principles of positivist approach more directly than do the other research techniques. Researchers in the natural sciences (e.g. chemistry and physics), related applied fields (e.g. engineering, agriculture, and medicines) and the social sciences conduct experiments. The logic that guides an experiment on plant growth in biology or testing a metal in engineering is applied in experiments on human social behavior. Although it is most widely used in psychology, the experiment is found in education, criminal justice, journalism, marketing, nursing, political science, social work, and sociology.

The purpose of experimental research is to allow the researcher to *control* the research situation so that *causal* relationships among variables may be evaluated. The experimenter, therefore, manipulates a single variable in an investigation and holds constant all other, extraneous variables. (Events may be controlled in an experiment in a way that is not possible in a survey.) The goal of the experimental design is the confidence that it gives the researcher that his experimental treatment is the cause of the effect he measures.

Experiment is a research design in which conditions are controlled so that one or more variables can be manipulated in order to test a hypothesis. Experimentation is a research design that allows evaluation of causal relationship among variables.

Experiments differ from other research methods in terms of degree of control over the research situation. In a typical experiment one variable (the *independent variable*) is manipulated and its effect on another variable (the *dependent variable*) is measured, while all other variables that may confound such relationship are eliminated or controlled. The experimenter either creates an artificial situation or deliberately manipulates a situation.

Once the experimenter manipulates the independent variable, changes in the dependent variable are measured. The essence of a behavioral experiment is to do something to an individual and observe his or her reaction under conditions where this reaction can be measured against a known baseline.

To establish that variable X cause's variable Y, *all three* of the following conditions should be met:

1. Both X and Y should co-vary (i.e. when one goes up, the other should also simultaneously go up (or go down).
2. X (the presumed causal factor) should precede Y. In other words, there must be a time sequence in which the two occur.
3. No other factor should possibly cause the change in the dependent variable Y.

It may thus be seen that to establish causal relationships between two variables in an organizational setting, several variables that might co-vary with the dependent variable have to be controlled. This would then allow us to say that variable X and variable X alone causes the dependent variable Y. Useful as it is to know the cause-and-effect relationships, establishing them is not so easy, because several other variables that co-vary with the dependent variable have to be controlled. It is not always possible to control all the co-variables while manipulating the causal factor (the independent variable that is causing the dependent variable) in organizational settings, where events flow or occur naturally and normally. It is, however, possible to first isolate thee effects of a variable in a tightly controlled artificial setting (the lab setting), and after testing and establishing the cause-and-effect relationship under these tightly controlled conditions, see how generalizable such relationships are to the field setting.

The Language of Experiments

Experimental research has its own language or set of terms and concepts. One important term frequently used is *subjects* or *test units*. In experimental research, the cases or people used in research projects and on whom variables are measured are called thee *subjects* or *test units*. In other words these are those

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entities whose responses to the experimental treatment are measured or observed. Individuals, organizational units, sales territories, or other entities may be the test units. Similar terminology is used on different component parts of the experiments.

Parts of Experiments: We can divide the experiments into seven parts and for each part there is a term. Not all experiments have all these parts, and some have all seven parts plus others. The following seven usually make up a true experiment.

1. Treatment or independent variable.
2. Dependent variable.
3. Pretest.
4. Posttest.
5. Experimental group.
6. Control group.
7. Assignment of subjects.

Treatment or independent variable: The experimenter has some degree of control over the independent variable. The variable is independent because its value can be manipulated by the experimenter to whatever he or she wishes it to be. In experimental design the variable that can be manipulated to be whatever the experiment wishes. Its value may be changed or altered independently of any other variable.

In most experiments, a researcher creates a situation or enters into an ongoing situation, then modifies it. The treatment (or the stimulus or manipulation) is what the researcher modifies. The term comes from medicine, in which a physician administers a treatment to patients; the physician intervenes in a physical or psychological condition to change it. It is the independent variable or the combination of independent variables.

In experiments, for example, the researcher creates a condition or situation. Look at “the degree of fear or anxiety”; the levels are high-fear or low-fear situation. Instead of asking the subjects, as we do in surveys, whether they are fearful, experimenter puts the subjects into either in a high-fear or low-fear situation. They measure the independent variable by manipulating conditions so that some subjects feel a lot of fear and others feel little.

Researchers go to great lengths to create treatments. They want the treatment to have an impact and produce specific reactions, feelings, or behaviors.

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It is also possible the researchers look at the alternative manipulations of the independent variable being investigated. In business research, the independent variable is often categorical or classificatory variable, representing some classifiable or qualitative aspects of management strategy. To determine the effects of training, for example, the experimental treatment that represents the independent variable is the training program itself.

Dependent Variable: The criterion or standard by which the results are judged. It is assumed that the changes in the dependent variable are consequence of changes in the independent variable. For example, measures of turnover, absenteeism, or morale might be alternative choices for the dependent variable, depending on the purpose of the training.

The outcomes in the experimental research are the physical conditions, social behaviors, attitudes, feelings, or beliefs of subjects that change, in response to treatment. Dependent variables can be measured by paper-and-pencil indicators, observations, interviews, or physiological responses (e.g. heartbeat, or sweating palms).

Selection of dependent variable is crucial decision in the design of an experiment.

Pretests and Posttests: Frequently a researcher measures the dependent variable more than once during an experiment. The *pretest* is the measurement of the dependent variable prior to the introduction of the treatment. The *posttest* is the measurement of the dependent variable after the treatment has been introduced into the experimental situation.

Experimental and Control Groups: Experimental researchers often divide subjects into two or more groups for purposes of comparison. A simple experiment has only two groups, only one of which receives the treatment. The experimental group is the group that receives the treatment or in which the treatment is present.

The group that does not receive the treatment is called the “control group.” When the independent variable takes on many different values, more than one experimental group is used.

In the simplest type of experiment, only two values of the independent variable are manipulated. For example, consider measuring the influence of a change in work situation, such as playing music over an intercom during working hours, on employee productivity. In the experimental condition (the treatment administered to the **experimental group**), music is played during working hours. In the control condition (the treatment administered to the **control group**), the work situation remains the same, without change. By holding conditions constant in the control group, the researcher controls for potential sources of error in the experiment. Productivity, (the dependent variable) in the two groups is compared at the end of the experiment to determine whether playing the music (the independent variable) has any effect.

Several experimental treatment levels can also be used. The music/productivity experiment, with one experimental and one control group, may not tell the researcher everything he or she wishes to know about the music/productivity relationship. If the researcher wished to understand the functional nature of the relationship between music and productivity at several treatment levels, additional experimental groups with music played for only 2 hours, only for 4 hours, and only for 6 hours might be studied. This type of design would allow the experimenter to get a better idea about the impact of music on productivity.

Assignment of Subjects/Test Units: Social researchers frequently want to compare. When making comparisons, the researchers want to compare the cases that do not differ with regard to variables that offer alternative explanations. Therefore the groups should be similar in characteristics in such a way that the change in the dependent variable is presumably the outcome of the manipulation of the independent variable, having no alternative explanations.

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Random assignment (Randomization) is a method for assigning the cases (e.g. individuals, organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one's confidence that the groups do not differ in a systematic way. It is a mechanical method; the assignment is automatic, and the researcher cannot make assignments on the basis of personal preference or the features of specific cases.

Random assignment is random in statistical/mathematical sense, not in everyday sense. In everyday speech, random means unplanned, haphazard, or accidental, but it has a special meaning in mathematics. In probability theory, random describes a process in which each case has a known chance of being selected. Random selection allows the researcher calculate the odds that a specific case will be sorted into one group or the other. A random process is the one in which all cases have an exactly equal chance of ending up in one or the other group.

Random assignment or randomization is unbiased because a researcher's desire to confirm a hypothesis or a research subject's personal interests does not enter into the selection process. It also assures the researcher that repetitions of an experiment – under the controlled conditions – will show true effects, if

they exist. **Random assignment of subjects allows the researcher to assume that the groups are identical with respect to all variables except for experimental treatment.**

Random assignment of subjects to the various experimental groups is the most common technique used to prevent test units from differing from each other on key variables; it assumes that all the characteristics of these subjects have been similarly randomized. If the experimenter believes that certain extraneous variable may affect the dependent variable, he or she may make sure that the subjects in each group are matched on these characteristics. **Matching** the subjects on the basis of pertinent background information is another technique for controlling assignment errors.

Matching presents a problem: What are the relevant characteristics to match on, and can one locate exact matches? Individual cases differ in thousands of ways, and the researcher cannot know which might be relevant.

Three Types of Controls

1. **Manipulation of the Independent Variable:** In order to examine the causal effects of an independent variable on a dependent variable, certain manipulations need to be tried. Manipulation simply means control over the stimulus that is we create different levels of the independent variable to assess the impact on the dependent variable. Let us say we want to test the effects of lighting on worker production levels among sewing machine operators. To establish cause and effect relationship, we must measure the production levels of all the operators over a 15 day period with the usual amount of light they work with – say 60 watt bulbs. We might then want to split the group of 60 operators into three groups of 20 members each, and while allowing the subgroup to continue to work under the same conditions as before (60-watt electric light bulbs). We might want to manipulate the intensity of the light for the other two subgroups, by making one group work with 75 watt and the other with 100 watt light bulbs. After the different groups have worked with these varying degrees of light exposure for 15 days, each group's total production for these 15 days may be analyzed to see the difference between the pre-experimental and the post experimental productions among the groups is directly related to the intensity of the light to which they have been exposed. If our hypothesis that better lighting increases the production levels is correct, the subgroups that did not have any change in the lighting (control group), should have no increase in production and the other two groups should show increases, with the one having the most light (100 watts) showing greater increases than those who had the 75 watt lighting.

In this case the independent variable, lighting, has been manipulated by exposing different groups to different degrees of changes in it. This manipulation of the independent variable is also known treatment, and the results of the treatment are called treatment effects.

2. **Holding Conditions Constant:** When we postulate cause-and-effect relationships between two variables X and Y, it is possible that some other factor, say A, might also influence the dependent variable Y. In such a case, it will not be possible to determine the extent to which Y occurred only because of X, since we do not know how much of the total variation of was caused by the presence of the other factor A. If the true effect of the X is to be assessed, then the effect of A has to be controlled. This is also called as controlling the effect of contaminating factors or confounding factors.
3. **Control over the Composition of Groups:** If the experimental and control groups have such characteristics that could contaminate the results then the researcher may have to take note of such factors, if there are any. The group differences should not confound the effect of X variable that happens to be under study. The experimental and control groups need to be balanced. For this purpose the researcher may use **random selection** of the subjects and allocating to different groups. Finally the experimental and control groups should also be selected randomly. Another way to have identical groups is by following the procedure of

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matching. One could look at the possible characteristics of the subjects that could contaminate the effect of X variable, and try to distribute these evenly in all the groups. So pick up one subject and try to match it with another subject on the specified characteristics (age, gender, education, marital status) and put one subject in one group and the other in the other group. After the formation of groups, the researcher may randomly decide about experimental and control groups.

Random Assignment

Social researchers frequently want to compare. For example, a researcher has two groups of 15 students and wants to compare the groups on the basis of key differences between them (e.g. a course that one group completed). Or a researcher has five groups of customers and wants to compare the groups on the basis of one characteristic (e.g. geographic location). “Compare apples with apples, don’t compare apples with oranges.” It means that a valid comparison depends on comparing things that are fundamentally alike. Random assignment facilitates comparison in experiments by creating similar groups.

Random assignment is a method for assignment cases (e.g. individuals, organizations) to groups for the purpose of making comparisons. It is a way to divide or sort a collection of cases into two or more groups in order to increase one’s confidence that the groups do not differ in a systematic way. It is mechanical method; the assignment is automatic, and the researcher cannot make assignments on the basis of personal preference or the features of specific cases.

Random assignment is random in a statistical or mathematical sense, not in an everyday sense. In everyday speech, random means unplanned, haphazard, or accidental, but it has a specialized meaning in mathematics. In probability theory, *random* describes a process in which each case has a known chance of being selected. Random assignment lets a researcher calculate the odds that a specific case will be sorted into one group over another.

Random assignment or randomization is unbiased because a researcher’s desire to confirm a hypothesis or a research subject’s personal interest does not enter into selection process.

Matching

It implies to match the characteristics (such as age, sex) of the cases in each group. Matching is an alternative to random assignment, but it is an infrequently used one.

Matching presents a problem: What are the relevant characteristics to match on, and can one locate exact matches. Individual cases differ in thousands of ways, and the researcher cannot know which might be relevant. Therefore, randomization is preferred over matching. It takes care of the contaminating factors.

LESSON 34

EXPERIMENTAL RESEARCH (Cont.)

Steps in Conducting an Experiment

Following the basic steps of the research process, experimenters decide on a topic, narrow it into a testable research problem or question, and then develop a hypothesis with variables. Once a researcher has the hypothesis, the steps of experimental research are clear. Broadly there are about 12 steps in conducting an experiment, which are as below:

1. Begin with a straightforward hypothesis that is appropriate for experimental research.
2. Decide on an experimental design that will test the hypothesis within practical limitations. The researcher decides the number of groups to use, how and when to create treatment conditions, the number of times to measure the dependent variable, and what the groups of subjects will experience from beginning till end.
3. Decide how to introduce the treatment or create a situation that induces the independent variable.
4. Develop a valid and reliable measure of the dependent variable.
5. Set up an experimental setting and conduct a pilot test of the treatment and dependent variable measures.
6. Locate appropriate subjects or cases.
7. Randomly assign subjects to groups (if random assignment is used in the chosen research design) and give careful instructions.
8. Gather data for the pretest measure of the dependent variable for all groups (if pretest is used in the chosen design).
9. Introduce the treatment to the experimental group only (or to the relevant groups if there are multiple experimental groups) and monitor all groups.
10. Gather data for posttest measure of the dependent variable.
11. Debrief the subjects by informing them of the true purpose and reasons for the experiment. Ask subjects what they thought was occurring. Debriefing is crucial when subjects have been deceived about some aspect of the treatment.
12. Examine data collected and make comparisons between different groups. Where appropriate, use statistics and graphs to determine whether or not the hypothesis is supported.

Types of Designs

Researchers combine parts of experiment (e.g. pretests, control groups, etc.) together into an experimental design. For example some designs lack pretests, some do not have control groups, and others have many experimental groups. Certain widely used standard designs have names.

Classical Experimental Design: All designs are variations of the classical experimental design, which has random assignment of subjects, a pretest and a posttest, an experimental group, and a control group.

Quasi-Experimental Designs:

One-shot Case Study Design: Also called the one-group posttest-only design, the one-shot case study design has only one group, a treatment, and a posttest. Because it is only one group, there is no random assignment. For example, a researcher shows a group of students a horror film, then measures their attitude with a questionnaire. A weakness of this design is that it is difficult to say for sure that the treatment caused the dependent variable. If subjects were the same before and after the treatment, the researcher would not know it.

One Group Pretest-posttest Design: This design has one group, a pretest, a treatment, and a posttest. It lacks a control group and random assignment. Continuing with the previous example, the researcher gives a group of students an attitude questionnaire to complete, shows a horror film, then has them

complete the same questionnaire second time. This is an improvement over the one-shot case study because the researcher measures the dependent variable both before and after the treatment. But it lacks the control group for comparison. The researcher cannot know whether something other than the treatment occurred between the pretest and the posttest to cause the outcome.

Two Groups Posttest-only Design: It has two groups, a random assignment of subjects, a posttest, and a treatment. It has all parts of the classical design except a pretest. Continuing with our previous example, the researcher forms two groups through randomization process. He shows group a horror film to one group i.e. the experimental group. The other group is not shown any film. Both groups then complete the questionnaire. The random assignment reduces the chance that the groups differed before the treatment, but without a pretest, a researcher cannot be as certain that the groups began the same on the dependent variable.

True Experimental Designs

Experimental designs, which have at least two groups, a random assignment of subjects to experimental and control groups, only experimental group is exposed to treatment, both groups record information before and after the treatment, are known as ex-post facto experimental designs.

Pretest and Posttest Experimental and Control Group Design: Two groups, one control group and the other experimental group, are formed randomly. Both the groups are exposed to pretest and posttest. The experimental group is exposed to treatment while the control group is not. Measuring the difference between the differences in the post- and pretests of the two groups would give the net effects of the treatment.

Experimental Group: Pretest (O1) X Posttest (O2)
Control Group: Pretest (O3) - Posttest (O4)

Randomization used for setting up the group.

$[(O2 - O1) - (O4 - O3)] = \text{Treatment effect (could be anywhere between 0 to -1 or +1).}$

Solomon's Four Group Design: To gain more confidence in internal validity in experimental designs, it is advisable to set up two experimental groups and two control groups. One experimental group and one control group can be given the both pretest and the posttest. The other two groups will be given only the posttest. Here the effects of treatment can be calculated in several different ways as shown in figure 1:

Figure 1: Solomon's four group design

Group	Pretest	Treatment	Posttest
1. Experimental	O1	X	O2
2. Control	O3	-	O4
3. Experimental	-	X	O5
4. Control	-	-	O6

$(O2 - O1) = E$

$(O4 - O3) = E$

$(O5 - O6) = E$

$(O5 - O3) = E$

$[(O2 - O1) - (O4 - O3)] = E$

E = Effect

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If all Es are similar, the cause and effect relationship is highly valid.

Interaction Effect

The effect of two variables together is likely to be greater than the individual effect of each put together. The idea of an interaction effect is familiar, especially in the area of medicine or illness. As an example, imagine that for a given population of 100 persons, all of the same age and sex, it was found that if all 100 smoked cigarettes the effect would be a lung cancer rate of 20 percent. Assume that for an identical group of 100 persons who did not smoke but lived in a smoggy environment, 10 percent would get lung cancer. Now consider a third identical group of 100 persons all of whom smoke and also live in a smoggy environment. The additive effect of both smoking and smog would be 20 percent plus 10 percent, or a total of 30 percent (30 people) having cancer. However, imagine that an actual medical survey of the population shows a cancer rate of 37 percent among persons experiencing both smoking and smog. This extra 7 percent can be computed residually as:

$$\begin{aligned}\text{Interaction Effect} &= \text{Total effect} - (\text{smoking effect} + \text{smog effect}) = 37 \text{ percent} \\ &= 37 \text{ percent} - (20 \text{ percent} + 10 \text{ percent}) \\ &= 37 \text{ percent} - 30 \text{ percent} \\ &= 7 \text{ percent}\end{aligned}$$

In experiments we have the pretests and posttests, in which case we use the same instrument for measuring the dependent variable, for example racial prejudice as an effect of a movie. In pretest is a questionnaire in which items forming the prejudice scale are dispersed at random among other items so that the subject does not know that his or her level of racial prejudice is being measured. Nevertheless, the measurement of this variable (prejudice) itself, by presenting questions about race relations may stimulate the subject's thinking and actually cause a change in his or her level of racial prejudice. Any pretest effect that occurs will be visible as part of extraneous change (change caused by the test stimulus) in the control group, as the pretest is also presented to the control group. Any change between the pretest and posttest for measuring the dependent variable in the control group may be attributed to the sensitization of the subjects with the instrument. In the experimental group of course a movie (an X variable) was shown due to which we expect a change in the racial prejudice of the subjects. But that is not all. The subjects in the experimental group were also exposed to the instrument for measuring the racial prejudice, hence they were also sensitized. Their posttest results include the combined effect of exposure to a movie and that of sensitization to the instrument. In other words the racial prejudice of the subjects in the experimental group exhibits the interaction effect of the treatment plus that of sensitization of the instrument.

In order to calculate the interaction effect in the experiment we shall have two experimental groups and one control group created by using the randomization process. It may look like this:

Experimental group 1: Pretest (O1)	X	Posttest (O2)
Control group: Pretest (O3)	-	Posttest (O4)

Why O4 be different from O3? The difference may be due to sensitization. So let us figure it out. Let us take another experimental group and we do not pretest i.e. no sensitization with the instrument.

Experimental group 2: No pretest	X	Posttest (O5)
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Let us work out the results:

$$(O2 - O1) = D$$

$$(O4 - O3) = D/$$

$$(O5 - O3) = D// \text{ (Since all groups are identical, so we can use the pretest of any of the Other two groups)}$$

$$\begin{aligned}\text{Interaction effect} &= D - [D/ + D//]. \text{ Substituting it with our example of lung cancer } \rightarrow \\ &37 - [10 + 20] = 37 - 30 = 7\end{aligned}$$

There are many other experimental designs like the randomized block design, Latin square design, natural group design, and factorial design.

LESSON 35

EXPERIMENTAL RESEARCH (Cont.)

Validity in Experiments

Experiments are judged by two measures. The first, internal validity indicates whether the independent variable was the sole cause of the change in the dependent variable. It implies the researcher's ability to eliminate alternative explanations of the dependent variable. Variables, other than the treatment, that affect the dependent variable are threats to internal validity. They threaten the researcher's ability to say that the treatment was the true causal factor producing change in the dependent variable. The second measure, external validity, indicates the extent to which the results of the experiment are applicable in the real world.

Internal validity is high in the laboratory experiment, reason being the control over all the confounding factors. External validity (generalisability) is not sure because of the effect of variety of factors. Field experiments have more external validity but less internal validity because it is closer to the real situations.

Factors Affecting Internal Validity

In choosing or evaluating experimental research design, researchers must determine whether they have internal and external validity. There are eight major types of extraneous variables that may jeopardize internal validity: History effect, maturation effect, testing effect, instrumentation effect, selection bias effect, selection bias effect, statistical regression, mortality, and mechanical loss.

1. History Effect: A specific event in the external environment occurring between the first and second measurement that is beyond the control of the experimenter and that affects the validity of an experiment. Advertisement of a particular product (mineral water) and its sale is affected by an event in the society (contamination of drinking water). The researcher does not have control on such happenings which have an impact on the X and Y relationship.

2. Maturation Effect: Cause and effect relationship can also be contaminated by the effects of the passage of time – another uncontrollable variable. Such contamination is called maturation effect. The maturation effects are a function of the processes – biological and psychological – operating within the subjects as a result of the passage of time. Examples of maturation processes could include *growing older, getting tired, feeling hungry, and getting bored*. In other words there could be maturation effect on the dependent variable purely because of the passage of time. For example, let us say that an R & D director intends that an increase in the efficiency of workers would result within three months' time if advanced technology is introduced in the work setting. If at the end of three months increased efficiency is indeed found, it will be difficult to claim that the advanced technology (and it alone) increased the efficiency of workers, because with the passage of time, employees would also gain experience, resulting in better performance and therefore improved efficiency. Thus, the internal validity also gets reduced owing to the effects of maturation in as much as it is difficult to pinpoint how much of the increase is attributable to the introduction of the enhanced technology alone.

3. Testing Effects: Frequently, to test the effects of treatment, subjects are given what is called a **pretest** (say, a short questionnaire eliciting their feelings and attitudes). That is, a measure of the dependent variable is taken (pretest), then the treatment given, and after that a second test, called **posttest**, administered. The difference between the posttest and the pretest scores is then attributed to the treatment. However, the very fact that the subjects were exposed to the pretest might influence their responses on the posttest, which will adversely impact on internal validity. It is also called sensitization through previous testing.

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4. Instrumentation Effects: Instrumentation effects are yet another source of threat to internal validity. These might arise because of a change in the measuring instrument between pretest and posttest, and not because of the instrument's differential impact at the end. For example, in a weight-loss experiment, the springs on the scale weaken during the experiment, giving lower readings in the posttest.

A change in the wording of questions (may be done to avoid testing effects), change in interviewers, or change in other procedures to measure the dependent variable can cause instrumentation effect.

Performance of the subjects measured by the units of output in the pretest, but when measuring the output in posttest the researcher measures it by “the number of units rejected, and the amount of resources expended to produce the units.

5. Selection Bias Effect: Selection bias is the threat that subjects will not form equivalent groups. It is a problem in design without random assignment, hence differential selection of the subjects for the comparison groups. It occurs when subjects in one experimental group have a characteristic that affects the dependent variable. For example, in an experiment on physical aggressiveness, the experimental group unintentionally contains subjects who are sportsmen, whereas the control group is made up of musicians, chess players, and painters.

6. Statistical Regression: Statistical regression is not easy to grasp intuitively. It is a problem of extreme values or a tendency for random error to move group results towards the average. If extremes are taken then they tend to regress toward the mean. Those who are on either end of the extreme would not truly reflect the cause and effect relationship.

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One situation arises when subjects are unusual with regard to dependent variable. Because they begin as unusual or extreme, subjects are likely to respond further in the same direction. For example, a researcher wants to see whether violent films make people act violently. The researcher chooses a group of violent criminals from a high security prison, gives them a pretest, shows violent films, and then administers a posttest. To the researcher's surprise, the criminals are slightly less violent after the film, whereas a control group of non-prisoners who did see the film are slightly more violent than before. Because the violent criminals began at an extreme, it is unlikely that a treatment could make them more violent; by random chance alone, they appear less extreme when measured a second time.

If participants chosen for experimental group have extreme scores on the dependent variable to begin with then the laws of probability say that those with very low scores on a variable have a greater probability to improve and scoring closer to mean on the posttest after treatment. This phenomenon of low scorers tending to score closer to the mean is known as “regressing toward the mean.”

Likewise, those with high scores have a greater tendency to regress toward the mean – will score lower on the posttest than on pretest. Thus the extremes will not “truly” reflect the causal relationship – a threat to internal validity.

7. Mortality: Mortality, or attrition, arises when some subjects do not continue throughout the experiment. Although the word mortality means death, it does not necessarily mean that subjects have died. If a subset of subjects leaves partway through an experiment, a researcher cannot whether the results would have been different had the subjects stayed. Even with departure of few subjects, the groups do not remain balanced.

Consider for example of a training experiment that investigates the effects of close supervision of salespersons (high pressure) versus low supervision (low supervision). The high pressure condition may misleadingly appear to be superior if those subjects who completed the experiment did very well. If, however, the high-pressure condition caused more subjects to drop-out than the other condition, this apparent superiority may be due to a self-selection bias (those who could not bear the pressure had left –

mortality) – perhaps only very determined and/or talented salespersons made it through the end of the experiment.

8. Mechanical Loss: A problem may be experienced due to equipment failure. For example, in an experiment if the subjects are told that their behavior is being video taped, and during the experiment the video equipment failed to work for some subjects, then the validity of the results could become doubtful.

9. Experimenter Expectancy: In addition to the usually listed eight factors affecting the internal validity some times **experimenter expectancy** may threaten the causal logic of the relationship between the variables. A researcher may threaten internal validity, not purposefully unethical behavior but by indirectly communicating experimenter expectancy to the subjects. Researchers may highly committed to the hypothesis and indirectly communicate desired findings to subjects. For example, a researcher studying reactions towards disabled deeply believes that females are more sensitive toward the disabled than the males are. Through eye contact, tone of voice, pauses, and other nonverbal communication, the researcher unconsciously encourages female subjects to report positive feelings toward the disabled; the researcher's nonverbal behavior is the opposite for male subjects.

The **double-blind experiment** is designed to control experimenter expectancy. In it, people who have direct contact with subjects do not know the details of the hypothesis or the treatment. It is **double** blind because both the subjects and those in contact with them are blind to details of the experiment. For example a researcher wants to see if new drug is effective. Using capsules of three colors – green, yellow, and pink -- the researcher puts the new drug in the yellow capsule, puts an old drug in the pink one, and take the green capsule a **placebo** – a false treatment that appears to be real (e.g., a sugar capsule without any physical effects). Assistants who give the capsules and record the effects do not know which color contains the new drug. Only another person who does not deal with subjects directly knows which colored capsule contains the drug and examines the results.

External Validity

Even if the researcher eliminates all concerns for internal validity, external validity remains a potential problem. External validity is the ability to generalize experimental findings to real life situations. Without external validity, the findings are of little use for both basic and applied research i.e. we shall not be able to develop any theories that could be applicable to similar other situations.

Reactivity: A Threat to External Validity

Subjects may react differently in an experiment than they would in real life; because they know they are in a study. *The Hawthorn Effect*, a specific kind of reactivity to the experimental situation is a good example in this respect. The experiment was conducted in the Hawthorn Electric Company where the performance of the participants was supposed to change due to the change in the environmental conditions i.e. improvement on the environmental conditions will have a positive effect on thee performance. The researchers modified many aspects of the working conditions and measured productivity. Productivity rose after each modification. Productivity rose even if there was no real modification but it was announced that there is a modification. The behavior change was simply a reaction to the announcement of modification and some other factors like the participants were being watched and had a feeling of being 'very important persons.'

Here the workers did not respond to treatment (modification of working conditions) but to the additional attention they received (being in the experiment ad being the focus of attention).

Demand characteristic (discussed earlier) is another type of reactivity. Here the participants change their behavior as a reaction to the demands of the experimenter who may have inadvertently told the subjects about the expected outcome of the treatment. They change their behavior as demanded by the experimenter.

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Ethical Issues in Lab Experiments

We have already discussed the ethical issues in research. Just for the sake of emphasis, it may be appropriate to very briefly repeat some of those which are specifically relevant to experimental designs. The following actions may be unethical:

- Putting pressure on individuals to participate in experiments through coercion, or apply social pressure.
- Asking demeaning questions from the subjects that hurt their self respect or giving menial task to subjects that diminish their self respect.
- Deceiving subjects by deliberately misleading them as to the true purpose of research.
- Exposing participants to physical or mental stress.
- Not allowing subjects to withdraw from the experiment when they want to.
- Using research results to disadvantage the participants, or for purposes not to their liking.
- Not explaining the procedures to be followed in the experiment.
- Exposing subjects to hazardous and unsafe environments.
- Not debriefing the participants fully and accurately after the experiment is over.
- Not preserving the privacy and confidentiality of the information given by the participants.
- Withholding benefits from the control group.

Human Subjects Committee

In order to protect the rights of participating subjects the research institutions have usually set up Ethics Committees. Sometime project specific ethics committees are also formed. Such committees try to look after the rights of the subjects participating in the experiments, as well as in other research techniques.

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LESSON 36

NON-REACTIVE RESEARCH

Experiments and surveys research are both **reactive**; that is, the people being studied are aware of the fact that they are being studied. In **non-reactive** research, those being studied are not aware that they are part of a research project. Such a research is largely based on positivistic principles but is also used by interpretive and critical researchers.

The Logic of Non-Reactive Research

The critical thing about non-reactive or *unobtrusive measures* (i.e. the measures that are not obtrusive or intrusive) is that the people being studied are not aware of it but leave evidence of their social behavior or actions ‘naturally.’ The researcher infers from the evidence to behavior or attitudes without disrupting those being studied. Unnoticed observation is also a type of non-reactive measure. For example, a researcher may be observing the behavior of drivers from a distance whether drivers stopped at red sign of the traffic lights. The observations can be made both at the day time and at night. It could also be noted whether the driver was a male or a female; whether the driver was also or with passengers; whether other traffic was present; and whether the car came to a complete stop, a slow stop, or no stop.

Varieties of Non-Reactive Observations

Non-reactive measures are varied, and researchers have been creative in inventing indirect ways to measure behaviors. Because the measures have little in common except being non-reactive, they are best learned through examples like:

Physical Traces:

- **Erosion:** Wear and tear suggests a greater use. For example, a researcher examines children’s toys at a children’s play centre that were purchased at the same time. Worn out toys suggest greater interest of children in them.
- **Accretion:** Accumulation of physical evidence suggests behavior. A researcher examines the soft drink cans or bottles in the garbage collection. That might indicate the brands and types of soft drinks that are very popular.

Archives:

- **Running Records:** Regularly produced public records may reveal lot of information. For example, a researcher may examine marriage records for brides’ and grooms’ recorded ages. The differences might indicate that males marrying younger females are greater than the other way around.
- **Other Records:** Irregular or private records can reveal a lot. For example, a researcher may look into the number of reams of paper purchased by a college principal’s office for the last 10 years and compare it with students’ enrollment.

Observations:

- **External Appearance:** How people appear may indicate social factors. For example, a researcher watches students to see whether they are more likely to wear their college’s colors and symbols after the college team won or lost.
- **Count Behaviors:** Counting how many people do something can be informative. For example a researcher may count the number of men and women who come to a full stop and those who come to a rolling stop at a traffic stop sign. This suggests gender difference in driving behavior.
- **Time Duration:** How long people take to do things may indicate their intention. For example a researcher may measure how long men and women pause in front of a particular painting. Time taken may indicate their interest in the painting.

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Recording and Documentation

Creating non-reactive measures follows the logic of quantitative measurement, although qualitative researchers also use non-reactive observations. **A researcher first conceptualizes a construct, and then links the construct to non-reactive empirical evidence, which is its measure.** The operational definition of the variable includes how the researcher systematically notes and records observations.

Content Analysis

Content analysis is a technique for gathering and analyzing the content of a text. The *content* refers to words, meanings, pictures, symbols, ideas, themes, or any message that can be communicated. The *text* is anything written, visual, or spoken that serves as a medium of communication. **Possible artifacts for study** could be books, newspaper or magazine articles, advertisements, poems, letters, laws, constitutions, dramas, speeches, official documents, films or videotapes, musical lyrics, photographs, articles of clothing, or works of arts. All these works may be called as documents. The documents can be:

- Personal – letters, diary, autobiography.
- Non-personal – interoffice memos, official documents, proceedings of a meeting.
- Mass media – newspapers, magazines, fiction, films, songs, poems, works of arts.

Content analysis goes back nearly a century and is used in many fields – literature, history, journalism, political science, education, psychology, sociology, and so on. It is also called a study of communication, which means who says what, to whom, why, how, and with what effect.

In content analysis, the researcher uses objective and systematic counting and recording procedures to produce a quantitative description of the symbolic content in a text. It may also be called “**textual coding.**” There are qualitative versions of content analysis. The emphasis here is quantitative data about a text’s content.

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Content Analysis is Non-Reactive: It is non-reactive because the placing of words, messages, or symbols in a text to communicate to the reader or receiver occurs without influence from the researcher who analyzes its contents. There is no interaction between the researcher and the creator of the text under analysis.

Content analysis lets a researcher reveal the contents (i.e. messages, meanings, symbols, etc.) in a source of communication (i.e. a book, article, movie, etc.). It lets him/her probe into and discover content in a different way from ordinary way of reading a book or watching a television program.

With content analysis, a researcher can compare content across many texts and analyze it with quantitative techniques (table, charts). In addition, he or she can reveal aspects of the text’s content that are difficult to see. For example, you might watch television commercials and feel that women are mostly portrayed working in the house, cooking food, using detergents, looking after children. Content analysis can document – in objective, quantitative terms – whether or not your vague feelings based on unsystematic observation are true. It yields repeatable, precise results about the text.

Content analysis involves random sampling, precise measurement, and operational definitions for abstract constructs. Coding turns aspects of content that represent variables into numbers. After a content analysis researcher gathers the data, he or she enters them into computers and analyzes them with statistics in the same way that an experiment or survey researcher would.

Measurement and Coding

Careful measurement is crucial in content analysis because a researcher takes different and murky symbolic communication and turns it into precise, objective, quantitative data. He or she carefully

designs and documents the procedures for coding to make replication possible. For example, a researcher wants to determine how frequently television dramas portray elderly characters in terms of negative stereotypes. He or she develops a measure of the construct “negatively stereotypes of the elderly.” The conceptualization may result in a list of stereotypes or negative generalizations about older people (e.g., senile, forgetful, frail, hard of hearing, slow, ill, inactive, conservative, etc.) that do not accurately reflect the elderly. Another example could be negative stereotypes about women.

Constructs in content analysis are operationalizing with a *coding system*, a set of instructions or rules on how to systematically observe and record content from text. Look at the construct of “leadership role;” for measuring this construct written rules should be provided telling how to classify people. Same is about the concept of “social class.” In case the researcher has three categories of upper, middle, and lower class then the researcher must tell what are the characteristics that are associated with upper class, middle class, and the lower class so that the coders could easily classify people in the three proposed categories.

Observations can be structured: Measurement in content analysis uses *structured observation* i.e. systematic, careful observation based on written rules. The rules explain how to categorize and classify observations in terms of:

- **Frequency:** Frequency simply means counting whether or not something occurs and how often (how many times). For example how many elderly people appear on a television program within a given week? What percentage of all characteristics are they, or in what percentage of programs do they appear.
- **Direction:** Direction is noting the direction of messages in the content along some continuum (e.g., positive or negative, supporting or opposed). For example the researcher devises a list of ways an elderly television character can act. Some are positive (e.g., friendly, wise, considerate) and some are negative (e.g., nasty, dull, selfish).
- **Intensity:** Intensity is the strength or power of a message in a direction. For example, the characteristic of forgetfulness can be minor (e.g. not remembering to take the keys when leaving home, taking time to recall the name of someone whom you have not seen in years) or major (e.g., not remembering your name, not recognizing your children).
- **Space:** A researcher can record the size of the text message or the amount of space or volume allocated to it. Space in written text is measured by counting words, sentences, paragraphs, or space on a page (e.g. square inches) for video or audio text, space can be measured by the amount of time allocated. For example, a TV character may be present for a few seconds or continuously in every scene of a two hour program.

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The unit analysis can vary a great deal in content analysis. It can be a word, a phrase, a theme, a plot, a news paper article, a character, and so forth.

Coding

The process of identifying and classifying each item and giving labels to each category. Later on each category may be assigned a numerical value for its entry into the computer. In content analysis one can look at the manifest coding and latent coding.

Manifest Coding: Coding the visible, surface content in a text is called manifest coding. For example, a researcher counts the number of times a phrase or word (e.g. red) appears in the written text, or whether a specific action (e.g. shaking hands) appears in a photograph or video scene. The coding system lists terms or actions or characters that are then located in text. A researcher can use a computer program to search for words or phrases in the text and have a computer do the counting work.

Manifest coding is highly reliable because the phrase or the word either is or is not present. However, manifest coding does not take the connotation of word into account. The same word can take on

different meanings depending on the context. The possibility that there are multiple meanings of a word limits the measurement validity of manifest coding.

Latent Coding: A researcher using *latent coding* (also called *semantic analysis*) looks for the underlying meaning in the content of a text. For example, the researcher reads the entire paragraph and decides whether it contains vulgar themes or a romantic mood. His or her coding system has general rules to guide his or her interpretation of the text and for determining whether particular themes or mood are present.

Latent coding tends to be less reliable than the manifest coding. It depends on a coder's knowledge of language and its social meaning. Training, practice, and written rules improve reliability, but still it is difficult to consistently identify themes, moods, and the like.

Keeping in view the amount of work, often a number of coders are hired. The researcher trains the coders in coding system. Coders should understand the variables, follow the coding system, and ask about ambiguities. A researcher who uses several coders must always check for consistency across coders. He or she does this by asking coders to code the same text independently and then checking for consistency across coders. The researcher measures inter-coder reliability, a type of equivalence reliability, with a statistical coefficient that tells the degree of consistency across among coders. The coefficient is always reported with the results of content analysis research.

How to Conduct Content Analysis Research

Question Formulation: As in most research, content analysis researchers begin with a research question. When the question involves variables that are messages or symbols, content analysis may be appropriate. For example, how women are portrayed in advertisements? The construct here is the portrayal of women which may be measured by looking at the activities they are shown to be doing, the occupations in which they are employed, the way decision making is taking place, etc.

Unit of Analysis: A researcher decides on the unit of analysis (i.e. the amount of text that is assigned a code). In the previous example each advertisement may be a unit of analysis.

Sampling: Researchers often use random sampling in content analysis. First, they define the population and the sampling element. For example, the population might be all words, all sentences, all paragraphs, or all articles in certain type of documents over a period of specified period. Likewise, it could include each conversation, situation, scene, or episode of a certain type of television program over a specified time period. Let us consider that we want to know how women are portrayed in weekly news magazines. The unit of analysis is the article. The population includes all articles published in weekly news magazines during 2001 to 2007. Make a list of English magazines that were published during the said period. Define what is a news magazine? Define what is an article? Decide on the number of magazines. Decide on the sample size. Make a sampling frame. Here the sampling frame shall be all the articles published in the selected magazines during 2001 to 2007. Finally draw the random sample using table of random numbers.

Variables and Constructing Coding Categories: Say a researcher is interested in women's portrayal in significant leadership roles. Define "significant leadership role" in operational terms and put it as written rules for classifying people named in the articles. Say the researcher is further interested in positive leadership roles, so the measure will indicate whether the role was positive or negative. Researcher has to make a list of adjectives and phrases reflective of the leadership role being positive or negative. If someone in the article is referred to with one of the adjective, then the direction is decided. For example, the terms *brilliant* and *top performer* are positive, whereas *drug kingpin* and *uninspired* are negative. Researcher should give written rules to classify role of women as portrayed in the articles.

In addition to written rules for coding decisions, a content analysis researcher creates a recording sheet (also called a *coding form* or *tally sheet*) on which to record the information. Each unit should have a separate recording sheet.

Inferences: The inference a researcher can or cannot make on the basis of results is critical in content analysis. Content analysis describes what is in the text. It cannot reveal the intentions of those who created the text or the effects that messages in the text have on those who receive them.

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LESSON 37

USE OF SECONDARY DATA

Existing statistics/documents

Prior to the discussion of secondary data, let us look at the advantages and disadvantages of the use of content analysis that was covered in the last lecture. In a way content analysis is also the study of documents through which the writers try to communicate, though some of the documents (like population census) may simply contain figures.

Advantages

1. Access to inaccessible subjects: One of the basic advantages of content analysis is that it allows research on subjects to which the researcher does not have physical access. These could be people of old civilizations, say their marriage patterns. These could also be the documents from the archives, speeches of the past leaders (Quaid-e-Azam) who are not alive, the suicide notes, old films, dramas, poems, etc.

2. Non-reactivity: Document study shares with certain types of observations (e.g., indirect observation or non participant observation through one-way mirror) the advantage of little or no reactivity, particularly when the document was written for some other purpose. This is unobtrusive. Even the creator of that document, and for that matter the characters in the document, is not in contact with the researcher, who may not be alive.

3. Can do longitudinal analysis: Like observation and unlike experiments and survey, document study is especially well suited to study over a long period of time. Many times the objective of the research could be to determine a trend. One could pick up different periods in past and try to make comparisons and figure out the changes (in the status of women) that may have occurred over time. Take two martial periods in Pakistan, study the news papers and look at the reported crime in the press.

4. Use Sampling: The researcher can use random sampling. One could decide on the population, develop sampling frame and draw sample random sample by following the appropriate procedure. For example how women are portrayed in weekly English news magazines. One could pick up weekly English news magazines, make a listing of articles that have appeared in the magazines (sampling frame), and draw a simple random sample.

5. Can use large sample size: Larger the sample closer the results to the population. In experimentation as well as in survey research there could be limitations due to the availability of the subjects or of the resources but in document analysis the researcher could increase the sample and can have more confidence in generalization. Let us assume that a researcher is studying the matrimonial advertisements in the newspapers over a long period of time, there should be no problem in drawing a sample as large as several thousand or more.

6. Spontaneity: The spontaneous actions or feelings can be recorded when they occurred rather than at a time specified by the researcher. If the respondent was keeping a diary, he or she may have been recording spontaneous feelings about a subject whenever he or she was inspired to do so. The contents of such personal recording could be analyzed later on.

7. Confessions: A person may be more likely to confess in a document, particularly one to be read only after his or her death, than in an interview or mailed questionnaire study. Thus a study of documents such as diaries, posthumously published autobiographies, and suicide notes may be the only way to obtain such information.

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8. Relatively low cost: Although the cost of documentary analysis can vary widely depending on the type of document analyzed, how widely documents are dispersed, and how far one must travel to gain access to them, documentary analysis can be inexpensive compared to large-scale surveys. Many a time's documents are gathered together in a centralized location such as library where the researcher can study them for only the cost of travel to the repository.

9. High quality: Although documents vary tremendously in quality, many documents, such as news paper columns, are written by skilled commentators and may be more valuable than, for example, poorly written responses to mailed questionnaires.

Disadvantages

1. Bias: Many documents used in research were not originally intended for research purposes. The various goals and purposes for which documents are written can bias them in various ways. For example, personal documents such as confessional articles or autobiographies are often written by famous people or people who had some unusual experience such as having been a witness to a specific event. While often providing a unique and valuable research data, these documents usually are written for the purpose of making money. Thus they tend to exaggerate and even fabricate to make good story. They also tend to include those events that make the author look good and exclude those that cast him or her in a negative light.

2. Selective survival: Since documents are usually written on paper, they do not withstand the elements well unless care is taken to preserve them. Thus while documents written by famous people are likely to be preserved, day-to-day documents such as letters and diaries written by common people tend either to be destroyed or to be placed in storage and thus become inaccessible. It is relatively rare for common documents that are not about some events of immediate interest to the researcher (e.g., suicide) and not about famous occurrence or by some famous person to be gathered together in a public repository that is accessible to researchers.

3. Incompleteness: Many documents provide incomplete account to the researcher who has had no prior experience with or knowledge of the events or behavior discussed. A problem with many personal documents such as letters and diaries is that they were not written for research purposes but were designed to be private or even secret. Both these kinds of documents often assume specific knowledge that researcher unfamiliar with certain events will not possess. Diaries are probably the worst in this respect, since they are usually written to be read only by the author and can consist more of "soul searching" and confession than of description. Letters tend to be little more complete, since they are addressed to a second person. Since many letters assume a great amount of prior information on the part of the reader.

4. Lack of availability of documents: In addition to the bias, incompleteness, and selective survival of documents, there are many areas of study for which no documents are available. In many cases information simply was never recorded. In other cases it was recorded, but the documents remain secret or classified, or have been destroyed.

5. Sampling bias: One of the problems of bias occurs because persons of lower educational or income levels are less likely to be represented in the sampling frames. The problem of sampling bias by educational level is more acute for document study than for survey research. It is a safe generalization that a poorly educated people are much less likely than well educated people to write documents.

6. Limited to verbal behavior: By definition, documents provide information only about respondent's verbal behavior, and provide no direct information on the respondent's nonverbal behavior, either that of the document's author or other characters in the document.

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7. Lack of standardized format: Documents differ quite widely in regard to their standardization of format. Some documents such as newspapers appear frequently in a standard format. Large dailies always contain such standard components as editorial page, business page, sports page, and weather report. Standardization facilitates comparison across time for the same newspapers and comparison across different newspapers at one point in time. However, many other documents, particularly personal documents have no standard format. Comparison is difficult or impossible, since valuable information contained in the document at one point in time may be entirely lacking in an earlier or later documents.

8. Coding difficulties: For a number of reasons, including differences in purpose for which the documents were written, differences in content or subject matter, lack of standardization, and differences in length and format, coding is one of the most difficult tasks facing the content analyst. Documents are generally written arrangements, rather than numbers are quite difficult to quantify. Thus analysis of documents is similar to analysis of open-ended survey questions.

9. Data must be adjusted for comparability over time: Although one of the advantages of document study is that comparisons may be made over a long period of time, since external events cause changes so drastic that even if a common unit of measure is used for the entire period, the value of this unit may have changed so much over time that comparisons are misleading unless corrections are made. Look at the change in measuring distance, temperature, currency, and even literacy in Pakistan.

Use of Secondary Data: Existing Statistics/Documents

Secondary Data

Secondary data refer to information gathered by someone other than the researcher conducting the present study. Secondary data are usually historical, already assembled, and do not require access to respondents or subjects. Many types of information about the social and behavioral world have been collected and are available to the researcher. Some information is in the form of statistical documents (books, reports) that contain numerical information. Other information is in the form of published compilations available in a library or on computerized records. In either case the researcher can search through collections of information with a research question and variables in mind, and then reassemble the information in new ways to address the research question.

Secondary data may be collected by large bureaucratic organization like the Bureau of Statistics or other government or private agencies. These data may have been collected for policy decisions or as part of public service.

The data may be a time bound collection of information (population census) as well as spread over long periods of time (unemployment trends, crime rate). Secondary data are used for making comparisons over time in the country (population trends in the country) as well as across the countries (world population trends).

Selecting Topic for Secondary Analysis

Search through the collections of information with research question and variables in mind, and then reassemble the information in new ways to address the research question.

It is difficult to specify topics that are appropriate for existing statistics research because they are so varied. Any topic on which information has been collected and is publicly available can be studied. In fact, existing statistics projects may not neatly fit into a deductive model of research design. Rather researchers creatively recognize the existing information into the variables for a research question after first finding what data are available.

Experiments are best for topics where the researcher controls a situation and manipulates an independent variable. Survey research is best for topics where the researcher asks questions and learns about reported attitudes and behavior. Content analysis is for topics that involve the content of messages in cultural communication.

Existing statistics research is best for topics that involve information collected by large bureaucratic organizations. Public or private organizations systematically gather many types of information. Such information is collected for policy decisions or as a public service. It is rarely collected for purposes directly related to a specific research question. Thus existing statistics research is appropriate when a researcher wants to test hypotheses involving variables that are also in official reports of social, economic and political conditions. These include descriptions of organizations or people in them. Often, such information is collected over long periods. For example, existing statistics can be used by researcher who wants to see whether unemployment and crime rates are associated in 100 cities across a 20 year period.

As part of the trends, say in development, researchers try to develop social indicators for measuring the well being of the people. A *social indicator* is any measure of wellbeing used in policy. There are many specific indicators that are operationalization of well-being. It is hoped that information about social well being could be combined with widely used indicators of economic performance (e.g., gross national product) to better inform government and other policy making officials.

The main sources of existing statistics are government or international agencies and private sources. An enormous volume and variety of information exists. If you plan to conduct existing statistics research, it is wise to discuss your interests with an information professional – in this case, a reference librarian, who can point you in the direction of possible sources.

Many existing documents are “free” – that is, publicly available at libraries – but the time and effort it takes to research for specific information can be substantial. Researchers who conduct existing statistics research spend many hours in libraries or on the internet.

There are so many sources of existing statistics like: UN publications, UNESCO Statistical Yearbook, UN Statistical Yearbook, Demographic Yearbook, Labor Force Survey of Pakistan, and Population Census Data.

Secondary Survey Data

Secondary analysis is a special case of existing statistics; it is reanalysis of previously collected survey or other data that was originally gathered by others. As opposed to primary research (e.g., experiments, surveys, and content analysis), the focus is on analyzing rather than collecting data.

Secondary analysis is increasingly used by researchers. It is relatively inexpensive; it permits comparisons across groups, nations, or time; it facilitates replication; and permits asking about issues not thought by the original researchers. There are several questions the researcher interested in secondary research should ask: Are the secondary data appropriate for the research question? What theory and hypothesis can a researcher use with the data? Is the researcher already familiar with the substantive area? Does the researcher understand how the data were originally gathered and coded?

Large-scale data collection is expensive and difficult. The cost and time required for major national surveys that uses rigorous techniques are prohibitive for most researchers. Fortunately, the organization, preservation, and dissemination of major survey data sets have improved. Today, there are archives of past surveys open to researchers (e.g., data on Population Census of Pakistan, Demographic Survey of Pakistan).

Reliability and Validity

Existing statistics and secondary data are not trouble free just because a government agency or other source gathered the original data. Researchers must be concerned with validity and reliability, as well as with some problems unique to this research technique.

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A common error is the *fallacy of misplaced concreteness*. It occurs when someone gives a false impression of accuracy by quoting statistics in greater detail than warranted by how the statistics are collected and by overloading detail. For example, in order to impress an audience, a politician might say that every year 3010,534 persons, instead of saying 3 million persons, are annually being added to the population of Pakistan.

Validity: Validity problems occur when the researcher's theoretical definition does not match that of the government agency or organization that collected the information. Official policies and procedures specify definitions for official statistics. For example, a researcher defines a *work injury* as including minor cuts, bruises, and sprains that occur on the job, but the official definition in government reports only includes injuries that require a visit to a physician or hospital. Many work injuries as defined by the researcher will not be in the official statistics. Another example occurs when a researcher defines people *unemployed* if they would work if a good job was available, if they have to work part-time when they want full-time work, and if they have given up looking for work. The official definition, however, includes only those who are now actively seeking work (full or part-time) as unemployed. The official statistics exclude those who stopped looking, who work part-time out of necessity, or who do not look because they believe no work is available. In both the cases the researcher's definition differs from that in official statistics.

Another validity problem arises when official statistics are a proxy for a construct in which the researcher is really interested. This is necessary because the researcher cannot collect original data. For example, the researcher wants to know how many people have been robbed, so he or she uses police statistics on robbery arrests as a proxy. But the measure is not entirely valid because many robberies are not reported to the police, and reported robberies do not always result in an arrest.

Another validity problem arises because the researcher lacks control over how information is collected. All information, even that in official government reports, is originally gathered by people in bureaucracies as part of their job. A researcher depends on them for collecting, organizing, reporting, and publishing data accurately. Systematic errors in collecting the initial information (e.g., census people who avoid poor neighborhoods and make-up information, or people who put a false age on their ID card); errors in organizing and reporting information (e.g., police department that is sloppy about filing crime reports and loses some); errors in publishing information (e.g., a typographical error in a table) all reduce measurement validity.

Reliability: Stability reliability problems develop when official definition or the method of collecting information changes over time. Official definitions of work injury, disability, unemployment, literacy, poverty, and the like change periodically. Even if the researcher learns of such changes, consistent measurement over time is impossible.

Equivalence reliability can also be a problem. For example, studies of police department suggest that political pressures to increase arrests are closely related to the number of arrests. It could be seen when political pressures in one city may increase arrests (e.g., a crackdown on crime), whereas pressures in another city may decrease arrests (e.g., to show drop in crime shortly before an election in order to make officials look better).

Researchers often use official statistics for international comparisons but national governments collect data differently and the quality of data collection varies.

Inferences from Non-Reactive Data:

A researcher's ability to infer causality or to test a theory on the basis of non-reactive data is limited. It is difficult to use unobtrusive measures to establish temporal order and eliminate alternative explanations. In content analysis, a researcher cannot generalize from the content to its effects on those who read the text, but can only use the correlation logic of survey research to show an association among variables. Unlike the case of survey research, a researcher does not ask respondents direct questions to measure variables, but relies on the information available in the text.

LESSON 38

OBSERVATION STUDIES/FIELD RESEARCH

Observation studies are primarily part of **qualitative research**. Though qualitative and quantitative researches differ yet they complement each other. Qualitative research produces *soft data*: impressions, words, sentences, photos, symbols. Usually it follows an **interpretive approach**, the goal of which is to develop an understanding of social life and discover how people construct meanings in natural settings. The research process follows a **non-linear approach** (spiral).

Quantitative research produces *hard data*: numbers. It follows a **positivist approach** to research in which the researcher speaks the language of variables and hypotheses. There is a much emphasis on precise measurement of variables and the testing of hypotheses. The researcher tries to establish causality. In most of the case there is a linear approach i.e. it follows sequential steps in doing research.

Participant/Non-Participant Observation

Observation studies can be participant or non-participant. In participant observation the researcher directly observes and participates in small scale social settings in the present time. Such a study is also referred to as **field research**, ethnography, or anthropological study. Here the researchers:

- Study people in their natural settings, or in situ.
- Study people by directly interacting with them.
- Gain an understanding of the social world and make theoretical statements about members' perspective.

The people could be a group who interact with each other on regular basis in a field setting: a street corner, a tea shop, a club, a nomad group, a village, etc.

Non-participant studies are such where the research tries to observe the behavior of people without interacting with them. It could be observing the behavior of shoppers in a departmental store through a mirror or on a closed circuit TV. Some body might be counting the number of vehicles crossing a particular traffic light.

Field researchers study people in a location or setting. It has been used to study entire communities. Field research has a distinct set of methodologies. Field researchers directly observe and interact with community members in natural settings to get inside their perspectives. They embrace an activist or social constructionist perspective on social life. They do not see people as a neutral medium through which social forces operate, nor do they see social meanings as something “out there” to observe. Instead they believe that people create and define the social world through their interactions. Human experiences are filtered through a subjective sense of reality, which affects how people see and act on events. Thus they replace the positivist emphasis on “objective facts” with a focus on the everyday, face-to-face social processes of negotiation, discussion, and bargaining to construct social meaning.

Ethnography and Ethno-methodology

Two modern extensions of field research, ethnography and ethno-methodology, build on the social constructionist perspective.

Ethnography comes from cultural anthropology. *Ethno* means people or a folk distinct by their culture and *graphy* refers to describing something. Thus ethnography means describing a culture and understanding another way of life from the native point of view. It is just an understanding the culture of people from their own perspective.

Ethno-methodology implies how people create reality and how they interpret it. Ethno-methodologists examine ordinary social interaction in great detail to identify the rules for constructing social reality and

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common sense, how these rules are applied, and how new rules are created. They try to figure out how certain meanings are attached to a reality.

Logic of Field Research

It is difficult to pin down a specific definition of *field research* because it is more of an orientation toward research than a fixed set of techniques to apply. A field researcher uses various methods to obtain information. A field researcher is a 'methodological pragmatist,' a resourceful, talented individual who has ingenuity and an ability to think on his or feet while in the field.

Field research is based on naturalism, which involves observing ordinary events in natural settings, not in contrived, invented, or researcher created settings.

A field researcher examines social meanings and grasps multiple perspectives in natural setting. He or she gets inside the meanings system of members and goes back to an outside or research viewpoint. Fieldwork means involvement and detachment, loyalty and betrayal, both openness and secrecy, and most likely, love and hate. The researcher switches perspectives and sees the setting from multiple pints of view simultaneously. Researchers maintains membership in the culture in which they were reared (research culture) while establishing membership in the groups which they are studying.

The researcher's direct involvement in the field often has an emotional impact. Field research can be fun and exciting, but it can also disrupt one's personal life, physical security, or mental well being. More than other types of research, it reshapes friendship, family life, self identity, or personal values.

What Do the Field Researchers Do?

A field researcher does the following:

1. Observes ordinary events and everyday activities as they happen in natural settings, in addition to unusual occurrences.
2. Becomes directly involved with people being studied and personally experiences the process of daily life in the field setting.
3. Acquires an insider's point of view while maintaining the analytic perspective or distance of an outsider.
4. Uses a variety of techniques and social skills in a flexible manner as the situation demands.
5. Produces data in the form of extensive, written notes, as well as diagrams, maps, pictures to provide very detailed descriptions.
6. Sees events holistically (as a whole, not in pieces) and individually in their social context.
7. Understands and develops empathy for members in a field setting, and does not just record 'cold' objective facts.
8. Notices both explicit (recognized, conscious, spoken) and tacit (less recognized, implicit, unspoken) aspects of culture.
9. Observes ongoing social processes without upsetting, or imposing an outside point of view.
10. Copes with high levels of personal stress, uncertainty, ethical dilemmas, and ambiguity.

Steps in Field Research

Naturalism and direct involvement mean that field research is more flexible or less structured than quantitative research. This makes it essential for a researcher to be well organized and prepared for the field. It also means that the steps of project are not entirely predetermined but serve as an approximate guide or road map. Here is just the listing of these steps:

1. Prepare yourself, read the literature and defocus.
2. Select a site and gain access.

3. Enter the field and establish social relations with members.
4. Adopt a social role, learn the ropes, and get along with members.
5. Watch, listen, and collect quality data. .
6. Begin to analyze data, generate and evaluate working hypothesis.
7. Focus on specific aspects of the setting and use theoretical sampling.
8. Conduct field interviews with member informants.
9. Disengage and physically leave the setting.
10. Complete the analysis and write the report.

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LESSON 39

OBSERVATION STUDIES (Contd.)

Steps in Field Research

Background

Naturalism and direct involvement mean that field research is more flexible or less structured than quantitative research. This makes it essential for a researcher to be well organized and prepared for the field. It also means that the steps of project are not entirely predetermined but serve as an approximate guide or road map. These guideline steps are:

1. **Prepare yourself, read the literature and defocus.** As with all social and behavioral research, reading the scholarly literature helps the researcher learn concepts, potential pitfalls, data collection methods, and techniques for resolving conflicts. In addition field researcher finds diaries, novels, journalistic accounts, and autobiographies useful for gaining familiarity and preparing emotionally for the field. Field research begins with a general topic, not specific hypotheses. A researcher does not get locked into any initial misconceptions. He or she needs to be well informed but open to discovering new ideas.
A researcher first empties his or her mind of preconceptions and defocuses. There are two types of defocusing. The first is casting a wide net in order to witness a wide range of situations, people, and setting – getting a feel of the overall setting before deciding what to include or exclude. The second type of defocusing means not focusing exclusively on the role of researcher. It may be important to extend one's experience beyond a strictly professional role. Another preparation for field research is self knowledge. A field researcher needs to know him or herself and reflect on personal experiences. He or she can expect anxiety, self doubt, frustration, and uncertainty in the field. Also all kinds of stereotypes about the community should be emptied.
2. **Select a site and gain access.** Although a field research project does not proceed by fixed steps, some common concerns arise in the early stages. These include selecting a site, gaining access to the site, entering the field, and developing rapport with members in the field.

Field site is the context in which events or activities occur, a socially defined territory with shifting boundaries. A social group may interact across several physical sites. For example, a college football team may interact on the playing field, in the dressing room, at a training camp or at the place where they are staying. The team's field site includes all four locations.

Physical access to a site can be an issue. Sites can be on a continuum, with open and public areas (e.g., public restaurants, airport waiting rooms) at one end and closed and private settings (e.g., private firms, clubs, activities in a person's home) at the other end. A researcher may find that he or she is not welcome or not allowed on the site, or there are legal and political barriers to access.

Look for the gate keepers for getting an entry. A gatekeeper is someone with the formal authority to control access to a site. It can be a thug at the corner, an administrator of a hospital, or the owner of a business. In formal public areas (e.g., sidewalks, public waiting rooms) rarely have gatekeepers; formal organizations have authorities from whom permission must be obtained. Field researchers expect to negotiate with gatekeepers and bargain for access. Entry and access can be visualized as an *access ladder*. A researcher begins at the bottom rung, where access is easy and where he or she is an outsider looking for public information. The next access rung requires increased access. Once close on-site observations begin, he or she becomes a passive observer, not questioning what members of community say. With time in the field, the researcher observes specific activities that are potentially sensitive or seeks clarification of what he or she sees or hears. Reaching this access rung is more difficult. Finally, the researcher may try to shape interaction so that it reveals specific information, or he or she may want to see highly sensitive material. This highest rung of access ladder is rarely

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attained and requires deep trust. Such a situation may be applicable to a site of a public or private organization. In other situations just like entering the village community, the researcher may have to use different kind of access ladder. He or she may have to use local influential and some other contact persons who could introduce the researcher to local leaders and help building the rapport.

3. **Enter the field and establish social relations with members.** Present yourself in the field the way it is acceptable to the people to be studied. Develop relations and establish rapport with individual members. Here the researcher may have to learn the local language. A field researcher builds rapport by getting along with members in the field. He or she forges a friendly relationship, shares the same language, and laughs and cries with members. This is a step toward obtaining an understanding of members and moving beyond understanding to empathy – that is seeing and feeling events from another’s perspective.

4. **Enter the field: Adopt a social role, learn the ropes, and get along with members.** At times, a researcher adopts an existing role. Some existing roles provide access to all areas of the site, the ability to observe and interact with all members, the freedom to move around, and a way to balance the requirements of researcher and member. There could be some limitations for the adoption of specific roles. Such limitations may be because of researcher’s age, race, gender, and attractiveness. At other times, a researcher creates new roles or modifies the existing one. The adoption of field role takes time, and a researcher may adopt several different field roles over time.

The role may also depend upon the level of involvement in the community’s activities. The researcher may be a complete observer, observer as participant, participant as observer, and complete participant.

As a researcher learns the ropes on the field site, he or she learns how to cope with personal stress, how to normalize the social research, and how to act like an “acceptable incompetent.” A researcher is in the field to learn, not to be an expert. Depending on the setting, he or she appears to be friendly but naïve outsider, an acceptable incompetent who is interested in learning about social life of the field. An acceptable incompetent is one who is partially competent (skilled or knowledgeable) in the setting but who is accepted as a non-threatening person

5. **Observing and collecting data: Watch, listen, and collect quality data.** A great deal of what field researchers do in the field is to pay attention, watch, and listen carefully. They use all the senses, noticing what is seen, heard, smelled, tasted, or touched. The researcher becomes an instrument that absorbs all sources of information.

Most field research data are in the form of field notes. Good notes are the brick and mortar of field research. Full field notes can contain maps, diagrams, photographs, interviews, tape recordings, videotapes, memos, objects from the field, notes jotted in the field, and detailed notes written away from the field. A field researcher expects to fill many notebooks, or the equivalent in computer memory. He or she may spend more time writing notes than being in the field.

Writing notes is often boring, tedious work that requires self discipline. The notes contain extensive descriptive detail drawn from memory. The researcher makes it a daily habit or compulsion to write notes immediately after leaving the field. The notes must be neat and organized because the researcher will return to them over and over again. Once written, the notes are private and valuable. A researcher treats them with care and protects confidentiality.

Field researcher is supposed to collect quality data. What does the term high-quality data mean in the field research, and what does a researcher do to get it? For a quantitative researcher, high quality data are reliable and valid; they give precise, consistent measures of the “objective” truth for all researchers. An interpretive approach suggests a different kind of data quality. Instead of assuming one single, objective truth, field researchers hold that members subjectively interpret experiences within social context. What a member takes to be true results from social interaction and interpretation. Thus high quality field data capture such processes and provide an understanding of the member’s viewpoint.

A field researcher does not eliminate subjective views to get quality data: rather, quality data include his or her subjective responses and experiences. Quality field data are detailed descriptions from the researcher's immersion and authentic experiences in the social world of members.

6. ***Begin to analyze data generate and evaluate working hypothesis.*** Right in the field try to look into the research questions and the kind of answers the researcher is getting. The analysis of the answers might help in the generation of hypotheses. Over time are such hypotheses being supported by further field research?

7. ***Focus on specific aspects of the setting and use theoretical sampling.*** Field researcher first gets a general picture, and then focuses on a few specific problems or issues. A researcher decides on specific research questions and develops hypotheses only after being in the field and experiencing it first hand. At first, everything seems relevant; later, however, selective attention focuses on specific questions and themes.

Field research sampling differs from survey sampling, although sometime both use snowball sampling. A field researcher samples by taking a smaller, selective set of observations from all possible observations. It is called theoretical sampling because it is guided by the researcher's developing theory. Field researchers sample times, situations, types of events, locations, types of people, or context of interest.

For example field researcher samples time by observing a setting at different times. He or she observes at all time of the day, on every day of the week, and in all seasons to get a full sense of how the field site stays the same or changes. Another example, when the field researcher samples locations because one location may give depth, but narrow perspective. Sitting or standing in different locations helps the researcher to get a sense of the whole site. Similarly the field researchers sample people by focusing their attention or interaction on different kinds of people (young, adult, old).

8. ***Conduct field interviews with member informants.*** Field researchers use unstructured, non directive, in-depth interviews, which differs from formal survey research interviews in many ways. The field interview involves asking question, listening, expressing interest, and recording what was said.

Field interview is a joint production of a research and a member. Members are active participants whose insights, feelings, and cooperation are essential parts of a discussion process that reveals subjective meaning. The interviewer's presence and form of involvement – how he or she listens, attends, encourages, interrupts, disagrees, initiates topics, and terminates responses – is integral to the respondent's account.

Field research interviews go by many names: unstructured, depth, ethnographic, open ended, informal, and long. Generally, they involve one or more people being present, occur in the field, and are informal and nondirective.

A comparison of the field research interview and a survey interview could be as below:

Survey Interview	Field Interview
1. It has clear beginning and end.	1. The beginning and end are not clear. The interview can be picked up later.
2. The same standard questions are asked of all respondents in the same sequence.	2. The questions and the order in which they are asked are tailored to specific people and situations.
3. The interviewer appears neutral at all times.	3. The interviewer shows interest in responses, encourages elaboration.
4. The interviewer asks questions, and the respondent answers.	4. It is like a friendly conversational exchange but with more interviewer questions.

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| 5. It is almost always with one respondent alone. | 5. It can occur in group setting or with others in area, but varies. |
| 6. It has a professional tone and businesslike focus, diversions are ignored. | 6. It is interspersed with jokes, aside, stories, diversions, and anecdotes, which are recorded. |
| 7. Closed-ended questions are common, with rare probes. | 7. Open-ended questions are common, and probes are frequent. |
| 8. The interviewer alone controls the pace and direction of interview. | 8. The interviewer and member jointly control the pace and direction of the interview. |
| 9. The social context in which the interview occurs is ignored and assumed to make little difference. | 9. The social context of the interview is noted and seen as important for interpreting the meaning of responses. |
| 10. The interviewer attempts to mold the communication pattern into a standard framework. | 10. The interviewer adjusts to the member's norms and language usages. |

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9. ***Disengage and physically leave the setting.*** Work in the field can last for a few weeks to a dozen years. In either case at some point of work in the field ends. Some researchers suggest that the end comes naturally when the theory building ceases or reaches a closure; others feel that fieldwork could go on without end and that a firm decision to cut off relations is needed. Experienced field researchers anticipate a process of disengaging and exiting the field. Depending on the intensity of involvement and the length of time in the field, the process can be disruptive or emotionally painful for both the researcher and the members. Once researcher decides to leave – because the project reaches a natural end and little new is being learned, or because external factors force it to end (e.g., end of job, gatekeepers order the researcher out) – he or she chooses a method of exiting. The researcher can leave by quick exit (simply not return one day) or slowly withdraw, reducing his or her involvement over weeks. He or she also needs to decide how to tell members and how much advance warning to give. The best way to exist is to follow the local norms and continuing with the friendly relations.
10. ***Complete the analysis and write the report.*** After disengaging from the field setting the researcher writes the report. The researcher may share the written report with the members observed to verify the accuracy and get their approval of its portrayal in print. It may help in determining the validity of the findings. However, it may not be possible to share the findings with marginal groups like addicts, and some deviant groups.

Ethical Dilemmas of Field research

The direct personal involvement of a field researcher in the social lives of other people raises many ethical dilemmas. The dilemmas arise when the researcher is alone in the field and has little time to make a moral decision. Although he or she may be aware of general ethical issues before entering the field, they arise unexpectedly in the course of observing and interacting in the field. Let us look at some of these dilemmas:

Deception: Deception arises in several ways in field research: The research may be covert; or may assume a false role, name, or identity; or may mislead members in some way. The most hotly debated of the ethical issues arising from deception is that of covert versus overt field research. Some support it and see it as necessary for entering into and aiming a full knowledge of many areas of social life. Others oppose it and argue that it undermines a trust between researchers and society. Although its

moral status is questionable, there are some field sites or activities that can only be studied covertly. One may have to look into the cost and benefit equation; where the researcher is the best judge.

Covert research is never preferable and never easier than overt research because of the difficulties of maintaining a front and the constant fear of getting caught.

Confidentiality: A researcher learns intimate knowledge that is given in confidence. He or she has a moral obligation to uphold the confidentiality of data. This includes keeping information confidential from others in the field and disguising members' names in field notes.

Involvement with deviants: Researchers who conduct research on deviants who engage in illegal behavior face additional dilemmas. They know of and are sometimes involved in illegal activity. They might be getting 'guilty knowledge.' Such knowledge is of interest not only to law enforcement officials but also to other deviants. The researcher faces a dilemma of building trust and rapport with the deviants, yet not becoming so involved as to violate his or her basic personal moral standards. Usually, the researcher makes an explicit arrangement with the deviant members.

The powerful: Field researchers tend to study those without power in society (e.g., street people, the poor, children, and lower level workers). Powerful elites can block access and have effective gatekeepers. Researchers are criticized for ignoring the powerful, and they are also criticized by the powerful for being biased toward the less powerful.

Publishing field reports: The intimate knowledge that a researcher obtains and reports creates a dilemma between the right of privacy and the right to know. A researcher does not publicize member secrets, violate privacy, or harm reputations. Yet if he or she cannot publish anything that might offend or harm someone, some of what the researcher learned will remain hidden, and it may be difficult for others to believe the report if critical details are omitted.

Some researchers suggest asking members of the group under study to look at a report to verify its accuracy and to approve of their portrayal in print. For marginal groups (addicts), this may not be possible, but the researchers must always respect member privacy. On the other hand, censorship or self-censorship can be a danger. A compromise position is that truthful but unflattering material may be published only if it is essential to the researchers' larger arguments.

LESSON 40**HISTORICAL COMPARATIVE RESEARCH**

History has several meanings; one of which could refer to ‘the events of the past.’ *Historiography* is the method of doing historical research or of gathering and analyzing historical evidence.

Historical-comparative research is a collection of techniques and approaches. It is a distinct type of research that puts historical time and /or cross-cultural variation at the center of research – that is, which treats what is studied as part of the flow of history and situated in cultural context.

Major questions

Historical comparative research is a powerful method for addressing big questions: How did major societal change take place? What fundamental features are common to most societies? Why did current social arrangements take a certain form in some societies but not in others? For example, historical-comparative researchers have addressed the questions of what caused societal revolutions in china, France, and Russia; how major social institutions, medicine, have developed and changed over two centuries; how basic relationships, like feelings about the value of children, change; why public policy toward the treatment of elderly developed in one way instead of another way in an industrial country; why South Africa developed a system of greater racial separation as the United States moved toward a greater racial integration.

Historical-comparative research is suited for examining the combination of societal factors that produce a specific outcome (e.g., civil war). It is also appropriate for comparing entire social system to see what is common across societies and what is unique, and to study long term change. An H-C researcher may apply a theory to specific cases to illustrate its usefulness. And he or she compares the same social processes and concepts in different cultural or historical contexts.

Researchers also use H-C method to reinterpret data or challenge old explanations. By asking different questions, finding new evidence, or assembling evidence in a different way, the H_C researcher raises questions about old explanations and finds support for new ones by interpreting the data in its cultural-historical context.

Historical-comparative research can strengthen conceptualization and theory building. By looking at historical events or diverse cultural contexts, a researcher can generate new concepts and broaden his or her perspective. Concepts are less likely to be restricted to a single historical time or to a single culture; they can be grounded in the experiences of people living in a specific cultural and historical context.

Historical-Comparative research focuses on:

- Tracing the development of social forms (patterns) overtime as well as its broad its broad historical processes, and
- Comparing those forms and its developmental processes across cultures (countries/nations).

Historical-Comparative research follows scientific approach:

- Can be a survey of events in history – could be through the study of documents.
Organizations generally document themselves, so if one is studying the development of some organization he/she should examine its official documents: charters, policy statements, speeches by the leaders, and so on. Often, official government documents provide the data needed for analysis. To better appreciate the history of race relations in the United States one could examine 200 years of laws and court cases involving race.
One could also do the communication analysis of different documents related to a particular issue (like the communication among the leaders of Pakistan movement through their letters,

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communication between the migrants to a new country and their relatives back in their country of origin)

Researcher could also get lot of information by interviewing people who may recall historical events (like interviewing participants in the Pakistan movement).

- Historical-Comparative researchers mostly do a longitudinal analysis i.e. look into the developmental processes of the issues under reference.
- Historical –Comparative researchers make cross-cultural comparisons of the social forms or economic form as well as the developmental processes of those forms, aiming at making generalizations.

Examples:

Social forms: Several researchers have examined the historical development of ideas about different forms of society. They have looked at the progression of social forms from simple to complex, from rural, from rural-agrarian to urban-industrial. The US anthropologist Lewis Morgan, for example, saw a progression from “savagery” to “barbarism” to “civilization.” Robert Redfield, another anthropologist, has more recently written of a shift from “folk society” to “urban society.” Emile Durkheim saw social evolution largely as a process of ever-greater division of labor. Ibn-e-Khaldun looked at the cyclical process of change in the form of societies from nomadic (Al-badawi) to sedentary (Al-hadari). These researchers discuss the forces that produce changes as well as the characteristics of each form of society. The historical evidence collected by researchers from different sources about different societies supports the whole discussion.

Forms of economic systems: Karl Marx examined the forms of economic systems progressing historically from primitive to feudal to capitalistic. All history, he wrote in this context, was a history of class struggle – the “haves” struggling to maintain their advantages and the “have-nots” struggling for a better lot in life. Looking beyond capitalism, Marx saw the development of a ‘classless’ society. In his opinion the economic forces have determined the societal system.

Not all historical studies in the social sciences have had this evolutionary flavor. Some social scientific readings of the historical record, in fact point to grand cycles rather than to linear progression (Ibn-e-Khaldun, P. Sorokin).

Economic forms and ideas: In his analysis of economic history, Karl Marx put forward a view of economic determinism. That is, he felt that economic factors determined the nature of all other aspects of society. Without denying that economic factors could and did affect other aspects of society, Max Weber argued that economic determinism did not explain everything. Indeed, Weber said, economic forms could come from non-economic ideas. In his research in the sociology of religion, Weber examined the extent to which religious institutions were the source of social behavior rather than mere reflection of economic conditions. His most noted statement of this side of the issue is found in *The Protestant Ethic and the Spirit of Capitalism*. John Calvin, a French theologian, was an important figure in the Protestant reformation of Christianity. Calvin thought that God had already decided the ultimate salvation or damnation of every individual; this idea is called *predestination*. Calvin also suggested that God communicated his decisions to people by making them either successful or unsuccessful during their earthly existence.

God gave each person an earthly “calling” – an occupation or profession – and manifested his or her success or failure through that medium. Ironically, this point of view led Calvin’s followers to seek proof of their coming salvation by working hard, saving for economic success.

In Weber’s analysis, Calvinism provided an important stimulus for the development of capitalism. Rather than “wasting” their money on worldly comforts, the Calvinists reinvested it in economic enterprises, thus providing the *capital* necessary for the development of capitalism. In arriving at this interpretation of the origin of capitalism, Weber researched the official doctrines of the early Protestant churches, studied the preaching of Calvin and other church leaders, and examined other historical documents.

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In three other studies, Weber conducted detailed analyses of Judaism, and the religions of China and India. Among other things, Weber wanted to know why capitalism had not developed in the ancient societies of China, India, and Israel. In none of the three religions did he find any teaching that would have supported the accumulation and reinvestment of capital – strengthening his conclusion about the role of Protestantism in that regard.

Logic of Historical-Comparative Research

Confusion over terms reigns H_C research. Researchers call what they do historical, comparative or historical-comparative, but mean different things. The key question is: Is there a distinct historical-comparative method and logic, or is there just social research that happens to examine social life in the past or in several societies? Some researchers use positivist, quantitative approach to study historical or comparative issues, while others rely on qualitative approach.

Quantitative approach: Positivist researchers reject the idea that there is a distinct H-C method. They measure variables, test hypotheses, analyze quantitative data, and replicate research to discover generalizable laws that hold across time and societies. They see no fundamental distinction between quantitative social research and historical-comparative research. They apply quantitative research techniques, with some minor adjustments, to study the past or other cultures.

- The researcher can focus on the issue in one society few societies or multiple societies.
- The researcher can focus on the issue in one time in the past or examine the issue across many years/periods in the past.
- The researcher can focus on the issue in the present or a recent past period.
- The researcher's analysis could be based primarily on quantitative data or qualitative data.
- Nevertheless, the debate continues.

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H-C researchers sometimes use *time-series* data to monitor changing conditions over time, such as data on population, crime rates, unemployment, infant mortality rates, and so forth. The analysis of such data sometimes requires sophistication for purposes of comparability. In case the definitions of the concept vary, it becomes difficult to make comparisons. The definitions not only could vary across nations but also these could vary within the same country over time (In Pakistan the definition of literacy changed from what it was in first population census of 1951 and what we had later on).

Qualitative approach:

There are no easily listed steps to follow in the analysis of historical data. Max Weber used the German term *verstehen* –“understanding” – in reference to an essential quality of research in behavioral sciences. He meant that the researcher must be able to take on, mentally, the circumstances, views, and feelings of those being studied to interpret their actions appropriately.

The historical-comparative researcher must find patterns among the voluminous details describing the subject matter of study. Often this takes the form of what Weber called ideal types: conceptual models composed of the essential characteristics of the phenomena. Thus, for example, Weber himself conducted lot of research on bureaucracy. Having observed numerous bureaucracies, Weber detailed those qualities essential to bureaucracies in general: jurisdictional areas, hierarchically structured authority, written files, and so on. Weber did not merely list those characteristics common to all bureaucracies he observed. Rather, he needed to understand fully the essentials of bureaucratic operation to create a theoretical model of the “perfect” (ideal type) bureaucracy.

A distinct, qualitative historical-comparative research differs from the positivist approach. Historical-comparative researchers who use case studies and qualitative data may depart from positivist approach. Their research is an intensive investigation of a limited number of cases in which the social meaning and context are critical. Case studies even in one nation, can be very important. Without case studies, scholars “would continue to advance theoretical arguments that are inappropriate, outdated, or totally irrelevant for a specific region”.

Historical-comparative researcher focuses on culture (patterns of behavior), tries to see through the eyes of those being studied, reconstructs the lives of the people studied, and examines particular individuals or groups.

A distinct H-C approach borrows from ethnography and cultural anthropology, and some varieties of H-C are close to “thick description” in their attempt to recreate the reality of another time or place.

A Distinct Historical-Comparative Approach

A distinct historical-comparative research method avoids the excesses of the positivist and interpretive approaches. It combines sensitivity to specific historical or cultural contexts with theoretical generalization. Historical-comparative researches may use quantitative data to supplement qualitative data and analysis. The logic and goals of H-C research are closer to those of field research than to those of traditional positivist approaches.

Similarities to Field Research:

First, both H-C research and field research recognize that the researcher’s point of view is an avoidable part of research. Both involve interpretation, which introduces the interpreter’s location in time, place, and world-view. H-C research does not try to produce a single, unequivocal set of objective facts. Rather, it is a confrontation of old with new or different world-views. It recognizes that the researcher’s reading of historical or comparative evidence is influenced by an awareness of the past and by living in the present. Our present day consciousness of history is fundamentally different from the manner in which the past appeared to any foregoing people.

Second, both field and H-C research examine a great diversity of data. In both, the researcher becomes immersed in data to gain an emphatic understanding of events and people. Both capture subjective feelings and note how everyday, ordinary activities signify important social meaning. The researcher inquires, selects, and focuses on specific aspects of social life from the vast array of events, actions, symbols, and words. An H-C researcher organizes data and focuses attention on the basis of evolving concepts. He or she examines rituals and symbols and dramatize culture and investigates the motives, reasons, and justifications for behaviors.

Third, both field and H-C researchers often use *grounded theory*. Theory usually emerges during the process of data collection. Both examine data without beginning with fixed hypotheses. Instead, they develop and modify concepts and theory through a dialogue with the data, then apply theory to reorganize evidence. [Historically grounded theory means that concepts emerge from the analytic problem of history: ordering the past into structures, conjectures and events. History and theory can thus be simultaneously constructed.]

Fourthly, both field and H-C research involve a type of translation. The researcher’s meaning system usually differs from that of people he or she studies, but he or she tries to penetrate and understand their point of view. Once the life, language, an perspective of the people being studied have been mastered, the researcher “translates” it for others who read his or her report.

Fifth, both field and H-C researchers focus on action, process, and sequence and see time process as essential. Both say that people construct a sense of social reality through actions that occur over time. Both see social reality simultaneously as something created and changed by people and as imposing a restriction on human choice.

Sixth, generalizations and theory are limited in field and H-C research. Historical and cross-cultural knowledge is incomplete and provisional, based on selective facts and limited questions. Neither deduces propositions or tests hypotheses in order to uncover fixed laws. Likewise replication is unrealistic because each researcher has a unique perspective and assembles a unique body of evidence. Instead, researchers offer plausible accounts and limited generalizations.

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Unique Features of H-C Research: Despite its many similarities to field research, some important differences distinguish H-C research. Research on past and on an alien culture share much in common with each other, and what they share distinguishes them from other approaches.

First, the evidence of H-C research is usually limited and indirect. Direct observation and involvement by a researcher is often impossible. A H-C researcher reconstructs what occurred from the evidence, but he or she cannot have absolute confidence in his reconstruction. Historical evidence in particular depends on the survival of data from the past, usually in the form of documents (e.g., letters and newspapers). The researcher is limited to what has not been destroyed and what leaves a trace, record, or other evidence behind.

Second, H-C researchers interpret the evidence. Different people looking at the same evidence often ascribe different meanings to it, so a researcher must reflect on evidence. An understanding of it based on a first glance is rarely possible. The researcher becomes immersed in and absorbs details about a context. For example, a researcher examining the family in the past or a distant country needs to be aware of the full context (e.g., the nature of work, forms of communication, transportation technology, etc.).

Another feature is that a researcher's reconstruction of the past or another culture is easily distorted. Compared to the people being studied, H-C researchers is usually more aware of events occurring prior to the time studied, events occurring in places other than the location studied, and events that occurred after the period studied. This awareness gives the researchers a greater sense of coherence than was experienced by those living in the past or in an isolated social setting. Historical explanation surpasses any understanding while events are still occurring. The past we reconstruct is more coherent than the past when it happened.

A researcher cannot see through the eyes of those being studied. Knowledge of the present and changes over time can distort how events, people, laws, or even physical objects are perceived. When the building was newly built (say in 1800) and standing among similar buildings, the people living at the time saw it differently than people do in the 21st century.

H-C researcher does not use deterministic approach. H-C research takes an approach to causality that is more contingent than determinist. A H-C researcher often uses combinational explanations. They are analogous to a chemical reaction in which several ingredients (chemicals, oxygen) are added together under specified conditions (temperature, pressure) to produce an outcome (explosion). This differs from a linear causal explanation. H-C research focuses on whole cases and on comparisons of complex wholes versus separate variables across cases. The logic is more “A, B, and C appeared together in time and place, then D resulted” than “A caused B, and B caused C, and C caused D.”

H-C researcher has the ability to shift between a specific context and a generalized context for purposes of comparison. A researcher examines several specific contexts, notes similarities and differences, then generalizes. He or she looks again at the specific context using the generalization. H-C researchers compare across cultural-geographic units. They develop trans-cultural concepts for purposes of comparative analysis. In comparative research, a researcher translates the specifics of a context into a common, theoretical language. In historical research theoretical concepts are applied across time.

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LESSON 41**HISTORICAL-COMPARATIVE RESEARCH (Contd.)**

Conducting historical-comparative research does not involve a rigid set of steps and, with only a few exceptions; it does not use complex or specialized techniques. Nevertheless, some guideline for doing historical-comparative research may be provided.

Conceptualizing the Object of Inquiry

An H-C researcher begins by becoming familiar with the setting and conceptualizes what is being studied. He or she may start with a loose model or set of preliminary concepts and apply them to specific setting. The provisional concepts contain implicit assumptions or organizing categories that he or she uses to see the world, “package” observations, and search through evidence.

Decide on the historical era or comparative settings (nations or units). If the researcher is not already familiar with historical era or comparative settings, he or she conducts an orientation reading (reading several general works). This will help the researcher grasp the specific setting, assemble organizing concepts, subdivide the main issue, and develop lists of questions relating to specific issue.

Locating Evidence

The researcher locates and gathers evidence through extensive bibliographic work. A researcher uses many indexes, catalogs, and reference works that list what libraries contain. For comparative research, this means focusing on specific nations or units and on particular kinds of evidence within each. The researcher frequently spends weeks searching for sources in libraries, travels to several different specialized research libraries, and reads dozens of books and articles. Comparative research often involves learning one or more foreign languages.

As the researcher masters the literature and takes numerous detailed notes, he or she completes many specific tasks: creating a bibliography list (on cards or on computer) with complete citations, taking notes that are neither too skimpy nor too extensive, leaving margins on note cards for adding themes later on, taking all note in the same format, and developing a file on themes or working hypothesis.

A researcher adjusts initial concepts, questions, or focus on the basis of what he or she discovers in the evidence. New issues and questions arise as he or she reads and considers a range of research reports at different levels of analysis (e.g., general context and detailed narratives on specific topic), and multiple studies on a topic, crossing topic boundaries.

Evaluating Quality of Evidence

As the H-C researcher gathers evidence, he or she asks two questions: How relevant is the evidence to emerging research questions and evolving concepts? How accurate and strong is the evidence?

The question of relevance is difficult one. All documents may not be equally valuable in reconstructing the past. As the focus of research shifts, evidence that was not relevant can become relevant. Likewise, some evidence may stimulate new avenues of inquiry and search for additional confirming evidence.

Accuracy of evidence may be looked at for three things: the implicit conceptual framework, particular details that are required and empirical generalizations. H-C researcher evaluates alternative interpretations of evidence and looks for “silences,” of cases where the evidence fails to address an event, topic, or issue.

Researchers try to avoid possible fallacies in the evidence. For example, a fallacy of pseudo proof is failure to place something into its full context. The evidence might state that there was a 50 percent increase in income taxes, but it is not meaningful outside of a context. The researcher must ask: Did other taxes decline? Did income increase? Did the tax increase apply to all income? Was everyone affected equally?

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Organizing Evidence

As a researcher gathers evidence and locates new sources, he or she begins to organize the data. Obviously, it is unwise to take notes madly and let them pile up haphazardly. A researcher usually begins a preliminary analysis by noting low-level generalizations or themes. For example, in a study of revolution, a researcher develops a theme: The rich peasants supported the old regime. He or she can record this theme in his or her notes and later assign to significance.

Researcher organizes evidence, using theoretical insights to stimulate new ways to organize data and for new questions to ask of evidence. The interaction of data and theory means that a researcher goes beyond a surface examination of the evidence based on theory. For example, a researcher reads a mass of evidence about a protest movement. The preliminary analysis organizes the evidence into a theme: People who are active in protest interact with each other and develop shared cultural meanings. He or she examines theories of culture and movements, then formulates new concept: “oppositional movement subculture.” The researcher then uses this concept to re-examine the evidence.

Synthesizing

The researcher refines concepts and moves toward a general explanatory model after most of the evidence is in. Old themes or concepts are discussed or revised, and new ones are created. Concrete events are used to give meaning to concepts.

The researcher looks for patterns across time or units, and draws out similarities and differences with analogies. He or she organizes divergent events into sequences and groups them together to create a larger picture. Plausible explanations are then developed that subsume both concepts and evidence as he or she organizes the evidence into a coherent whole. The researcher then reads and rereads notes and sorts and resorts them into piles or files on the basis of organizing schemes. He or she looks for and writes down the links or connections he or she sees while looking at evidence in different ways.

Synthesis links specific evidence with an abstract model of underlying relations or causal mechanism. A researcher often looks for new evidence to verify specific links that appear only after an explanatory model is developed. He or she evaluates how well the model approximates the evidence and adjusts it accordingly.

Historical-comparative researchers also identify critical indicators and supporting evidence for themes or explanations. A *critical indicator* is unambiguous evidence, which is usually sufficient for inferring a specific theoretical relationship. Researchers seek these indicators for key parts of an explanatory model. Indicators critically confirm a theoretical inference and occur when many details suggest a clear interpretation.

Writing a Report

Combine evidence, concepts, and synthesis into a research report. The way in which the report is written is key in H-C research. Assembling evidence, arguments, and conclusions into a report is always a crucial step; but more than in quantitative approaches, the careful crafting of evidence and explanation makes or breaks H-C research. A researcher distills mountains of evidence into exposition and prepares extensive footnotes. She or he weaves together evidence and arguments to communicate a coherent, convincing picture to readers.

Data and Evidence in Historical context

Historical-comparative researchers draw on four types historical evidence or data:

1. Primary sources;
2. Secondary sources;
3. Running records; and
4. Recollections.

Traditional historians rely heavily on primary sources. H-C researchers often use secondary sources or the different data types in combination.

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1. Primary Sources: The letters, diaries, newspapers, movies, novels, articles of clothing, photographs, and so forth are those who lived in the past and have survived to the present are the primary sources. They are found in archives (a place where documents are stored), in private collections, in family closets, or in museums. Today's documents and objects (our letters, television programs, commercials, clothing, and automobiles) will be primary sources for future historians. An example of a classic primary source is a bundle of yellowed letters written by a husband away at war to his wife and found in a family closet by a researcher.

Published and unpublished written documents are the most important type of primary source. Researchers find them in their original form or preserved in microfilm or on film. They are often the only surviving record of the words, thoughts, and feelings of people in the past. Written documents are helpful for studying societies and historical periods with writing and literate people. A frequent criticism of written sources is that elites or those in official organizations largely wrote them; thus the views of the illiterate, the poor, or those outside official social institutions may be overlooked.

The written word on paper was the main medium of communication prior to the widespread use of telecommunications, computers, and video technology to record events and ideas. In fact, the spread of forms of communication that do not leave a permanent physical record (e.g., telephone conversation), and which have largely replaced letters, written ledgers, and newspapers, make the work of future historians difficult.

Potential Problems with Primary Sources: The key issue is that only a fraction of everything written or used in the past has survived into present. Moreover, whatever is survived is nonrandom sample of what once existed.

H-C researchers attempt to read primary sources with the eyes and assumptions of a contemporary who lived in the past. This means “bracketing,” or holding back knowledge of subsequent events and modern values. “If you do not read the primary sources with an open mind and an intention to get inside the minds of the writings and look at things the way *they* saw them, you are wasting time.” For example, when reading a source produced by a slaveholder, moralizing against slavery or faulting the author for not seeing its evil is not worthwhile. The H-C researcher holds back moral judgments and becomes a moral relativist while reading primary sources. He or she must think and believe like subjects under study, discover how they performed in their own eyes.

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Another problem is that locating primary documents is a time consuming task. A researcher must search through specialized indexes and travel to archives or specialized libraries. Primary sources are often located in dusty, out-of-the-way room full of stacked cardboard boxes containing masses of fading documents. These may be incomplete, unorganized, and various stages of decay. Once the documents or other primary sources are located, the researcher evaluates them subjecting them to external and internal criticism.

External criticism means evaluating the authenticity of a document itself to be certain that it is not a fake or a forgery. Criticism involves asking: Was the document created when it is claimed to have been, in the place where it was supposed to be, and by the person who claims to be its author? Why was the document produced to begin with, and how did it survive? Once the document passes as being authentic, a researcher uses *internal criticism*, an examination of the document's contents to establish credibility. A researcher evaluates whether what is recorded was based on what the author directly witnessed or is secondhand information.

Many types of distortions can appear in primary documents. One is *bowdlerization* – a deliberate distortion designed to protect moral standards or furnish a particular image. For example, photograph is taken of the front of a building. Trash and empty bottles are scattered all around the building, and the paint is faded. The photograph, however, is taken of the one part of the building that has little trash and is framed so that the trash does not show; dark room techniques make the faded paint look new.

2. Secondary Sources: Social researchers often use secondary sources, the books and articles written by specialist historians and other researchers, as an evidence of past conditions. It has its own limitations.

Potential Problems with Secondary Sources: The limitations of secondary historical evidence include problems of inaccurate historical accounts and lack of studies in areas of interest. Such sources cannot be used to test hypotheses. Post facto explanations cannot meet positivist criteria of falsifiability, because few statistical controls can be used and replication is not possible.

The many volumes of secondary sources present a maze of details and interpretations for an H-C researcher. He or she must transform the mass of specialized descriptive studies into an intelligible picture. This picture needs to be consistent with the reflective of the richness of the evidence. It also must bridge the many specific time periods and locals. The researcher faces potential problems with secondary sources.

One problem is reading the works of historians. Historians do not present theory-free, objective “facts.” They implicitly frame raw data, categorize information, and shape evidence using concepts. The historian’s concepts are a mixture drawn from journalism, language of historical actors, ideologies, Philosophy, everyday language in the present, and social science. Most lack a rigorous definition, are vague, are applied inconsistently, and are not mutually exclusive, nor exhaustive.

Second problem is that historian’s selection procedure is not transparent. They select some information from all possible evidence. From the infinite oceans of facts historian selects those, which are significant for his purpose. Yet, the H-C researcher does not know how this was done. Without knowing the selection process, a historical-comparative researcher must rely on the historian’s judgments, which can contain biases.

A third problem is in the organization of the evidence. Historians organize evidence as they write works of history. They often write *narrative history*. This compounds problems of undefined concepts and the selection of evidence. In the historical narrative, the writer organizes material chronologically around a single coherent “story.” The logic is that of a sequence of unfolding action. Thus, each part of the story is connected to each other part by its place in the time order of events. Together all the parts form a unity or whole. Conjecture and contingency are the key elements of the narrative form. The contingency creates a logical interdependency between earlier and later elements.

With its temporal logic, the narrative organization differs from how the social researchers create explanations. It also differs from quantitative explanation in which the researcher identifies statistical patterns to infer causes. A major difficulty of the narrative is that the organizing tool – time order or position in a sequence of events – does not alone denote theoretical or historical causality. In other word, the narrative meets only one of the three criteria for establishing causality – that of temporal sequence.

Fourth and the last problem is that historiographic schools, personal beliefs, social theories influence a historian, as well as current events at the time research were conducted. Historians writing today examine primary material differently from how those writing in the 1920s did. In addition, there are various schools of historiography (diplomatic, Marxist) that have their own rules for seeking evidence and asking questions. It is also said history gets written by the people in power; it may include what the people in power want to be included.

3. Running Records: Running records consist of files or existing statistical documents maintained by organizations. An example of a running record is keeping of vital statistics by the government departments in Pakistan; vital statistics relating to births, marriage, divorce, death, and other statistics of vital events. We also have so many documents containing running records relating to demographic statistics, and economic statistics being maintained by different agencies of UNO.

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4. Recollections: The words or writing of individuals about their past lives or experiences based on memory are recollections. These can be in the form of memoirs, autobiographies, or interviews. Because memory is imperfect, recollections are often distorted in ways that primary sources are not.

In gathering *oral history*, a type of recollection, a researcher conducts unstructured interviews with people about their lives or events in the past. This approach is especially valuable for non-elite groups or the illiterate.

Evaluating the Documents

Historical-comparative researchers often use secondary sources or different data types in combination. For secondary sources they often use existing documents as well as the data collected by other organizations for research purposes. While looking into the authenticity of these document researchers often want answers to the questions like: Who composed the documents? Why were these written? What methods were used to acquire the information? What are some of the biases in the documents? How representative was the sample? What are the key categories and concepts used? What sorts of theoretical issues and debates do these documents cast light on?

Problems in Comparative Research

Problems in other types of research are magnified in a comparative study. In principle, there is no difference between comparative cross-cultural research and research conducted in a single society. The differences lie, rather, in the magnitude of certain types of problems.

The Units being compared:

For convenience, comparative researchers often use nation-state as their unit of analysis. The nation-state is the major unit used in thinking about the divisions of people across globe today. The nation-state is a socially and politically defined unit. In it, one government has sovereignty over populated territory. The nation-state is not the only unit for comparative research, but also frequently used as a surrogate for culture, which more difficult to define as a concrete, observable unit. The boundaries of nation-state may not match those of a culture. In some situations a single culture is divided into several nations (Muslim culture); in other cases, a nation-state contains more than one culture (Canada). The nation-state is not always the best unit for comparative research. A researcher should ask: What is the relevant comparative unit for my research question – the nation, the culture, a small region, or a subculture?

Problems of Equivalence: Equivalence is a critical issue in all research. It is the issue of making comparisons across divergent contexts, or whether a researcher, living in a specific time period and culture, correctly reads, understands, or conceptualizes data about people from different historical era or culture. Without equivalence, a researcher cannot use the same concepts or measures in different cultures or historical periods, and this makes comparison difficult, if not impossible. It is similar to the problems of validity in quantitative research. Look at the concept of a *friend*. We ask some body how many friends do you have? People living in different countries may have different meanings attached to it. Even in Pakistan, we have variations in its meaning across the Provinces, and between rural and urban areas.

Ethical problems are less intense in H-C research than in other types of social research because a researcher is less likely to have direct contact with people being studied. Historical-comparative research shares the ethical concerns found in other non-reactive research techniques.

LESSON 42**FOCUS GROUP DISCUSSION**

A visitor to a locality stops by a house and inquires about the address of a resident he wants to see. May be he starts talking with a couple of persons asking for their help. In the meantime, some other passersby, or coming out of other houses join, showing their curiosity about the issue. They ask for some more information about the resident concerned, and then start discussing among them to come up with the exact identification of the resident. As an outcome of this discussion they would guide the visitor to reach the destination. This is quite a common feature in a folk society (village, neighborhood in a city) where we may start talking with a couple of persons and others come and join the conversation. This is an example of *informal* focus group discussion, which is built upon the social networks that operate in a natural setting. These social networks include both kinsfolk and other neighbors. In some cases the participants may be the local decision makers.

In research, focus group discussions (FGD) are a more formal way of getting groups of people to discuss selected issues. A focus group discussion is a group discussion of 6-12 persons guided by a facilitator, during which group members talk freely and spontaneously about a certain topic. There may be some disagreement about the exact number of participants in the discussion, as one comes across variations in numbers (6 to 10, 6 to 12, 6 to 15, 8 to 10, 5 to 7) in different books on research methods. The trend has been toward smaller groups due to some problems with the larger groups, which like:

- In a bigger group each participant's speaking time is substantially restricted. Dominant/submissive relationships are almost inevitable.
- Frustration or dissatisfaction among group members is likely to result because of some members' inability to get a turn to speak. This produces lower quality and quantity of data.
- Participants are often forced into long speeches, often containing irrelevant information, when they get to speak only infrequently.
- The tendency for side conversations between participants increases.

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In contrast, smaller group sessions are felt to provide greater depth response for each participant. The group is often more cohesive and interactive, particularly when participants are professionals, such as physicians or pharmacists.

The key factor concerning group size is generally the of group purpose. If the purpose of the group is to generate as many ideas as possible, a larger group may be most useful. If the purpose of the group is to maximize the depth of expression from each participant, a smaller group works better.

The Purpose of FGD

The purpose of an FGD is to obtain in-depth information on concepts, perceptions, and ideas of the group. An FGD aims to be more than a question-answer interaction (Focus group interview is different). Here the idea is that group members discuss the topic among themselves.

Formal Focus Groups

Formal groups are formally constituted, that is these are organized in advance by inviting the selected individuals to participate in the discussion on a specific issue. They are structured groups brought together in which the participants are expected to have similar background, age, sex, education, religion, or similar experiences. Similarity in background is likely to make them comfortable where they could express their viewpoint frankly and freely. If the big boss and his junior officer working in an organization together participate in an FGD, the junior officer may not be able to express his or her opinion freely in the presence of his/her boss. Similarly, in some situations the children may experience some inhibitions in expressing their views on a sensitive issue in the presence of their parents. A lot depends on the kind of issue that is to be discussed.

The group is guided by a moderator/facilitator. The participants address a specific issue (talk freely, agree or disagree among them) within a specified time in accordance with clearly spelled out rules of procedure.

Designing a Focus group Study

As with other approaches to studying social phenomena, designing a focus group study requires careful thought and reflection. Given that focus groups can be used for a variety of purposes within social research, the design of focus group study will depend on its purpose. At one extreme, FGD is used at the exploratory stage of the study (FGD may help in the identification of variables, formulation of questions and response categories) and at the other extreme, when qualitative information is needed on issues about which the researchers have substantial background knowledge and a reasonable grasp of the issues. Here we are focusing on the latter type of design.

How to conduct FGD?

The following guideline may be provided for conducting FGD.

1. Preparation:

- Selection of topic, questions to be discussed. It is appropriate to define and clarify the concepts to be discussed. The basic idea is to lay out a set of issues for the group to discuss. It is important to bear in mind that the moderator will mostly be improvising comments and questions within the framework set by the guidelines. By keeping the questions open-ended, the moderator can stimulate useful trains of thought in the participants that were not anticipated.
- Selecting the study participants: Given a clear idea of the issues to be discussed, the next critical step in designing a focus group study is to decide on the characteristics of the individuals who are to be targeted for sessions. It is often important to ensure that the groups all share some common characteristics in relation to the issue under investigation. If you need to obtain information on a topic from several different categories of informants who are likely to discuss the issue from different perspectives, you should organize a focus group for each major category. For example a group for men and a group for women, or a group for older women and group for younger women. The selection of the participants can be on the basis of purposive or convenience sampling. The participants should receive the invitations at least one or two days before the exercise. The invitations should explain the general purpose of the FGD.
- Physical arrangements: Communication and interaction during the FGD should be encouraged in every way possible. Arrange the chairs in a circle. Make sure the area will be quite, adequately lighted, etc., and that there will be no disturbances. Try to hold the FGD in a neutral setting that encourages participants to freely express their views. A health center, for example, is not a good place to discuss traditional medical beliefs or preferences for other types of treatment. Neutral setting could also be from the perspective of a place where the participants feel comfortable to come over and above their party factions.

Conducting the session:

- One of the members of the research team should act as a “facilitator” or “moderator” for the focus group. One should serve as “recorder.”
- **Functions of the Facilitator:** The facilitator should not act as an expert on the topic. His or her role is to stimulate and support discussion. He should perform the following functions:
 - o **Introduce the session:** He or she should introduce himself/herself as facilitator and introduce the recorder. Introduce the participants by name or ask them to introduce themselves (or develop some new interesting way of introduction). Put the participants at ease and explain the purpose of the FGD, the kind of information needed, and how the information will be used (e.g., for planning of a health program, an education program, etc.).
 - o **Encourage discussion:** The facilitator should be enthusiastic, lively, and humorous and show his/her interest in the group’s ideas. Formulate questions and encourage as many participants as

possible to express their views. Remember there are no “right” or “wrong” answers. Facilitator should react neutrally to both verbal and nonverbal responses.

- o **Encourage involvement:** Avoid a question and answer session. Some useful techniques include: asking for clarification (can you tell me more?); reorienting the discussion when it goes off the track (saying: wait, how does this relate to the issue? Using one participant’s remarks to direct a question to another); bringing in reluctant participants (Using person’s name, requesting his/her opinion, making more frequent eye contact to encourage his participation); dealing with dominant participants (avoiding eye contact or turning slightly away to discourage the person from speaking, or thanking the person and changing the subject).
- o **Avoid being placed in the role of expert:** When the facilitator is asked for his/her opinion by a respondent, remember that he or she is not there to educate or inform. Direct the question back to the group by saying: “What do you think?” “What would you do?” Set aside time, if necessary, after the session to give participants the information they have asked.
Do not try to give comments on everything that is being said. Do not feel you have to say something during every pause in the discussion. Wait a little and see what happens.
- o **Control the timing of the meeting but unobtrusively:** Listen carefully and move the discussion from topic to topic. Subtly control the time allocated to various topics so as to maintain interest. If the participants spontaneously jump from one topic to the other, let the discussion continue for a while because useful additional information may surface and then summarize the points brought up and reorient the discussion.
- o **Take time at the end of the meeting to summarize, check for agreement and thank the participants:** Summarize the main issues brought up, check whether all agree and ask for additional comments. Thank the participants and let them know that their ideas had been a valuable contribution and will be used for planning the proposed research/intervention/or whatever the purpose of FGD was.

Listen to the additional comments made after the meeting. Sometime some valuable information surfaces, which otherwise may remain hidden.

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RESEARCH METHODS

Lecture 19

RESEARCH DESIGN: ELEMENTS

Research Design

- **A master plan specifying the methods and procedures for collecting and analyzing the data.**
- **A strategy or blueprint that plans the action for carrying through the research data.**

Elements

- **Series of components of research design**

1. Purpose of the Study

- **Study can be exploratory, descriptive, and explanatory.**
- **Present study can't be exploratory. We have already covered long distance.**
- **Can be either descriptive or explanatory.**
- **Who, what, where, when, or how much?**
- **Why? Explain the relationship.**

Purpose determines how rigorous the study will be

- **Decide about the purpose.**
- **Within explanatory →**
- **Correlational: identification of factors associated with the problem.**
- **Causal: establish definite cause-and-effect relationship.**
- **Help in deciding the mode of observation.**

2. Unit of Analysis

- Refers to the level of aggregation of the data during data analysis stage.
- Is it at the individual, or at group, or at organization level.
- Raise the motivation level of employees. Collect data from individual employees. Aggregate analysis of information collected from individuals. Unit of analysis is *individual*.

- **Study two person interaction** – then several two person groups will become the unit of analysis (husband-wife, supervisor-subordinate, teacher-student – dyads)
- **Group effectiveness** – unit of analysis is group. Comparing different departments in the work organization.
- **Research question determines the unit of analysis.** Affects the data collection methods, sampling.
- **Unit of *observation*.** Information provider. Sometimes observe indirectly.

3. Time Dimension

- **Make observation more than one time i.e. over a long period.**
- **Cross-sectional studies.**
- **Longitudinal studies. Repeated over an extended period. Employees behavior before and after top management change**
- **Panel studies**
- **Cohort studies**

4. Researcher Control of Variables

- **Researcher's ability to manipulate variables.**
- **Experimental design.**
- **a. Non-contrived: natural environment. Field experiment. Correlational studies.**
- **b. Contrived: artificial setting. Causal**
- **Ex-post-facto design. No control over the variables. Report only what has happened or what is happening. Survey research.**

5. Mode of observation

- Depending upon the type of study (qualitative or quantitative, descriptive or causal, cross-sectional or longitudinal, contrived or non-contrived) researcher decides about the mode of observation.
- Survey, experiment, field observation, case study, focus group, communication analysis.

6. Sampling Design

- **Selecting some of the elements in the population and draw conclusions about the total population.**
- **Population element is the subject on which measurement is being taken.**
- **Has its advantages and disadvantages.**
- **Selection of appropriate sampling design.**

7. Observation Tools

- Questionnaire
- Interview schedule
- Interview guide
- Check list
- Researcher will specify the instrument along with justification of its appropriateness.

8. Field Data Collection

- Depending on the mode of observation, the researcher will outline the procedure.
- Field team selection, training, supervision.

9. Data Processing and Data Analysis

- **Manual vs. mechanical data processing.**
- **Quantification of data. Data reduction (score index), tabulation.**
- **Data analysis plan: use of statistics, interpretations**

Survey Research:

An Overview

- **Method of gathering primary data based on communication with a representative sample of individuals.**
- **Survey requires asking people (respondents) for information, using either verbal or written questions.**
- **Collect data on telephone, face-to-face, or other communication media.**

Steps in Conducting Surveys

- **Researcher follows a deductive approach.**
- **Begins with theoretical or applied research problem and ends with empirical measurement and data analysis.**
- **Six broad steps:**

1. Decide on the Type of Survey

- **Type of survey → Mail, interview, telephone, computer assisted. What is appropriate?**
- **Develop the instrument: questionnaire or interview schedule. Both are list of questions.**
- **Questionnaire: respondents read the questions themselves and mark answers on the questionnaire.**
- **Interview schedule: questions read to the respondent by an interviewer, who also records the answers.**

2. Plan How to Record Data

- **Researcher thinks ahead how the data shall be recorded and organized for analysis.**
- **Pilot test survey instrument on similar population.**

3. Decide on Target Population

- **Get sampling frame**
- **Decide on sample size**
- **Select the sample**

4. Do the Field Work

- **Locate sampled respondents in person, telephone, or by mail.**
- **Conduct interviews or administer questionnaires**
- **Carefully record the data.**

5. Data Processing

- Enter data into computers
- Data cleaning
- Tabulation
- Perform statistical analysis on data.

6. Research Report

- **Describe methods and findings in research report**
- **Present findings to others for critique and evaluation**

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RESEARCH METHODS

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SURVEY RESEARCH

Two approaches to collect primary data

- **1. Observe → conditions, behavior, events, people, or processes**
- **2. Communicate with people.**
- **Selected approach determines the research design to be followed.**

Communication Approach

- Involves surveying people and recording their responses for analysis.
- Versatile approach: variety of communication media can be used.
- Traditional communication media – Human interactive media.
- Modern communication – use of digital technology – Electronic interactive media.
- Non-interactive media.

Human Interactive Media

- **Personal forms of communication.**
Face to face interaction of individuals and or groups. Can also be conversation on telephone

Electronic Interactive Media

- **Use of digital technology to reach people.**
 - **Respondents are actively involved.**
- Internet – new electronic interactive media**

Non-Interactive Media

- **Traditional questionnaire received by mail**
- **Self administered questionnaires**
- **Questionnaires delivered through other means – Fax machine, uploading on internet, printing in the newspaper.**

Choosing a Communication Media

- Once decided about doing survey then collect data by using different media:
 - - Personal interviewing
 - - Telephone interviewing
 - - Mail questionnaire
 - - Self-administered questionnaire
 - - Computer assisted medium
 - - Mixed media

Personal Interviewing

- Face to face interaction between strangers
- Respondent provides information with little hope to receive any immediate or direct benefits.
- Personal interview can take place anywhere.

Advantages of Personal Interviewing

1. The opportunity for feedback

- **Clarify respondent's apprehensions. Assurance for confidentiality.**
- **Clarify questions asked. Clarify instructions given for responses.**
- **Debriefing**

2. Probing Complex Questions

- Probing implies the verbal prompts made by the interviewer when:**
- Response is not clear.**
- Enlargement of the response is needed.**
- All the more needed in open ended questions.**

3. Length of Interview

- **Compared with other interviews personal interviews can be lengthy, if needed.**

4. High Completion Rate

- Interviewer to make sure that all applicable questions have been answered.
- In telephone, the respondent may just hang up in the middle.
- Personally administered questionnaire have hi incompleteness rate. Don't want to write long answer for open ended questions.
- Item non-response: failure to provide an answer to a question. Low in personal interviewing

5. Props and Visual Aids

- **Can see the new product sample, taste a product and give his/her evaluation. It is not possible for telephone interview or a mail survey.**

6. High Participation Rate

- The presence of the interviewer increases the % of people to participate.**
- No reading, no writing – only talking. Share their feelings with friendly and sympathetic interviewers.**

7. Observing the Non-Verbal Behavior

- Interviewer can catch the body language, facial expressions. Observe the environment as well.**

8. Non-Literates can Participate

- **Illiterate or functionally illiterate person can take part in the study.**

9. Interviewer can Prescreen Respondent

- Respondent to fit the sampling criteria.**
- Interviewer has some control over the environment.**
- No control in mail survey, telephone interview, or internet interview**

10. Computer Assisted Personal Interviewing

- **CAPI – Use of modern technology**
- **Microcomputers can be used for direct entry of responses. Reduces error and cost**

Disadvantages of Personal Interviewing

1. High Cost

- Geographical proximity, length of questionnaire, No. of recalls – influence cost**
- Training, supervision, logistical arrangements add to the cost.**
- Estimates show 15 times higher than mail survey.**

2. Scarcity of Highly Trained Interviewers

- For quality study we need quality interviewers. In short supply**

3. Callbacks –A Labor Intensive Work

- Callbacks are attempts to re-contact the missing sample cases.**
- Labor intensive work – increases cost.**

4. Interviewer Influence

- **Demographic characteristics of the interviewer can influence the respondent's responses.**

5. Interviewer Bias

- Biases in understanding, recording, and interpreting the responses

6. No Opportunity to Consult

- Interview may take place anywhere
– the respondent unable to consult
any record or person in case he
has to.**

7. Less Standardized Wording

- Interviewers may make alterations in wording, and sequencing of questions. Bias the data**

8. Limitations of respondents' availability and accessibility

- Some executive officers and VIPs are never available or not willing to talk to researchers.**

9. Some Neighborhoods Difficult to Visit

- **Security reasons**
- **Some neighborhoods may not allow outsiders. Some others may be scary for the interviewers.**

Door to Door Interviews

- **Personal interviews at respondent's home or place of work.**
- **More representative than others. We walk through. Can catch those without phones or having unlisted numbers.**
- **May exclude living in multiple dwellings, high rise buildings, executives.**
- **People who are at home are special type – moms, retired persons**

Intercept Interviews in Shopping Mall

- **Shoppers are interviewed at a central point in the mall or at the entrance.**
- **Low cost.**
- **Incidence of refusal is high.**
- **May not be a representative sample of the population.**
- **Each mall will have its own characteristics.**
- **Hi traffic areas – some fast food center**

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Telephone Interviewing

- Mainstay of commercial survey research.
- Data quality comparable to personal interviewing.
- Respondents may more willing to give their opinions on phone.
- TS can provide representative samples of general pop in industrialized countries.

Central Location Interviewing

- **Conduct all interviewing from central location.**
- **Wide-Area Telecommunication Service (WATS) lines provided by long distance service at fixed rates, unlimited calls.**
- **Hire professional interviewers**
- **Supervise and control the quality of interviewing.**
- **Research cost-effective.**

Computer Assisted Telephone Interviewing (CATI)

- **Responses to be entered directly to computer.**
- **Telephone interviewers seated at computer terminals.**
- **Monitor displays the questionnaire – one question at a time, precoded responses.**
- **Interviewer enters the responses in computer**
- **Programming facilitates interviewing.**

The Strengths of Telephone Interviewing

1. High Speed

- **Quickly reach the respondents.**
- **With CATI quick data processing**

2. Saves Cost

- **Cost of telephone interviewing is less than 25% of the door to door personal interviews.**

3. Callbacks Easier

- **Unanswered calls, busy signal, respondent not at home needs callback.**
- **Telephone callbacks are substantially easier and less expensive than personal interviews.**

4. Expanded Geographic Area Coverage without increasing the cost.

**5. Uses fewer but highly
Skilled Interviewers.**

6. Reduced Interviewer Bias

7. Better Access to hard-to-reach respondents

- Some people don't want to go for person to person interviewing. Prefer telephone interviewing.**
- Interviewers may be reluctant to visit certain neighborhoods, especially in the evening. Telephone interviewing overcomes such problems**

8. Use Computer Assisted Telephone Interviewing

- Responses can be directly entered into computer file to reduce error and cost**

Weaknesses of Telephone Interviewing

1. Absence of Face-to-Face Contact

- **More impersonal than face-to-face interviews.**
- **Absence of face-to-face contact may be a liability. Respondent and the interviewer don't see each other what they are doing. (Respondent still responding when he/she is thinking and not speaking. Has the interviewer finished recording)**

2. Response rate lower than for Personal Interviews

- Some individuals refuse to participate. (Executives easily refuse. Participants find it easier to terminate)**

3. Lack of Visual Medium

- **Research requiring visual material cannot be conducted by phone.**

4. Limited Duration

- **Length of interview limited.
Respondent may hang up when they feel spent up.**
- **(Plan telephone interviews approximately for 10 minutes)**

5. Many Numbers Unlisted or not Working

- **Can affect the sampling design.**

6. Less Participant involvement

- **Result In less thorough responses.**
- **Respondents' experiences less rewarding than a personal interview.**
- **Rapport building defficient.**

7. Distracting Physical Environment

- Distractions on the side of the respondent.**

Self Administered Questionnaires

- **Ubiquitous in modern living.**
- **Mail questionnaires + other modes**

Advantages of Mail Questionnaire

- **1 Geographic Flexibility:** Reach all corners of the country/world.
- **2. Sample Accessibility:** Contact those who may inaccessible.
- **3. Save Time:**
- **4. Saves Cost:**
- **Respondent Convenience:**
- **Anonymity:**
- **Standardized Questions**

Disadvantages of Mail Questionnaire

- **1. Low Response Rate**
- **2. Low Completion Rate**
- **3. Increases Cost**
- **4. Interviewer's Absence**
- **5. No Control on Question Order**
- **6. Cannot use Lengthy Questionnaire**
- **7. No Control Over Environment**
- **8. Cannot catch the non-verbal behavior**
- **9. Non-Literates cannot Participate**

Increasing Response Rate

- Cover Letter
- Money Helps
- Interesting Questions
- Follow-Ups
- Preliminary Notification
- Survey Sponsorship
- Other Techniques: Questionnaire format, facilitating the return, postage, personalization.

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Self Administered Questionnaires

(Continued)

Increasing Response Rate

- Response rate is the number of questionnaires returned or completed, divided by the total number contacted.
- Response rate of self administered questionnaires is low.
- Here are some guidelines for increasing response rate.

Cover Letter

- An important means of inducing a reader to complete and return the questionnaire.
- Establish rapport by telling: why this study, who is the sponsor, how was subject selected, assuring the anonymity.
- Make it a personal letter, individually typed on a personal letterhead.

Money Helps

- Offer monetary incentives or rewards.
- Attracts the attention and creates a sense of obligation

Interesting Questions

- Topic and related questions cannot be changed.
- Certain interesting questions may added, perhaps in the beginning, to stimulate the respondent's interest.

Follow-Ups

- **Reminders**
- **May include another copy of the questionnaire.**
- **Number of reminders to be sent.**

Preliminary Notification

- **Preparing the respondent through advance notice through letter/telephone.**
- **Notify closer to the questionnaire mailing time.**
- **Depends upon the infrastructure, nature of study, and the type of respondents.**

Survey Sponsorship

- Goodwill of the sponsor may influence the response rates.

Facilitate the Return of Questionnaire

- **Simplify the return.**
- **Inclusion of self addressed stamped envelope encourages the response.**

Postage

- Expedited delivery may be effective.
- Use first class mail.
- Use special stamps (if available) in place of metered mail

Personalization

- Personal inside address, personally typed cover letter, and individually signed letter may make some difference.

Size, Reproduction, and Color

- **Use A-4 size paper**
- **Good quality of paper**
- **Attractive printing.**
- **Use different colors for different parts.**
- **Do not fold the questionnaire.**

**Make use of all the
techniques simultaneously
(Total Design Effort – TDE)**

E-Mail Surveys

- **Relatively new method of communication.**
- **Quite frequently used where possible.**

Advantages of E-Mail Surveys

- **Speed of distribution.**
- **Lower distribution and processing cost.**
- **More flexibility.**
- **Less handling of paper questionnaires.**
- **Many respondents more comfortable in impersonal communication.**

Disadvantages of E-Mail surveys

- “Eves-dropping” of e-mails by organization
- Maintaining anonymity is difficult. Sender’s address is there.
- Not all e-mails have the same capacity: software limitations

Internet Surveys

- Self administered questionnaire posted on a Web-site.
- Respondents provide answers to questions displayed on screen by highlighting a phrase, clicking an icon, keying in the response

Advantages of Internet Surveys

Internet surveys are becoming increasingly popular for a variety of reasons. They offer several advantages over traditional paper-based surveys, making them a preferred method for data collection in many research fields.

- **Cost-Effectiveness:** Internet surveys are significantly cheaper than traditional surveys, as they eliminate the need for printing, postage, and physical distribution of questionnaires.
- **Speed and Efficiency:** Data collection is rapid, allowing researchers to gather large amounts of information in a short period of time.
- **Geographical Reach:** Surveys can be distributed globally, reaching a diverse and widespread audience that might be difficult to access through traditional methods.
- **Flexibility and Customization:** Surveys can be easily modified or updated to reflect changes in research questions or to tailor questions to specific demographic groups.
- **Real-time Data:** Responses are collected instantly, enabling researchers to monitor trends and adjust their analysis as the survey progresses.
- **Reduced Bias:** The anonymity of online surveys can encourage more honest and accurate responses, potentially reducing social desirability bias.
- **Convenience for Respondents:** Participants can complete surveys at their own pace and on their own schedule, increasing the likelihood of participation.

Speed and Cost Effectiveness

- Reach a large audience, personalize the messages, secure confidential answer quickly and cost effectively.
- Eliminates the cost of paper, postage, data entry, and other administrative costs.
- Incremental cost of reaching additional respondents is marginal. Hence enlarge the sample.

Visual Appeal and Interactivity

- Internet surveys can be interactive.
- Use color, sound, animation which may increase respondent motivation.
- Visual material can be presented.

Respondent Participation

- Some computer users navigate to a particular Web site where questions are displayed.
- Expect participation and cooperation

Accurate Real-Time Data Capture

- **Respondents responses entered as soon as the questionnaire is submitted. (software for data cleaning as well)**
- **Researcher can review up-to-the-minute sample size counts and tabulation of data.**

Callbacks

- Depending upon the sampling frame. Easier to callback.
- Those sending incomplete questionnaires can also be identified.

Personalized and Flexible Questioning

- No interviewer in Internet survey but the software interacts with the respondent.
- Computer asks question in a sequence determine by respondents previous answer.

Respondent Anonymity

- **Respondents feel comfortable in sharing sensitive information.**
- **No interviewer, no address**

Response Rate

- **Response rate can be increased by sending e-mail friendly reminders.**

Disadvantages of Internet Surveys

- Low response rates
- Sampling error
- Non-response bias
- Coverage error
- Question wording
- Lack of control over the survey environment
- Lack of anonymity
- Lack of confidentiality
- Lack of control over the survey process
- Lack of control over the survey data
- Lack of control over the survey results
- Lack of control over the survey interpretation
- Lack of control over the survey dissemination
- Lack of control over the survey impact
- Lack of control over the survey legacy

All People cannot Participate

- **Many people do not have access to Internet facility.**
- **All with access do not have the same level of technology.**
- **Many lack powerful compatible software.**
- **Some may have no skills to navigate.**

No Physical Incentive

- Physical incentives cannot be sent

Selecting the Appropriate Survey Research Design

- No “best” form of survey.
- Compare the research objectives with the strengths and weaknesses of each method. Also
- How much confidentiality is necessary? See which method can best deliver.

Ask question like:

- Is the assistance of interviewer necessary?
- Are respondents likely to be interested in the issues being investigated?
- Will cooperation be easily attained?
- How quickly the information be easily attained?
- How quickly the information is needed?
- Will the study require a long complex questionnaire?
- How large is the budget?

The Criteria

- The cost, speed, anonymity, and the like may be different for each project.
- If none is a good fit, then use *mixed method*.

RESEARCH METHODS

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